

**Spatiotemporal Variability and Trends in Climatic Parameters
and their Association with Extreme Precipitation, Land use,
and Land cover Change in Northwest Bangladesh**



B.Sc. Project

*This project is submitted in partial fulfillment of the requirements for the
Bachelor of Science (B.Sc.) degree in Environmental Science and
Geography under the Faculty of Sciences*

Submitted by

Roll No: 2031003

Reg. No: 2130

Session: 2020-21

August 2026

**Department of Environmental Science and Geography
Islamic University, Bangladesh.**

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Submitted by-

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Professor

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August 2026

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DECLARATION

This project is a presentation of my original research work. Wherever contributions from others are involved, every effort is made to indicate this clearly, with due reference to the literature and acknowledgement of the collaborative research and discussions. The work was done under the guidance of supervisor **Dr. Atish Kumar Joardar, Professor**, Dept. of Mathematics, Islamic University Bangladesh, Kushtia-7003.

It is hereby also declared that this dissertation or any part of it has not been submitted elsewhere for the award of any degree or diploma.

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CERTIFICATE

This is to certify that this project paper entitled “*Spatiotemporal Variability and Trends in Climatic Parameters and their Association with Extreme Precipitation, Land use, and Land cover Change in Northwest Bangladesh*” is a research work conducted by **Mohd. Takiuddin**, bearing roll no. 2031003, reg. no. 2130, under my supervision for partial fulfillment of the Degree of Bachelor of Science (BSc.) in Environmental Science and Geography requirements in the Department of Environmental Science and Geography, Islamic University Bangladesh, Kushtia-7003.

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ACKNOWLEDGEMENT

First and foremost, I express my deepest gratitude to Almighty Allah, the Most Gracious and the Most Merciful, for granting me the strength, patience, and perseverance to complete this research successfully.

I am profoundly grateful to my supervisor, Professor **Dr. Atish Kumar Joardar**, Department of Mathematics, Islamic University, Bangladesh, for his valuable guidance, constructive suggestions, and continuous encouragement throughout this work. His academic insight and careful supervision helped me develop the research framework, refine the analysis, and complete this project.

I would also like to express my sincere gratitude to **Dr. Md. Abdul Al Mohit**, Professor, Department of Mathematics, Islamic University, Bangladesh, for his generous support and valuable guidance throughout this project. His insightful suggestions and encouragement greatly contributed to the successful completion of this research.

I thank **Dr. Md. Anisul Kabir**, Chairman, Department of Environmental Science and Geography, Islamic University, Kushtia, for his kind cooperation and academic support. I am also thankful to Assistant Professor **Bipul Roy** for his helpful guidance and encouragement during the preparation of this thesis.

Special thanks go to Washim Shahed Sovon and Shihab Sharar Himel for their valuable assistance, technical support, and constructive discussions at different stages of this research. Their cooperation made several challenging aspects of the work easier. I would also like to thank all the teachers and staff of the Department of Environmental Science and Geography for their support throughout my undergraduate studies.

Finally, I express my deepest gratitude to my parents and family members for their unconditional love, prayers, sacrifices, and constant support. Their encouragement has been a continuous source of strength. I am also thankful to my friends, classmates, and well-wishers who supported me in several ways during the completion of this research.

The Author

ABSTRACT

This study assessed spatial and temporal variability and long-term trends in rainfall, temperature, relative humidity, and wind speed across four districts of northwestern Bangladesh (Rajshahi, Pabna, Natore and Bogura) using monthly records for the period 1973-2023. The work combined district-level spatial comparison, Mann-Kendall trend tests with Sen's slope, seasonal and monthly climatology, extreme-value analysis of annual maximum monthly rainfall through the Generalized Extreme Value (GEV) distribution, land-use/land-cover (LULC) mapping, and correlation analysis using Pearson, Spearman and Kendall coefficients. Station data were available for Rajshahi, Pabna and Bogura; values for Natore were obtained by inverse distance weighted interpolation from neighboring stations. Results show clear spatial differences rather than a single regional pattern. Annual rainfall declined in all four districts, with statistically significant decreases in Natore, Pabna and Rajshahi; the decline in Bogura remained non-significant. Temperature rose significantly only in Rajshahi (about 0.15 °C per decade). Relative humidity increased significantly in Natore, Pabna and Rajshahi, while wind speed fell significantly in Natore and Pabna but showed weak, non-significant upward tendencies in Bogura and Rajshahi. Rainfall was strongly concentrated in the monsoon, with July the wettest month in every district. The other three variables followed their own seasonal cycles. Annual maximum monthly rainfall also trended downward in all districts, again non-significantly in Bogura. GEV models fitted the annual-maximum series adequately; estimated return levels rose with longer return periods and carried increasing uncertainty. Substantial LULC changes occurred between 1985 and 2023, yet these were treated only as environmental context, not as evidence of causal influence on the climatic trends. Correlations among the four climatic variables likewise differed from district to district. Taken together, the findings indicate that climatic variability in this part of northwestern Bangladesh is both strongly seasonal and spatially heterogeneous. The study supplies a consistent district level baseline that can support environmental planning and future research, and it underlines the value of denser station networks, daily or sub-daily rainfall records, and formal attribution studies.

Keywords: Spatiotemporal climatic variability, Rainfall trends, Extreme precipitation, Generalized Extreme Value (GEV), Land use/land cover (LULC), Northwestern Bangladesh, Mann-Kendall trend analysis.

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Chapter 1

General introduction

1.1 Background of the study

Climate variability and long-term climatic change are increasingly important considerations for environmental and resource management because changes in temperature, precipitation, atmospheric moisture, and wind conditions can alter water availability, agricultural conditions, and the functioning of environmental systems. The significance of these changes is particularly high in Bangladesh, where agricultural production and many water-dependent activities remain closely linked to climatic conditions. The World Bank has identified rising temperature and increasingly erratic rainfall as important climate-related pressures on Bangladesh, with implications for agriculture, productivity, and living standards (World Bank, 2018a). The effects of climatic variability, however, are not spatially uniform. Bangladesh contains substantial regional differences in rainfall, temperature, moisture conditions, physiography, and water availability. These differences are particularly relevant in the northwestern part of the country, where comparatively dry climatic conditions and the presence of the Barind Tract create important water-management challenges. Recent research in the Barind Tract has identified clear spatial differences in rainfall, with lower annual rainfall generally occurring in the western part and higher values toward the northeast (Esha et al., 2026). The region's dependence on groundwater for agricultural activities further increases the importance of understanding long-term rainfall and temperature behavior. Changes in climatic conditions therefore need to be examined at a spatial scale that can capture differences among individual districts rather than relying only on national or broad regional averages. Rainfall is particularly important in this context because its environmental significance depends not only on the total amount received but also on its temporal distribution. Changes in annual rainfall can occur alongside changes in seasonal or monthly rainfall patterns. Temperature is similarly important because persistent changes in temperature can affect evapotranspiration, crop water requirements, and the overall moisture balance. Relative humidity provides information about atmospheric moisture conditions, while wind speed contributes to evaporation and atmospheric moisture transport. Considering these variables together can therefore provide a broader

representation of regional climatic behavior than analysis of a single climatic parameter. Long-term observations already indicate that climatic behavior varies within northwestern Bangladesh. Ferdous and Baten (2012), using 50 years of data from 1961-2010, examined temperature, rainfall, relative humidity, and sunshine across regions of the former Rajshahi Division and reported different trends among Rajshahi, Rangpur, and Dinajpur. For example, annual rainfall decreased in Rajshahi while increasing trends were observed in Rangpur and Dinajpur, demonstrating that climatic trends cannot necessarily be generalized across the broader northwestern region. Sunny et al. (2020) subsequently examined rainfall, relative humidity, and temperature in Rajshahi and Sylhet over a 50-year period and assessed their annual, seasonal, and monthly behavior. More recently, Shahriyar et al. (2026) examined rainfall variability in Rajshahi City for 1990-2024 using NASA POWER data and found that rainfall behavior varied across temporal scales. These studies demonstrate the value of long-term climatic analysis but also indicate that climatic behavior needs to be assessed at appropriate spatial and temporal scales. A district may experience a different climatic tendency from a neighboring district, while a trend at the annual scale may not represent changes during individual seasons or months. A consistent comparison across several districts can therefore provide information that is difficult to obtain from single-station or single-location studies.

Another important dimension is the behavior of extreme precipitation. Average rainfall and long-term rainfall trends do not necessarily describe unusually large precipitation events. Extreme rainfall can have implications for drainage, agriculture, water management, and infrastructure even where the long-term annual rainfall trend is weak. Extreme-value analysis provides a way of examining the magnitude and recurrence of such events. In this study, the available monthly record is used to construct an Annual Maximum Series (AMS), with the largest monthly rainfall observation selected for each year. The Generalized Extreme Value (GEV) distribution is then used to estimate rainfall return levels for selected recurrence periods. This approach provides information on the probability characteristics of high-magnitude monthly precipitation that cannot be obtained from ordinary trend analysis alone. The physical landscape has also changed during the period considered in this study. Land use and land cover (LULC) changes in Bangladesh have been associated with population growth, agricultural expansion, urbanization, and other socioeconomic processes (Xu et al., 2020). Such changes are particularly relevant in northwestern Bangladesh because alterations in vegetation, agricultural land, built-up areas, and other land-cover types can influence important

environmental processes. Siddik et al. (2022), for example, demonstrated that LULC change in northwestern Bangladesh has implications for groundwater recharge and found substantial increases in impervious built-up land alongside reductions in vegetated land cover. LULC is therefore considered in the present study as environmental context for interpreting climatic patterns, rather than as a variable through which climatic trends are causally attributed. Against this regional and environmental context, the present study assesses the spatiotemporal variability and trends of rainfall, temperature, relative humidity, and wind speed across Rajshahi, Pabna, Natore, and Bogura during 1973-2023. The study combines long-term trend analysis, seasonal and monthly climatology, spatial comparison, extreme precipitation analysis using AMS and GEV, LULC change assessment, and correlation analysis among the selected climatic variables. The combined approach is intended to provide a consistent assessment of climatic behavior across the selected districts and to place extreme precipitation and climatic variability within their broader environmental setting.

1.2 Problem statement

The principal problem addressed by this study is that climatic change and variability in northwestern Bangladesh cannot be adequately understood from a single climatic variable, a single location, or a single temporal scale. Existing studies have established that climatic variables in the region exhibit spatial and temporal variability. Ferdous and Baten (2012), for example, identified different rainfall, temperature, and relative-humidity trends across Rajshahi, Rangpur, and Dinajpur. More recent studies have examined rainfall trends in Rajshahi City and the wider Barind Tract, but these studies have different spatial coverage, periods, datasets, and analytical scopes. Shahriyar et al. (2026), for instance, analyzed 1990-2024 rainfall using NASA POWER data, while Esha et al. (2026) focused on annual rainfall in the Barind Tract using 18 BWDB stations for 2000-2024. What remains insufficiently addressed is a consistent, long-term comparative assessment across Rajshahi, Pabna, Natore, and Bogura that considers multiple dimensions of climatic behavior within the same analytical framework. In particular, there is a need to examine rainfall, temperature, relative humidity, and wind speed together over a common 51-year period; compare their spatial and temporal patterns; evaluate seasonal and monthly variability; quantify monotonic trends and their magnitudes; assess the recurrence characteristics of extreme monthly precipitation;

document major LULC changes over the same broad period; and examine statistical relationships among the climatic variables. The gap is therefore not simply the absence of another climate-trend study. Rather, it concerns the limited integration of these complementary components within a common four-district framework. Previous research has often concentrated on stations, selected climatic variables, rainfall alone, or different periods and data sources. The present study addresses this gap by bringing these components together using a consistent study period and analytical framework.

Extreme precipitation represents a further problem because ordinary rainfall averages and trend statistics do not describe the probability of high-magnitude events. The present study addresses this through GEV modelling of annual maximum monthly rainfall. This provides return-level estimates for 2-, 5-, 10-, 25-, 50-, and 100-year periods, together with uncertainty intervals. However, because the underlying meteorological dataset is monthly, the analysis does not represent standard daily precipitation extremes. The study also considers LULC change because the environmental setting of the selected districts has changed during the study period. Changes in built-up areas, agricultural land, vegetation, water bodies, and bare land provide useful spatial context for interpreting climatic patterns. The purpose is not to demonstrate that LULC change caused observed climatic trends, since the study does not employ a causal attribution framework. Instead, the LULC component documents major environmental changes occurring within the same broad period as the climatic observations. Thus, the problem addressed by this research is the need for an integrated, long-term, and spatially comparative assessment of climatic variability and trends, extreme monthly precipitation, LULC change, and interrelationships among climatic parameters across the selected districts of northwestern Bangladesh.

1.3 Motivation and significance of the study

The motivation for the study is closely related to the environmental conditions of northwestern Bangladesh, where climatic variability has direct relevance to agriculture and water availability. The Barind region is characterized by comparatively dry conditions and substantial dependence on groundwater for agricultural activities. Recent evidence of spatial rainfall differences across the Barind Tract further indicates that water-related climatic conditions cannot be assumed to be uniform across the region (Esha et al., 2026). At the national level, climate-related changes in temperature and

rainfall are already recognized as important pressures on Bangladesh's agricultural and socioeconomic systems (World Bank, 2018b). The selection of Rajshahi, Pabna, Natore, and Bogura is motivated by the need to examine these climatic conditions at a district level. Although the districts belong to the broader northwestern region, their climatic characteristics are not necessarily identical. A comparative analysis can therefore identify whether observed climatic tendencies are regionally consistent or spatially differentiated. The 1973-2023 period provides another important basis for the study. A 51-year record is sufficiently long to examine persistent climatic tendencies while also allowing seasonal and monthly behavior to be evaluated. The long record also makes it possible to compare climatic conditions across different portions of the study period rather than relying solely on beginning-to-end differences. The extreme precipitation component adds a further practical dimension. A long-term trend in rainfall does not necessarily indicate whether high-magnitude rainfall events are becoming important. Estimating return levels through GEV analysis provides a statistical description of the magnitude and recurrence of annual maximum monthly rainfall. Such information can contribute to broader consideration of extreme-rainfall preparedness and environmental planning, while remaining appropriately limited by the monthly nature of the dataset.

The LULC component is significant because the landscape itself has changed during the period over which climatic conditions are assessed. Bangladesh has experienced substantial LULC transitions, while research in northwestern Bangladesh has demonstrated that land-cover changes can affect groundwater-recharge processes (Siddik et al., 2022; Xu et al., 2020). Including LULC change therefore helps place the climatic findings within a changing environmental landscape. Overall, the significance of the study lies in its integrated district-level assessment. It does not rely on a single trend statistic or a single climatic variable. Instead, it combines spatial and temporal analysis, formal trend detection, seasonal and monthly climatology, extreme-value analysis, LULC assessment, and inter-variable correlation. The resulting evidence can support a more detailed understanding of climatic conditions in the selected districts and provide a baseline for future environmental and climate-related research. The findings may be relevant to environmental management, climate-sensitive agricultural planning, water-resource considerations, and preparedness for high-rainfall events. They may also provide a useful reference for future studies using longer records, daily meteorological observations, additional extreme-climate indices, or more advanced methods of climate attribution.

1.4 Aim and Objectives of the Study

1.4.1 Aim of the Study

To assess the spatiotemporal variability and long-term trends of selected climatic parameters and examine their relationship with extreme precipitation and land use/land cover change in Rajshahi, Pabna, Natore, and Bogura districts of northwestern Bangladesh during 1973-2023.

1.4.2 Specific Objectives

The study was conducted with the following specific objectives:

- ▶ To analyze the spatial and temporal variations of key climatic parameters across the selected districts and examine their association with changes in land use and land cover (LULC).
- ▶ To assess the long-term trends and magnitude of changes in the selected climatic parameters using the Mann-Kendall trend test and Sen's slope estimator.
- ▶ To characterize the seasonal and monthly variability of the selected climatic parameters and identify their temporal climatological patterns across the study area.
- ▶ To assess extreme precipitation characteristics and estimate the return levels of extreme rainfall events using Annual Maximum Series (AMS) and Generalized Extreme Value (GEV) analysis.
- ▶ To examine the relationships among the selected climatic parameters using correlation analysis and identify their patterns of association through correlation matrices and heat-map visualization.

1.5 Scope of the Study

The study covers four districts of northwestern Bangladesh: Rajshahi, Pabna, Natore, and Bogura. The climatic analysis spans 51 years, from 1973 to 2023, using monthly observations of four climatic parameters: rainfall, temperature, relative humidity, and wind speed. Meteorological data were obtained from the Bangladesh Agricultural Research Council (BARC). Direct station data were available for Rajshahi, Bogura, and Pabna (Ishurdi). Since a corresponding direct station record for Natore was unavailable in the dataset, climatic values for Natore were estimated using Inverse Distance Weighted

(IDW) interpolation based on surrounding stations. The climatic assessment covers annual, seasonal, and monthly time scales. The study period is divided into four intervals for spatial comparison, while the complete annual series is used for formal trend assessment. Long-term trends are evaluated using the Mann-Kendall test and Sen's slope estimator, supported by linear trend visualization. Seasonal analysis considers four seasons: winter (December-February), pre-monsoon (March-May), monsoon (June-September), and post-monsoon (October-November). Monthly climatology is calculated to describe the typical behavior of each climatic parameter throughout the year. The LULC assessment covers four reference years, 1985, 1998, 2011, and 2023, aligned with the same four sub-periods used for the climatic thematic mapping, and considers four major classes: built-up area, water bodies, vegetation/forest, and bare land. Agricultural land was not retained as a separate class because it could not be reliably distinguished from vegetation and bare land during visual interpretation of the Landsat imagery. The purpose is to document major changes in the physical landscape and provide environmental context for interpreting the climatic assessment. It does not attempt to establish a causal relationship between LULC change and observed climatic trends.

The extreme precipitation component is specifically limited by the temporal resolution of the meteorological dataset. Because complete daily observations were unavailable, an Annual Maximum Series was constructed using the highest monthly rainfall value recorded in each year. Accordingly, the GEV results describe the recurrence characteristics of annual maximum monthly rainfall, rather than daily extreme precipitation. The GEV model is used to estimate return levels for 2-, 5-, 10-, 25-, 50-, and 100-year return periods, with 95% bootstrap confidence intervals used to represent uncertainty. These return levels are statistical estimates derived from the available record and should not be interpreted as exact predictions of future rainfall events. Finally, relationships among rainfall, temperature, relative humidity, and wind speed are examined using Pearson, Spearman, and Kendall correlation coefficients. The correlation analysis identifies statistical associations among the variables; it does not establish causal relationships.

1.6 Organization of the study

The report is organized into five chapters.

Chapter 1: Introduction presents the background of the study, problem statement, motivation and significance, aim and objectives, scope, and organization of the report. It establishes the rationale for examining climatic variability, trends, extreme precipitation, and LULC change in the selected districts.

Chapter 2: Literature Review reviews previous studies related to climatic variability and change, climatic trends in Bangladesh and northwestern Bangladesh, seasonal and spatial variability, extreme precipitation and extreme-value analysis, LULC change, and relationships among climatic variables. The chapter also establishes the research gap addressed by the present study.

Chapter 3: Methodology describes the study area, data sources, data preparation, climatic analysis, spatial and temporal assessment, trend analysis, seasonal and monthly analysis, extreme precipitation and GEV modelling, LULC classification and change analysis, correlation analysis, software, and analytical workflow.

Chapter 4: Results and Discussion presents and interprets the findings of the study. It covers spatial and temporal climatic patterns, long-term trends, seasonal and monthly variability, LULC changes, extreme precipitation and GEV return levels, and correlations among the selected climatic parameters. The findings are compared with relevant previous studies.

Chapter 5: Conclusion and Recommendations summarizes the major findings in relation to the research objectives, presents the conclusions of the study, discusses the implications and limitations of the findings, and provides recommendations for environmental management and future research.

Chapter 2

Literature Review

2.1 Climate Change, Variability and Climatic Extremes

Climate is characterized by both persistent long-term changes and shorter-term fluctuations. Climate variability refers to variations in climatic conditions around their longer-term state, whereas climate change involves persistent changes in the mean state or variability of the climate system over extended periods. Distinguishing between these components is important when interpreting long-term meteorological records because substantial year-to-year variability can occur even within a period showing an overall trend. The assessment of climate change has increasingly moved beyond changes in mean climatic conditions to include changes in extremes. The Sixth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC) indicates that many land regions have experienced increases in the frequency and intensity of heavy precipitation events, although the magnitude and confidence of these changes vary spatially (IPCC, 2021). Changes in extreme events are particularly important because their environmental and socioeconomic consequences can be disproportionate to changes in average climatic conditions.

For Bangladesh, the distinction between climatic variability, long-term trends and extremes is particularly relevant. The country's climate is strongly influenced by the South Asian monsoon, producing substantial seasonal differences in rainfall and temperature. Consequently, a long-term change in annual rainfall does not necessarily imply a uniform change throughout the year. Similarly, a relatively stable annual average may coexist with substantial changes in individual seasons or unusually wet and dry years. Long-term observational research has demonstrated that climatic changes in Bangladesh are not spatially uniform. Khan et al. (2019), using meteorological observations from 1988-2017, reported significant increases in maximum and minimum temperature while identifying reductions in wind speed and precipitation in several parts of the country. The northwestern region showed particularly pronounced changes in wind speed. Their analysis therefore provides evidence of a regional climatic signal, but the 30-year study period and station-based national framework also mean that the results cannot fully describe longer-term changes at the district scale.

Similarly, Kamruzzaman et al. (2018) examined rainfall and temperature records from 12 meteorological stations across western Bangladesh over 1961-2011. The study applied Mann-Kendall, Spearman's rho, modified Mann-Kendall and Sen's slope methods, including adjustment for lag-1 serial correlation in relevant series. Its broad spatial coverage is an important strength, but its focus on rainfall and temperature leaves other climatic parameters less examined. More importantly, a regional station network covering a large proportion of western Bangladesh cannot resolve all district-level differences within the region. The existing evidence therefore establishes that climatic change in Bangladesh has a strong spatial component, but it also reveals an important limitation in the literature: the scale at which a trend is detected matters. National studies can identify broad patterns, while regional and station-level studies may reveal contrasting local behavior. Similar national-scale evidence is provided by Alam et al. (2023) and Haider et al. (2024), who each documented spatially heterogeneous long-term climatic trends across Bangladesh using multi-decadal records. A climatic tendency identified at one station or within one sub-region should consequently not be treated as representative of all neighboring districts. This issue is especially important for northwestern Bangladesh, where relatively dry conditions, strong seasonal dependence on rainfall and recurrent water stress make local climatic variability particularly relevant. Literature consequently supports a more spatially explicit assessment of climatic variables, while also demonstrating the need to examine long-term variability and extremes separately rather than treating them as interchangeable components of climate change.

2.2 Climatic Parameters and Their Spatial and Temporal Variability

Rainfall, temperature, relative humidity and wind speed describe different components of the atmospheric environment and are influenced by different physical processes. Their joint assessment can provide a more complete representation of climatic conditions than an analysis restricted to a single variable. At the same time, the four parameters should not be expected to exhibit identical trends because their responses to atmospheric circulation, moisture availability and local environmental conditions differ. Rainfall shows particularly strong spatial and seasonal variability in Bangladesh because of the dominance of the monsoon system. Studies in western Bangladesh have repeatedly reported differences in the direction and magnitude of rainfall trends among stations.

Kamruzzaman et al. (2018), for example, found spatially varying rainfall and temperature trends across 12 stations. Khan et al. (2019) likewise identified regional differences in precipitation and other climatic variables. These findings demonstrate that rainfall behavior is not adequately represented by a single national trend. Temperature trends have received more consistent attention in Bangladesh. Khan et al. (2019) reported significant increases in both maximum and minimum temperature, with maximum temperature increasing at a faster rate. However, their results also showed differences among seasons and regions, indicating that warming is not necessarily uniform throughout the year. More recent national-scale work has similarly identified differences in the direction and magnitude of temperature changes across months and stations. Relative humidity has received considerably less attention than rainfall and temperature. Fattah et al. (2023) specifically addressed this limitation by examining 40 years of relative-humidity observations across Bangladesh. Their analysis incorporated monthly, seasonal and annual trends and examined the influence of temperature, rainfall and wind speed using Pearson, Spearman and Kendall correlations. The study found substantial spatial variability in relative humidity and positive trends at the examined temporal scales. The importance of this study lies not only in its findings but also in what it reveals about literature. Relative humidity has often been treated as a secondary variable in climate assessments, despite its close relationship with temperature and atmospheric moisture. Fattah et al. (2023) therefore broadened the climatic perspective, but their national-scale analysis does not resolve the specific climatic relationships among neighboring districts in northwestern Bangladesh.

Wind speed represents another relatively underexamined component of regional climate. Khan et al. (2019) reported a significant decrease in wind speed across Bangladesh, with particularly strong reductions in the northwestern region. This finding is important because wind-speed changes occurred alongside changes in temperature, precipitation and humidity. However, because the study covered the country, the magnitude and consistency of these relationships at the scale of individual neighboring districts remain less certain. Recent local rainfall studies further demonstrate why spatial and methodological context matters. Shahriyar et al. (2026) examined rainfall in Rajshahi City for 1990-2024 using NASA POWER data. The study reported a statistically significant annual increase of 34.32 mm/year, with the strongest increases occurring during the pre-monsoon and monsoon periods. However, the dataset was derived from a gridded NASA POWER product rather than a conventional station record, and the study

was restricted to Rajshahi City and rainfall. The authors themselves note that gridded data can introduce spatial-resolution uncertainty at local scales. The findings of Shahriyar et al. (2026) are particularly interesting when compared with the contemporaneous study of Esha et al. (2026). Esha et al. analyzed annual rainfall from 18 BWDB stations across the Barind Tract during 2000–2024 and found a predominantly decreasing pattern, including statistically significant declines at Badalgachi, Nachole and Chapai Nawabganj. Rajshahi station itself showed a weak, statistically insignificant positive trend. These apparently contrasting results should not be treated simply as contradictory evidence. The studies differ in data source, spatial coverage, study period and analytical unit. Shahriyar et al. used a gridded dataset for Rajshahi City over 1990–2024, whereas Esha et al. used station observations from a much broader Barind network over 2000–2024. The comparison demonstrates a practical problem in regional climate research: conclusions may vary when different datasets and spatial scales are used. It also reinforces the need for a consistent dataset and common analytical framework when comparing neighboring districts. Esha et al. (2026) also provide particularly useful evidence of fine-scale spatial heterogeneity. Their 18-station network revealed a southwest–northeast rainfall gradient and substantial differences in Sen's slope among stations. However, their analysis was focused on annual rainfall and did not examine temperature, relative humidity or wind speed. Furthermore, the study itself acknowledges that annual aggregation limits investigation of seasonal and intra-annual rainfall behavior. Serial correlation was also not formally tested before the Mann-Kendall analysis. Thus, the recent literature has improved the spatial understanding of rainfall in northwestern Bangladesh, but it has also exposed the limitations of single-variable and study-specific approaches. The evidence is strongest for rainfall, while temperature, relative humidity and wind speed remain less integrated within the same spatial framework. This provides a clear basis for examining the four parameters together using a common long-term record.

2.3 Climate Trend Analysis and Statistical Approaches

Long-term climatic records contain substantial interannual variability, making visual changes difficult to distinguish from systematic trends. Statistical trend analysis is therefore necessary to determine whether an observed increase or decrease represents a persistent monotonic tendency. However, no single statistical method is sufficient for describing all aspects of a climatic trend. Linear regression is commonly used to visualize

the direction and approximate rate of change in a time series. Its usefulness lies primarily in presenting the overall direction of change in an accessible form. Climatic records, however, frequently depart from normality and may contain outliers or non-linear variability. For this reason, non-parametric methods are widely used in hydroclimatic research. The Mann–Kendall (MK) test provides a rank-based method for detecting monotonic trends without requiring the observations to follow a normal distribution (Mann, 1945). Sen's slope estimator complements the MK test by providing a robust estimate of the magnitude of a detected trend (Sen, 1968). The combination is therefore more informative than using statistical significance alone: the MK test addresses whether a monotonic tendency is present, while Sen's slope indicates its approximate rate of change. These methods have been widely applied in Bangladesh. Khan et al. (2019) combined linear regression, MK and Sen's slope to assess climatic trends, while Kamruzzaman et al. (2018) applied MK, Spearman's rho, modified MK and Sen's slope to rainfall and temperature observations from western Bangladesh. Sunny et al. (2020) similarly used MK and Sen's slope to investigate 50 years of rainfall, relative humidity and temperature data in Rajshahi and Sylhet. The use of these methods, however, requires attention to the properties of the underlying time series. Serial correlation is a particularly important issue because temporal dependence can alter the variance of the MK statistics and affect the assessment of statistical significance. Hamed and Rao (1998) demonstrated the importance of accounting for serial correlation in trend analysis of environmental data. Kamruzzaman et al. (2018) explicitly addressed lag-1 serial correlation through a modified MK procedure, illustrating that methodological treatment of dependence can influence the interpretation of climatic trends. The recent Barind study by Esha et al. (2026) is instructive in this regard. Although the study used MK and Sen's slope appropriately for non-parametric rainfall trend analysis, it acknowledged that serial correlation was not formally tested before the MK procedure. This does not invalidate the reported rainfall patterns, but it illustrates why trend estimates from different studies should be compared with attention to methodological details rather than only to the direction of the reported trend.

A second limitation concerns the length of the available record. Shorter records may capture temporary wet or dry periods that can influence the estimated trend. This is particularly relevant when comparing the recent 25-year Barind study with longer records extending over several decades. A longer common observation period can provide a more stable basis for assessing persistent climatic tendencies, although it does not eliminate

natural variability or potential changes in station characteristics. Accordingly, the literature supports the use of graphical trend representation together with MK significance testing and Sen's slope estimation, if results are interpreted with attention to temporal dependence, record length and data quality. The present study adopts this combined framework to assess the direction and magnitude of climatic changes across the selected districts.

2.4 Seasonal Variability and Climatological Analysis

Annual climatic statistics provide useful summaries, but they can obscure substantial differences within the year. This limitation is particularly important in Bangladesh because the monsoon produces strong seasonal contrasts in rainfall, while temperature, humidity and wind speed also exhibit pronounced annual cycles. Sunny et al. (2020) demonstrated this issue by examining climatic variables in Rajshahi and Sylhet at annual, seasonal and monthly scales. Their comparison of two periods showed that climatic behavior varied considerably according to both location and temporal scale. The study is therefore useful in demonstrating the value of climatological analysis, although its comparison of Rajshahi with Sylhet does not address differences among neighboring districts within northwestern Bangladesh. Khan et al. (2019) likewise found that climatic trends differed between seasons. Their national analysis showed, for example, stronger changes in some temperature and wind-speed characteristics during seasons and regions. Such findings indicate that annual trends can represent an average of substantially different seasonal processes. The recent Rajshahi study provides a more direct example. Shahriyar et al. (2026) found that rainfall increases were concentrated mainly between March and October, with the strongest annual contribution coming from the monsoon period, while winter changes were comparatively weak. Thus, the reported increase in annual rainfall does not represent an equivalent increase throughout the year. This contrasts with the annual-only emphasis of Esha et al. (2026). Although the Barind study provides valuable spatial information from 18 stations, its authors explicitly identify the use of annual data as a limitation because it restricts examination of seasonal and intra-annual rainfall changes. The comparison between these recent studies illustrates why temporal scale should be treated as an analytical issue rather than merely a presentation choice. Annual rainfall can increase while rainfall during a particular season remains unchanged or decreases. Similarly, a temperature trend calculated from annual means

may conceal opposing monthly tendencies. Monthly climatology therefore provides the baseline for understanding the normal seasonal cycle, while seasonal aggregation allows changes to be interpreted in relation to the major climatic periods of the region. For the present study, this approach is particularly relevant because winter, pre-monsoon, monsoon and post-monsoon conditions differ substantially in rainfall and atmospheric characteristics. The literature consequently supports a multi-scale assessment in which annual trends are examined alongside monthly climatology and seasonal variability. This does not duplicate trend analysis; rather, it provides the temporal context necessary to interpret where within the annual cycle the observed changes are occurring.

2.5 Extreme Precipitation and GEV Analysis

Long-term changes in average rainfall do not necessarily describe the behavior of unusually large precipitation events. Extreme rainfall is important because relatively infrequent events can produce substantial impacts on drainage, agriculture, water resources and infrastructure. Consequently, analysis of the upper tail of rainfall distributions provides information that ordinary mean and trend analyses cannot provide. Extreme-value theory offers a statistical framework for analyzing observations that occur near the tails of a distribution. One established approach is the block-maxima method, in which the largest observation within a defined block is selected for analysis. When annual blocks are used, the resulting series is commonly referred to as an Annual Maximum Series (AMS). The Generalized Extreme Value (GEV) distribution provides the standard limiting distribution for block maxima (Fisher & Tippett, 1928; Gumbel, 1958; Jenkinson, 1955) and is widely used in extreme-value analysis (Coles, 2001). The GEV framework allows extreme observations to be characterized through location, scale and shape parameters and provides a basis for estimating return levels. However, fitting a GEV distribution does not automatically guarantee that the resulting estimates are reliable. Model diagnostics, goodness-of-fit assessment and uncertainty estimation are necessary because extreme-value estimates are sensitive to sample size and model specification (Coles, 2001; Katz, Parlange, & Naveau, 2002). Return-period analysis is particularly sensitive to the length of the observed record. A return level associated with a long recurrence interval is estimated from the tail of the fitted distribution, where observations are sparse. Consequently, uncertainty generally increases with return period.

Confidence intervals should therefore accompany return-level estimates rather than presenting them as deterministic predictions.

Another important consideration is the temporal resolution of the rainfall data. Much of the extreme-precipitation literature uses daily or sub-daily observations, and many standard extreme-rainfall indices are explicitly designed around daily precipitation. Such analyses are not directly equivalent to analyses based on monthly totals. This distinction is important for the present study. The available rainfall data are monthly, so the AMS is constructed by selecting the maximum monthly rainfall total within each year. Therefore, the resulting GEV model describes the statistical behavior of extreme monthly rainfall totals rather than annual maximum daily or sub-daily rainfall. This limitation does not prevent extreme-value analysis, but it determines how the resulting return levels should be interpreted. Recent regional literature also demonstrates the importance of distinguishing ordinary rainfall variability from extreme-event analysis. Shahriyar et al. (2026) examined monthly, seasonal and annual rainfall trends in Rajshahi City but did not perform GEV analysis. Their study therefore provides evidence of changing rainfall totals but does not address the probability structure of extreme monthly rainfall. Similarly, Esha et al. (2026) provided detailed annual rainfall variability and trend analysis across 18 Barind stations but focused on annual rainfall totals, anomalies and monotonic trends rather than fitting an extreme-value model. The study itself identifies the need for future research to examine rainfall at finer temporal scales. This distinction reveals an important limitation in the recent local literature: improved knowledge of rainfall trends does not necessarily provide information about extreme-event magnitude or return levels. Trend analysis and extreme-value analysis answer different questions and are therefore complementary rather than interchangeable. For the present study, AMS–GEV analysis is consequently used as a separate component of the climatic assessment. Its purpose is not to replace ordinary rainfall trend analysis but to characterize the upper tail of the annual maximum monthly rainfall series and estimate corresponding return levels with explicit consideration of uncertainty.

2.6 LULC Context and Interrelationships among Climatic Variables

Land use and land cover (LULC) provide an important environmental context for interpreting climatic variability. Changes in vegetation, agricultural land, water bodies and built-up surfaces can modify surface characteristics and hydrological processes. At

the same time, climatic conditions and extreme events can influence land-use decisions. The relationship is therefore complex and should not be reduced to a simple assumption that LULC change directly causes climatic trends. At the national scale, Xu et al. (2020) used Landsat imagery and biophysical and socioeconomic information to examine LULC dynamics in Bangladesh between 2000 and 2010. The study identified major transitions involving agricultural land, water bodies, forest and shrubland and demonstrated that different types of LULC change have different drivers. Its national spatial coverage is a major strength, although the relatively short decadal period limits its usefulness for interpreting longer-term climatic relationships. More geographically relevant evidence comes from Siddik et al. (2022), who examined LULC change and groundwater recharge in northwestern Bangladesh. They reported an 80.3% increase in impervious built-up areas and a 16.4% decrease in vegetated land cover between 2006 and 2016. Their modelling demonstrated that LULC changes can influence hydrological processes, with stronger effects at regional scales and during different seasons. However, the study focused on groundwater recharge rather than climatic trends, so its findings cannot by themselves establish a climatic effect of LULC change. This distinction is important when LULC is incorporated into climate study. Spatial agreement between a climatic change and a land-cover transition may suggest an association, but it does not demonstrate causality. Establishing a causal relationship would require additional variables and attribution modelling. Therefore, LULC change is more appropriately treated as environmental context for interpreting climatic differences in the present study. The second component of this section concerns relationships among climatic variables. Temperature, rainfall, relative humidity and wind speed are physically interconnected, so changes in one variable may occur alongside changes in others. Statistical correlation can help identify the strength and direction of these associations, although it cannot establish a causal mechanism. Khan et al. (2019) examined relationships among climatic variables in Bangladesh using Pearson and Spearman correlation coefficients in addition to their trend analysis. Their work demonstrated that the relationships among climatic variables vary across regions and seasons. However, the national-scale design makes it difficult to determine whether the same relationships occur consistently among neighboring districts in the northwest. Fattah et al. (2023) provided a more focused treatment of climatic interrelationships by using Pearson, Spearman and Kendall correlations to investigate the influence of temperature, rainfall and wind speed on relative humidity. This multi-method approach is particularly relevant because the three correlation measures do not depend on

the same assumptions and can therefore provide complementary evidence about the consistency of an association.

Nevertheless, correlation results require cautious interpretation. Pearson correlation primarily measures linear association, while Spearman and Kendall assess rank-based monotonic association. A strong coefficient therefore indicates statistical co-variation rather than a direct physical mechanism. Climatic variables can also respond simultaneously to atmospheric processes that are not included in the correlation analysis. The existing literature thus provides a basis for combining LULC assessment with climatic-variable correlation, but it does not establish the specific relationships among all four selected variables across the four districts of the present study. This leaves room for a more consistent multi-variable assessment in which correlations are interpreted alongside, rather than independently of, the observed spatial and temporal climatic patterns.

2.7 Research Gap

Research on climate variability in Bangladesh is extensive, but it is fragmented across variables, locations, time periods, data sources and analytical methods. Existing studies have provided important evidence of changes in rainfall, temperature, humidity and wind speed, while regional studies have documented substantial spatial differences in climatic behavior. However, the available evidence does not provide a consistent long-term assessment of the four selected climatic variables across the specific four-district framework used in the present study. Earlier regional studies provide a strong foundation but also demonstrate the limitations of existing approaches. Kamruzzaman et al. (2018) examined 12 stations across western Bangladesh over 1961-2011 and incorporated serial-correlation considerations, but the study focused primarily on rainfall and temperature. Khan et al. (2019) expanded the analysis to several climatic variables across Bangladesh, but its 1988-2017 period and national station framework do not provide a detailed four-district comparison. Sunny et al. (2020) examined rainfall, relative humidity and temperature at Rajshahi and Sylhet over 50 years but did not include wind speed or extreme-value analysis and did not compare the four neighboring districts considered here.

The most recent studies make the regional gap more specific. Shahriyar et al. (2026) provides recent evidence of increasing rainfall in Rajshahi City using a 1990-2024 NASA

POWER dataset, with a reported annual Sen's slope of 34.32 mm/year. Their analysis is valuable for understanding recent rainfall behavior at the city scale, but it remains restricted to rainfall and a single urban location. The use of a gridded dataset also introduces a different data structure from station-based observations. Esha et al. (2026), in contrast, used 18 BWDB stations across the Barind Tract for 2000-2024 and found significant rainfall declines at several western and southwestern stations, while Rajshahi showed a weak and statistically insignificant increase. This study provides much stronger spatial representation than a single-station or single-grid analysis, but it is restricted to annual rainfall and therefore does not address the seasonal and monthly behavior of multiple climatic variables. Its use of annual data also limits assessment of intra-annual changes. The contrast between these two 2026 studies are particularly instructive. They do not simply provide two competing answers about Rajshahi rainfall; they demonstrate how data source, spatial scale, study period and analytical scope can influence the climatic picture obtained. A consistent dataset and common analytical framework are therefore important when comparing neighboring districts. A second gap concerns extreme precipitation. Existing local rainfall studies have concentrated primarily on variability and monotonic trends. Shahriyar et al. (2026) and Esha et al. (2026), for example, provide detailed rainfall trend assessments but do not fit a GEV model to an Annual Maximum Series. Consequently, their results do not provide return-level estimates for extreme rainfall. Studies that do apply GEV modelling to Bangladesh rainfall, such as Alam et al. (2018) and Dey and Patwary (2025), and the comparable Chittagong-region analysis by Rahman et al. (2025), provide useful methodological precedents but are geographically distinct from the four districts examined here. At the same time, much of the wider extreme-precipitation literature is based on daily or sub-daily observations, whereas the monthly data available for the present study requires a different interpretation of the extreme-value series.

A third gap concerns the integration of climatic variables with environmental change. Bangladesh has substantial LULC research, including foundational work by Dewan and Yamaguchi (2009) in Greater Dhaka, national-scale work by Xu et al. (2020), northwestern Bangladesh studies such as Siddik et al. (2022), Rajshahi-specific modelling by Kafy et al. (2020) linking LULC change to land surface temperature, and more recent machine-learning-based mapping such as Biswas et al. (2023) in Chattogram. However, these studies have primarily addressed LULC dynamics and hydrological consequences rather than integrating LULC change with long-term trends in rainfall, temperature,

relative humidity, and wind speed. A fourth gap concerns inter-variable relationships. Correlation-based climatic studies have demonstrated relationships among temperature, rainfall, relative humidity, and wind speed, but these relationships have been investigated at national or selected-station scales. Their consistency across the four neighboring districts of Rajshahi, Pabna, Natore and Bogura has not been established within a single long-term analytical framework. Therefore, the specific research gap addressed by the present study is: Despite useful earlier work, no study identified in the reviewed literature has provided a consistent long-term (1973-2023) comparative assessment of rainfall, temperature, relative humidity and wind speed across Rajshahi, Pabna, Natore and Bogura that simultaneously incorporates formal trend detection, seasonal and monthly climatology, extreme monthly precipitation through GEV analysis, LULC change as environmental context, and multi-method correlation analysis. Existing research remains fragmented by variable selection, spatial coverage, temporal period, data source, and analytical scope. The present study addresses this gap through a common 1973-2023 framework covering the four selected districts. It combines spatial and temporal assessment of four climatic parameters with linear trend visualization, Mann-Kendall testing, Sen's slope estimation, seasonal and monthly climatological analysis, Annual Maximum Series and GEV modelling of extreme monthly rainfall, LULC change assessment and Pearson, Spearman, and Kendall correlation analysis. The purpose is not to attribute observed climatic changes to a single cause, but to provide a consistent description of the climatic behavior of the selected districts and to examine LULC and inter-variable relationships as supporting environmental context. This integrated approach also responds to a practical limitation in the existing literature: individual studies often provide robust evidence for one component of regional climate change, but comparisons across studies are complicated when they use different periods, datasets, station networks, and analytical procedures. By applying a common framework to the same four districts, the present study provides a more directly comparable assessment of their long-term climatic characteristics.

Chapter 3

Methodology

3.1 Study Area

The present study was conducted in four selected districts of northwestern Bangladesh: Rajshahi, Pabna, Natore, and Bogura. These districts were selected to assess the spatial and temporal variation of climatic parameters and to examine their relationship with extreme precipitation and land use/land cover (LULC) changes across the region. The study area represents an environmentally important region of Bangladesh, where climatic conditions are closely associated with agricultural activities, water availability, land-use patterns, and other regional environmental processes. The region is generally characterized by a warm and humid climate, with predominantly flat to gently undulating terrain. A considerable portion of the study area falls within or is influenced by the Barind tract, which has distinctive environmental and climatic characteristics compared with other parts of the country. The regional climatic and environmental conditions are also influenced by major river systems, along with their associated floodplains and drainage networks.

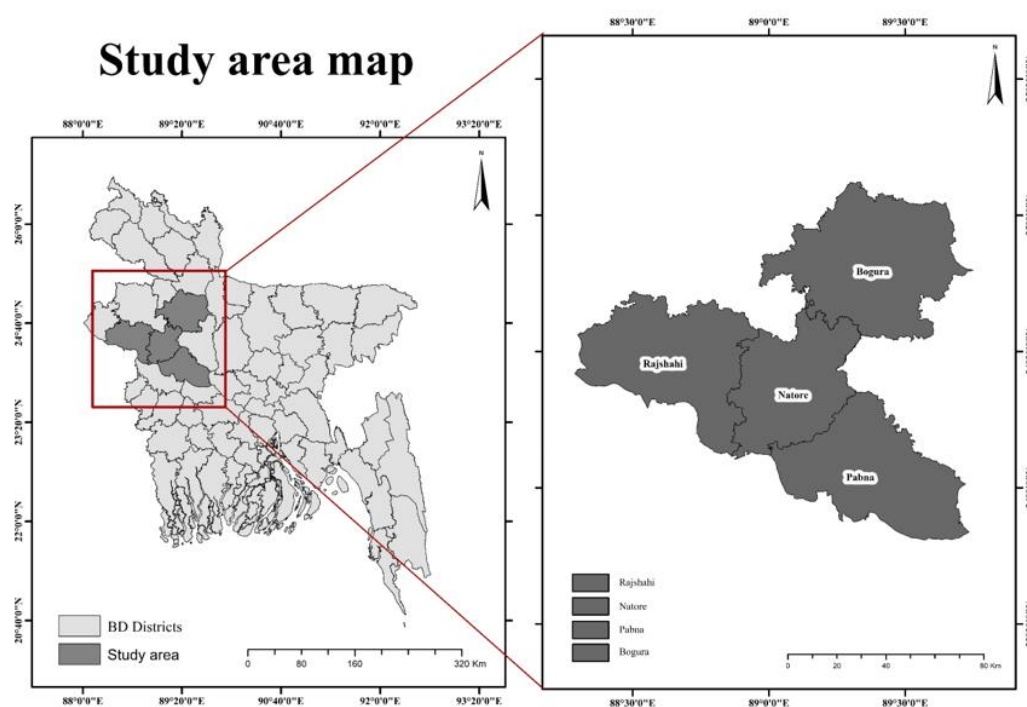


Fig.3.1. Study area map.

The selection of the four districts provides a comparative spatial framework for assessing climatic variability within northwestern Bangladesh. Although the districts fall within the same broad climatic region, differences in their geographical and environmental settings may contribute to variations in rainfall, temperature, relative humidity, and wind speed. Examining these variations at the district level therefore provides an opportunity to identify differences in the spatial distribution and temporal behavior of the selected climatic parameters. The study considered four major climatic parameters: (i) Rainfall, (ii) Temperature, (iii) Relative humidity, (iv) Wind speed. These parameters were analyzed over a 51-year period from 1973 to 2023. The analysis included spatial distribution mapping, temporal trend assessment, seasonal and monthly climatological analysis, extreme precipitation analysis, and correlation analysis. In addition, changes in LULC were examined to provide further environmental context for interpreting the observed climatic patterns and their spatial variation. A study-area map was prepared using district boundary shapefiles in ArcMap 10.8. The map illustrates the geographical extent and administrative boundaries of Rajshahi, Pabna, Natore, and Bogura and provides the spatial framework for the subsequent climatic and LULC analyses.

3.2 Data Sources and Preparation

3.2.1 Meteorological Data

Monthly meteorological data were used to examine the long-term variation of the selected climatic parameters. The dataset included rainfall, temperature, relative humidity, and wind speed for the period from 1973 to 2023, covering a total of 51 years. The meteorological data were obtained from the Bangladesh Agricultural Research Council (BARC). Monthly data were used because freely accessible daily meteorological observations were not available for the entire study period required for the research. Consequently, the analyses were conducted using the available monthly temporal resolution. The climatic variables were aggregated according to their respective physical characteristics. Rainfall was represented by accumulated values, while temperature, relative humidity, and wind speed were represented by mean values. The basic characteristics of the meteorological dataset are presented in Table 3.1.

Table 3.1: Data source by period for meteorological data.

Climatic parameter	Data source	Study period	Temporal resolution
Rainfall	BARC	1973-2023	Monthly
Temperature	BARC	1973-2023	Monthly
Relative humidity	BARC	1973-2023	Monthly
Wind speed	BARC	1973-2023	Monthly

The meteorological observations were associated with stations representing the study region. Direct station observations were available for Rajshahi, Bogura, and Pabna (Ishurdi). However, no direct meteorological station corresponding to Natore was available in the dataset used in this study. Therefore, climatic values for Natore were estimated using the Inverse Distance Weighted (IDW) interpolation method (Shepard, 1968), based on observations from the available surrounding stations. The IDW method estimates the value at an unsampled location using observations from nearby sampled locations, with greater weight assigned to stations located closer to the target location. The estimated values for Natore were subsequently incorporated into the district-level climatic analysis.

3.2.2 Data Quality and Preparation

Before the statistical and spatial analyses, the meteorological dataset was organized according to the temporal and spatial requirements of the study. The dataset contained no missing observations requiring removal or imputation. Therefore, no observations were excluded due to missing data. The original monthly observations were retained and reorganized to generate annual, seasonal, and monthly climatological datasets for subsequent analysis. The method of aggregation varied according to the climatic parameter and the specific analytical requirement. For annual analysis, annual rainfall was calculated by summing the twelve-monthly rainfall observations for each year. In contrast, annual temperature, relative humidity, and wind speed were calculated as the means of their respective monthly observations. Annual rainfall was calculated using the following equation:

$$P_{annual} = \sum_{m=1}^{12} P_m \quad (3.1)$$

where P_{annual} represents annual rainfall and P_m represents rainfall in month m . For temperature, relative humidity, and wind speed, the annual mean was calculated as:

$$X_{annual} = \frac{1}{12} \sum_{m=1}^{12} X_m \quad (3.2)$$

where (X) represents temperature, relative humidity, or wind speed, and X_m represents the monthly value of the respective parameter. The resulting annual datasets were subsequently used for temporal analysis, trend analysis, and correlation analysis. No additional standardization or transformation of the climatic variables was applied before the principal analysis.

3.2.3 Temporal Grouping of Climatic Data

To facilitate the comparison of climatic conditions across different portions of the study period, the 51-year dataset was divided into four consecutive periods: 1973-1985, 1986-1998, 1999-2011, 2012-2023. The first three periods each contain 13 years, whereas the final period contains 12 years. This grouping was used to compare climatic conditions between the beginning and end of the study period and to examine changes across successive time intervals. The period-wise climatic values were subsequently used in the thematic mapping component of the study. For each period, annual rainfall totals and the annual mean values of temperature, relative humidity, and wind speed were used to represent the climatic conditions of the respective period.

3.2.4 Land Use/Land Cover Data

Land use/land cover (LULC) data were included in the study to examine how the environmental characteristics of the study area changed over time. Landsat satellite imagery was used for the LULC assessment, with images from four reference years 1985, 1998, 2011, and 2023. These years correspond to the four sub-periods used in the climatic thematic mapping and allowed changes in land cover to be examined at different stages of the study period. For the earlier reference years, Landsat 1-5 imagery was used, while Landsat 8-9 imagery was used for 2023. The satellite images were processed in ArcMap 10.8, and a supervised classification approach (Congalton & Green, 2019) was applied to identify four major LULC classes: (i) Built-up area, (ii) Water bodies, (iii) Vegetation/forest and (iv) Bare land. Agricultural land was not considered a separate class because it could not be consistently distinguished from vegetation and bare land during visual interpretation of the Landsat imagery. This was particularly difficult because of seasonal variations in cropping patterns and the spectral similarity between

agricultural areas and the other two classes. The classified maps were then used to determine the spatial extent of each LULC class for the four reference years. The area covered by each class was calculated and expressed as a percentage of the total study area. Changes between successive reference years were also calculated to determine the direction and magnitude of LULC change over time. The LULC analysis was not intended to establish a direct cause and effect relationship between land-cover change and climatic change. Instead, the observed changes in LULC were compared with the climatic patterns to identify possible associations between changes in the environmental landscape and variations in the climatic parameters.

3.3 Spatial and Temporal Analysis of Climatic Parameters and LULC

3.3.1 Thematic Mapping of Climatic Parameters

The spatial variation of the selected climatic parameters was examined using thematic maps prepared in ArcMap 10.8. District boundary shapefiles of Rajshahi, Pabna, Natore, and Bogura were used as the spatial framework for the mapping. The climatic values corresponding to each district were assigned to the respective district polygons and displayed using graduated thematic maps. Four climatic parameters were mapped: (i) Annual rainfall, (ii) Annual mean temperature, (iii) Annual mean relative humidity, (iv) Annual mean wind speed. Maps were prepared for the selected temporal periods to illustrate changes in the spatial distribution of climatic conditions across the four districts over the study period. Manual classification was applied to group the climatic values into suitable categories. The classification ranges were determined from the observed minimum and maximum values of each climatic parameter and were selected to provide a clear visual comparison among the districts. Four classes were used for rainfall, whereas three classes were used for temperature, relative humidity, and wind speed. The thematic maps were mainly used to identify spatial differences among the districts and to observe changes in the distribution of climatic characteristics across the different periods of the study.

3.3.2 Linear Trend Analysis

Linear trend analysis was used to visualize the overall direction of change in the selected climatic parameters over time. For each district, annual climatic values were plotted against the corresponding year. Annual rainfall totals were used for rainfall, while annual

mean values were used for temperature, relative humidity, and wind speed. The linear relationship between the climatic variable and time was represented by the following equation:

$$Y_t = a + bt \quad (3.3)$$

Where, Y_t = value of the climatic variable at time (t), a = intercept of the fitted line, b = estimated linear trend coefficient, t = time in years. A positive value of the slope indicates an increasing linear tendency, whereas a negative value indicates a decreasing tendency over the study period. The fitted linear trend lines were used primarily to illustrate the general direction and rate of change in the climatic variables. They were not used as the formal test of statistical significance. The significance of trends and their magnitude were assessed separately using the Mann-Kendall test and Sen's slope estimator.

3.3.3 LULC Change Assessment

Changes in land use/land cover were assessed by comparing the LULC maps prepared for four reference years: 1985, 1998, 2011, and 2023. The comparison was carried out to quantify changes in the spatial extent of the major land-cover categories over the study period. For each LULC class, the area occupied in each of the four reference years was calculated. The percentage contribution of each class to the total study area was determined using the following equation:

$$L_i = \frac{A_i}{A_T} \times 100 \quad (3.4)$$

Where, L_i = percentage of the (i)th LULC class, A_i = area occupied by the (i)th LULC class, A_T = total area of the study area. The percentage change across the four reference years was then calculated to determine the extent to which each LULC category had increased or decreased. Changes in built-up areas, water bodies, vegetation/forest, and bare land were subsequently compared with the observed climatic patterns. This comparison was used to examine whether changes in the land-cover characteristics coincided with the observed changes in climatic variables. The interpretation was made with caution because the study did not apply a formal causal attribution framework. Therefore, changes in LULC were considered potentially associated with the observed climatic variations rather than being identified as direct causes of those changes.

3.4 Trend and Seasonal Variability Analysis

3.4.1 Mann-Kendall Trend Test

The Mann-Kendall (MK) test (Mann, 1945; Kendall, 1938) was used to determine whether the climatic variables showed a significant increasing or decreasing trend over the study period. It is a non-parametric statistical test that is commonly used in climate and environmental studies because it does not require the data to follow a normal distribution. For a time series consisting of $(X_1, X_2, \dots, X_n)$, the MK statistic was calculated by comparing the observations with each other:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(X_j - X_i) \quad (3.5)$$

$$\text{where, } \text{sgn}(X_j - X_i) = \begin{cases} 1 & X_j > X_i \\ 0 & X_j = X_i \\ -1 & X_j < X_i \end{cases}$$

A positive value of (S) indicates an increasing trend, while a negative value indicates a decreasing trend. The p-value obtained from the test was used to determine whether the trend was statistically significant. A significance level of $(\alpha = 0.05)$ was used in this study. Therefore, a trend was considered statistically significant when $(p < 0.05)$. The MK test was applied to the annual climatic data of the four selected districts to identify the direction and significance of long-term trends in rainfall, temperature, relative humidity, and wind speed.

3.4.2 Sen's Slope Estimator

Sen's slope estimator (Sen, 1968) was used along with the Mann-Kendall test to determine the rate of change in the climatic variables. While the Mann-Kendall test shows whether a trend is statistically significant, Sen's slope provides an estimate of how much the variable changes per year. The slope was calculated for all possible pairs of observations using the following equation:

$$Q_i = \frac{X_j - X_k}{j - k} \quad (3.6)$$

where X_j and X_k represent observations at times j and k , respectively. After calculating all possible slopes, the median value of the slopes was taken as the Sen's slope estimate. A positive value indicates an increasing trend, whereas a negative value indicates a

decreasing trend. The estimated slope was expressed in the appropriate unit per year. For example, rainfall was expressed as mm/year, temperature as °C/year, relative humidity as percentage points/year, and wind speed in its respective unit per year. Therefore, the Mann–Kendall test and Sen's slope were used together in the trend analysis. The MK test was used to determine whether a trend was statistically significant, while Sen's slope was used to show the direction and rate of change.

3.4.3 Seasonal Variability

Seasonal variability was examined by grouping the monthly climatic data into four seasons: winter, pre-monsoon, monsoon, and post-monsoon. The same seasonal classification was used for all four districts which is shown in Table 3.2.

Table 3.2: Seasonal Breakdown of the Annual Climate Cycle.

Season	Months
Winter	December - February
Pre-monsoon	March - May
Monsoon	June - September
Post-monsoon	October - November

For rainfall, the seasonal total was calculated by adding the rainfall values of all months within each season. For temperature, relative humidity, and wind speed, the seasonal value was calculated as the mean of the monthly observations for that season. The seasonal results were then used to describe how the selected climatic parameters varied throughout the year and to compare the seasonal characteristics of Rajshahi, Pabna, Natore, and Bogura.

3.4.4 Monthly Climatology

Monthly climatology was calculated to understand the usual monthly pattern of the selected climatic variables during the (1973-2023) study period. For each district, observations from the same month were averaged across all 51 years. For example, the January climatological value was obtained by averaging all January observations from 1973 to 2023. The same procedure was followed for the remaining months. The climatological value for a particular month was calculated using:

$$\bar{X}_m = \frac{1}{N} \sum_{y=1}^N X_{m,y} \quad (3.7)$$

Where, X_m = climatological mean of month (m), $X_{m,y}$ = observed value for month (m) in year (y), N = total number of years. The resulting twelve-month climatological values were used to describe the typical annual pattern of rainfall, temperature, relative humidity, and wind speed. These monthly patterns were also compared among the four selected districts to identify differences in their climatic characteristics.

3.5 Extreme Precipitation and GEV Analysis

3.5.1 Construction of Annual Maximum Series

Since daily meteorological data were not available for the entire study period, extreme precipitation was analyzed using an Annual Maximum Series (AMS) derived from the available monthly rainfall data. For each year, the highest rainfall value among the twelve-monthly observations was selected as the annual maximum. The AMS for year (y) was therefore calculated as:

$$AMS_y = \max(P_{y,1}, P_{y,2}, \dots, P_{y,12}) \quad (3.8)$$

Where, AMS_y = annual maximum monthly rainfall in year (y), $P_{y,m}$ = rainfall recorded in month (m) of year (y). This procedure produced one extreme-rainfall value for each year of the study period. It is important to note that the resulting AMS represents annual maximum monthly rainfall, rather than annual maximum daily rainfall. Separate AMS datasets were prepared for the four selected districts: Rajshahi, Pabna, Natore, Bogura. As the study period covered 1973-2023, each district had an AMS series containing approximately 51 annual observations.

3.5.2 Generalized Extreme Value Distribution

The Generalized Extreme Value (GEV) distribution was fitted to the AMS data for each district. The GEV distribution is widely used in extreme-value analysis for modelling block maxima and was used in this study to estimate the probability and recurrence characteristics of extreme rainfall. The cumulative distribution function of the GEV distribution is given by:

$$G(z) = \exp\{-[1 + \xi \left(\frac{z-\mu}{\sigma}\right)]^{-1/\xi}\} \quad (3.9)$$

Where, μ = location parameter, σ = scale parameter, with ($\sigma > 0$), ξ = shape parameter, z = observed extreme value. The location parameter describes the position of the distribution, while the scale parameter represents the spread of extreme values. The shape parameter determines the behavior of the upper tail of the distribution and therefore has an important role in estimating very large rainfall events. The GEV distribution was separately to the AMS data of each district using R. The estimated parameters were then used to calculate rainfall return levels for the selected return periods.

3.5.3 Goodness-of-Fit Assessment

The fitted GEV distributions were evaluated to determine how well they represented the observed annual maximum rainfall data. The Anderson-Darling (A-D) goodness-of-fit test (Anderson & Darling, 1954) was used for this purpose. The Anderson-Darling test measures the agreement between the empirical distribution of the observed extreme values and the fitted theoretical distribution. The test gives particular attention to differences in the tails, which is important in extreme-rainfall analysis. In addition to the numerical test, Q-Q plots were prepared to visually assess the fitted GEV models. The observed extreme-rainfall quantiles were compared with the theoretical quantiles obtained from the fitted GEV distribution. Gringorten plotting positions (Gringorten, 1963) were used to calculate empirical quantiles, following standard practice for quantile estimation in statistical packages (Hyndman & Fan, 1996). When the observed and theoretical quantiles show a close alignment with the reference line, the fitted distribution can be considered to provide a reasonable representation of the observed data. Noticeable or systematic departures from the reference relationship may indicate that the fitted distribution does not adequately describe some parts of the observed distribution, particularly the tails. The suitability of each GEV model was therefore assessed using both the Anderson-Darling test and Q-Q plots, rather than relying on a single diagnostic.

3.5.4 Return Period and Return-Level Analysis

Return-level analysis was carried out to estimate the magnitude of extreme rainfall associated with different return periods. The selected return periods were: 2 years, 5 years, 10 years, 25 years, 50 years, 100 years. The relationship between return period (T) and annual exceedance probability (p) is expressed as:

$$T = \frac{1}{p} \quad (3.10)$$

Thus, a 10-year return period corresponds to an annual exceedance probability of approximately 0.10 under the fitted GEV model. A return period should not be interpreted as an event occurring at a fixed interval. For example, a 10-year return period does not mean that an extreme rainfall event will occur exactly once every ten years. Instead, it represents the estimated probability that an event of that magnitude or greater will be exceeded in any particular year, based on the fitted model. Return levels were estimated separately for each of the four districts. Return-level plots were then prepared to show the relationship between the estimated rainfall magnitude and increasing return period. Since some of the selected return periods, particularly the 50-year and 100-year periods, extend beyond the approximately 51-year observational record, these estimates were interpreted with caution. They represent model-based extrapolations and should not be treated as directly observed rainfall values.

3.5.5 Uncertainty in GEV Return-Level Estimates

The uncertainty associated with the estimated GEV return levels was assessed using 95% bootstrap confidence intervals (Efron & Tibshirani, 1993). These intervals provide an indication of the range within which the estimated return levels may vary due to sampling uncertainty in the available AMS observations. The uncertainty is particularly relevant for longer return periods because each district had only approximately 51 annual maximum observations. As the return period increases, the corresponding return level is estimated farther into the upper tail of the fitted distribution, where fewer observations are available to support the estimate. Therefore, the 50-year and especially the 100-year return-level estimates were interpreted more cautiously than those for shorter return periods. The confidence intervals were presented together with the estimated return levels to show the associated uncertainty and to avoid treating the model estimates as exact values.

3.6 Correlation Analysis

Correlation analysis was performed to examine how the four climatic variables rainfall, temperature, relative humidity, and wind speed varied in relation to one another. The analysis used the annual climatic data for the period 1973-2023. Annual rainfall totals were considered for rainfall, while annual mean values were used for temperature, relative humidity, and wind speed. The relationships among the variables were examined using three correlation methods: Pearson, Spearman, and Kendall correlation. Pearson

correlation was considered for linear relationships, whereas Spearman and Kendall correlations were included to examine whether the variables showed a consistent increasing or decreasing pattern when ranked. Using more than one method also helped in comparing the results and identifying relationships that remained similar under different approaches.

3.6.1 Pearson Correlation

Pearson's correlation coefficient was calculated for each pair of climatic variables to determine the direction and strength of their linear relationship. It was calculated using the following equation:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}} \quad (3.11)$$

where X_i and Y_i are the observations of the two variables, and $(\bar{X})$ and $(\bar{Y})$ are their respective mean values. The coefficient ranges from $(-1$ to $+1)$. A positive value means that the two variables generally change in the same direction, while a negative value indicates that they tend to change in opposite directions. Values close to zero suggest a weak or negligible linear relationship. The Pearson coefficients obtained from the analysis were also arranged into a correlation matrix and presented as a heat map to make the relationships among the climatic variables easier to compare.

3.6.2 Spearman Correlation

Spearman's rank correlation was calculated to examine the relationships between the climatic variables without depending directly on their original values. Instead, the observations were converted into ranks before calculating the correlation. This approach was useful because a relationship between two climatic variables may not always follow a straight-line pattern. A relatively high Spearman coefficient can still occur when the variables generally move in the same direction, even if the rate of change is not constant. Because the method relies on ranks rather than the actual magnitudes, it is less sensitive to outliers and to departures from normality that can affect Pearson correlation. The Spearman results were therefore compared with the Pearson results to see whether the observed relationships remained similar when the data were considered in ranked form, and to confirm that any associations detected were not dependent on the assumption of linearity.

3.6.3 Kendall Correlation

Kendall's rank correlation (Kendall, 1938) was calculated in a similar way to Spearman correlation, but it is based on the agreement between the ordering of observations in the two variables. In simple terms, it examines whether higher values of one variable tend to occur with higher or lower values of the other variable. Unlike Pearson correlation, which assumes a linear relationship, both Kendall and Spearman coefficients are non-parametric and therefore more robust when the data do not follow a normal distribution or when the association is monotonic rather than strictly linear. Kendall correlation was included as an additional non-parametric measure of the relationships among the climatic variables. The results from Kendall and Spearman correlations were compared with the Pearson results to determine whether the direction of the relationships was consistent across the different methods and to identify associations that remained robust under all three approaches.

3.6.4 Correlation Matrix and Heat Map

A correlation matrix was prepared to summarize the pairwise relationships among rainfall, temperature, relative humidity, and wind speed. The Pearson correlation coefficients were used to construct the final correlation heat map. The heat map provided a visual representation of the correlation coefficients, making it easier to identify both the direction and relative strength of the relationships. Positive correlations represented variables that generally varied in the same direction, whereas negative correlations represented variables that tended to vary in opposite directions.

The statistical significance of the correlations was assessed at ($\alpha = 0.05$). Both the correlation coefficient and its corresponding significance level were considered when interpreting the results. The results were treated as statistical associations rather than evidence of causation. For instance, if temperature and relative humidity showed a negative correlation, it would mean that higher temperature values were generally associated with lower relative humidity values in the dataset. This relationship alone, however, would not be sufficient to conclude that a change in temperature directly caused the corresponding change in relative humidity.

3.7 Analytical Framework

3.7.1 Software used

Different software packages were used at different stages of the study depending on the type of analysis being performed. ArcMap 10.8 was mainly used for GIS based work, R was used for statistical and numerical analysis, and Microsoft Excel was used for data organization and preliminary processing.

ArcMap 10.8: ArcMap 10.8 was used for preparing the study area map and carrying out the spatial components of the research. This included district-level thematic mapping of the climatic variables, spatial visualization of climatic conditions, LULC classification, and assessment of LULC changes across the four reference years.

R: Statistical and numerical analyses were performed in R. The software was used to generate linear trend lines, conduct the Mann-Kendall trend test, estimate Sen's slope, fit the GEV distribution, examine GEV model diagnostics, calculate return levels, and perform the correlation analyses.

Microsoft Excel: Microsoft Excel was used mainly for organizing and preparing the datasets before analysis. It was also used for tabulation and, where necessary, for calculating and arranging derived datasets used in the subsequent analyses.

3.7.2 Overall Analytical Framework

The analysis followed a series of steps, beginning with the preparation of the meteorological data and ending with the assessment of climatic relationships and extreme rainfall characteristics. First, monthly observations of rainfall, temperature, relative humidity, and wind speed for the period 1973-2023 were compiled. The dataset was checked for completeness, and no missing observations requiring removal or imputation were found. The monthly data was then reorganized according to the requirements of the different analyses. Annual rainfall was obtained by summing the twelve-monthly rainfall values for each year, while annual temperature, relative humidity, and wind speed were calculated as the mean of their monthly values. Seasonal datasets and monthly climatological values were also prepared from the original monthly observations. Since no direct meteorological station observation for Natore was available in the dataset, its climatic values were estimated using the Inverse Distance Weighted (IDW) interpolation method based on the available surrounding stations.

The spatial distribution of the climatic variables was then examined through thematic maps prepared in ArcMap 10.8. The 51-year study period was divided into four periods to allow comparison of the spatial pattern of climatic conditions at different stages of the study period. Linear trend lines were also prepared to show the general direction of change in the annual climatic series. The changes in land use and land cover were assessed separately using Landsat imagery from four reference years: 1985, 1998, 2011, and 2023. A supervised classification was applied to the imagery to produce LULC maps for each reference year, and the changes in the major LULC categories across these years were calculated. These changes were then compared descriptively with the observed climatic patterns. For the statistical trend analysis, the Mann-Kendall test and Sen's slope estimator were applied to the annual climatic series. The MK test was used to determine the direction and statistical significance of the trends, while Sen's slope provided an estimate of their rate of change. Seasonal variability and monthly climatology were also examined to describe the seasonal and monthly behavior of the climatic variables.

Extreme precipitation was analyzed separately using the Annual Maximum Series (AMS). Since daily rainfall data were not available, the highest monthly rainfall value from each year was selected to construct an annual maximum monthly rainfall series for each district. The resulting AMS datasets were fitted to the Generalized Extreme Value (GEV) distribution. The fitted models were assessed using the Anderson-Darling goodness of fit test and Gringorten Q-Q plots. Return levels were then estimated for return periods ranging from 2 to 100 years, with 95% bootstrap confidence intervals used to represent the uncertainty of the estimates. Finally, the relationships among rainfall, temperature, relative humidity, and wind speed were examined using Pearson, Spearman, and Kendall correlation coefficients. The Pearson correlation matrix was presented as a heat map to provide a visual comparison of the relationships among the climatic variables.

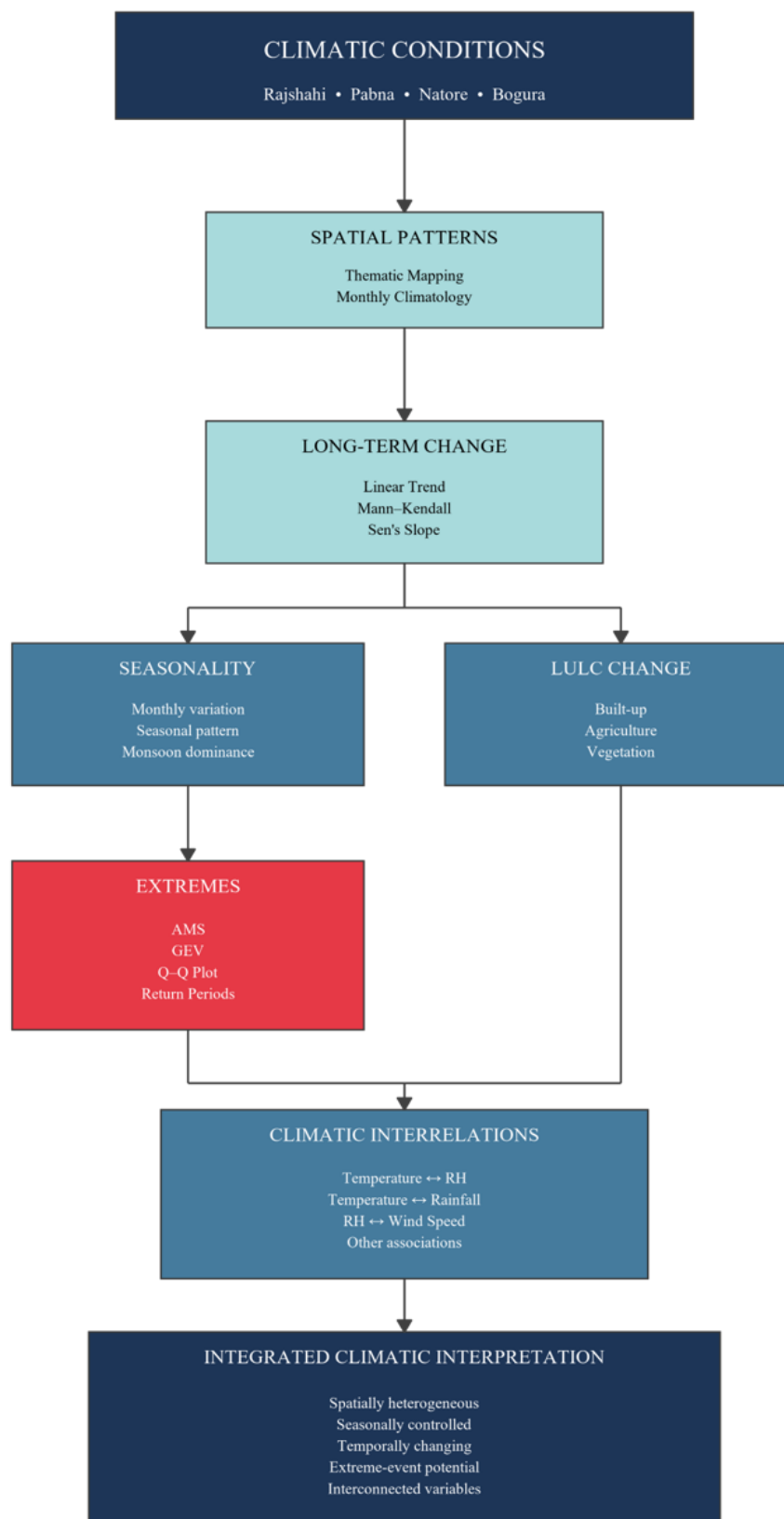


Fig.3.2. Integrated analytical framework.

Chapter 4

Results and Discussion

4.1 Spatial and Temporal Variation of Climatic Parameters and LULC Change

The spatial and temporal patterns of rainfall, temperature, relative humidity, and wind speed were examined for Rajshahi, Pabna, Natore, and Bogura over the 1973-2023 study period. To compare changes across the record, the 51 years were divided into four intervals: 1973-1985, 1986-1998, 1999-2011, and 2012-2023. District-level thematic maps were prepared for each period, while linear trend plots showed the overall direction of change across the full study period. LULC changes were also assessed for 1985, 1998, 2011, and 2023 to provide environmental context for the observed climatic variations.

4.1.1 Spatial and Temporal Variation of Rainfall

The spatial distribution of annual rainfall differed among the four districts and across the study periods, with Bogura and Pabna generally in the higher rainfall classes and Rajshahi and Natore in the comparatively lower classes throughout the study period.

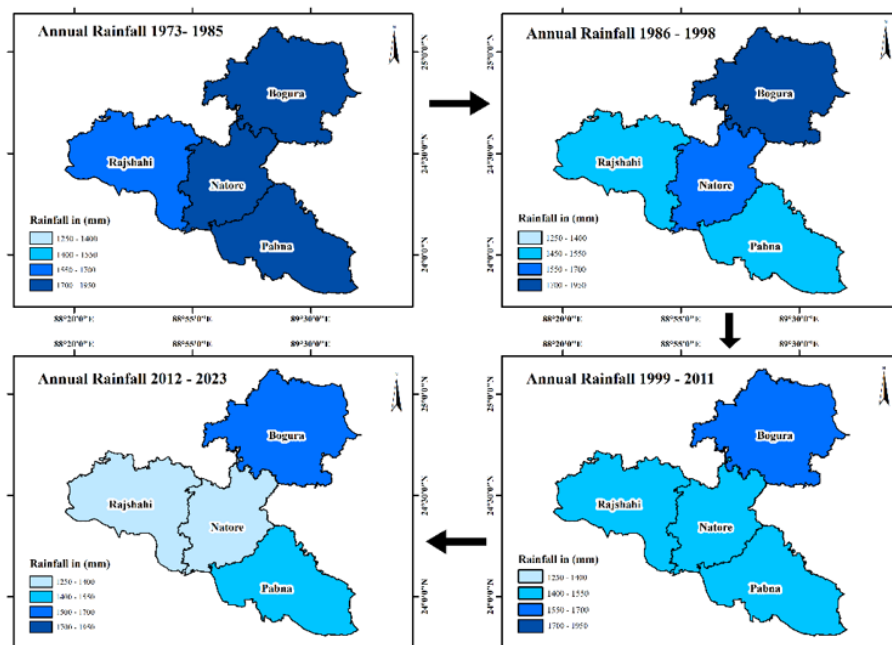


Fig.4.1. Spatial distribution of annual rainfall for 1973-1985, 1986-1998, 1999-2011 and 2012-2023.

Over the next three periods the contrast became more pronounced: Bogura remained in the higher rainfall class while Rajshahi, Natore and Pabna shifted toward lower classes, with Rajshahi and Natore occupying the lowest class and Pabna an intermediate class by 2012-2023. This indicates that rainfall is not spatially uniform across northwestern Bangladesh and that neighbouring districts can experience different regimes, consistent with earlier findings from western Bangladesh (Kamruzzaman et al., 2018) and the Barind Tract (Esha et al., 2026). The present maps confirm the same pattern at district scale, with Bogura consistently wetter than Rajshahi and Natore. Linear trend analysis further shows an overall decline in annual rainfall during 1973-2023, with negative trend lines in all four districts although the apparent magnitude of decline varies.

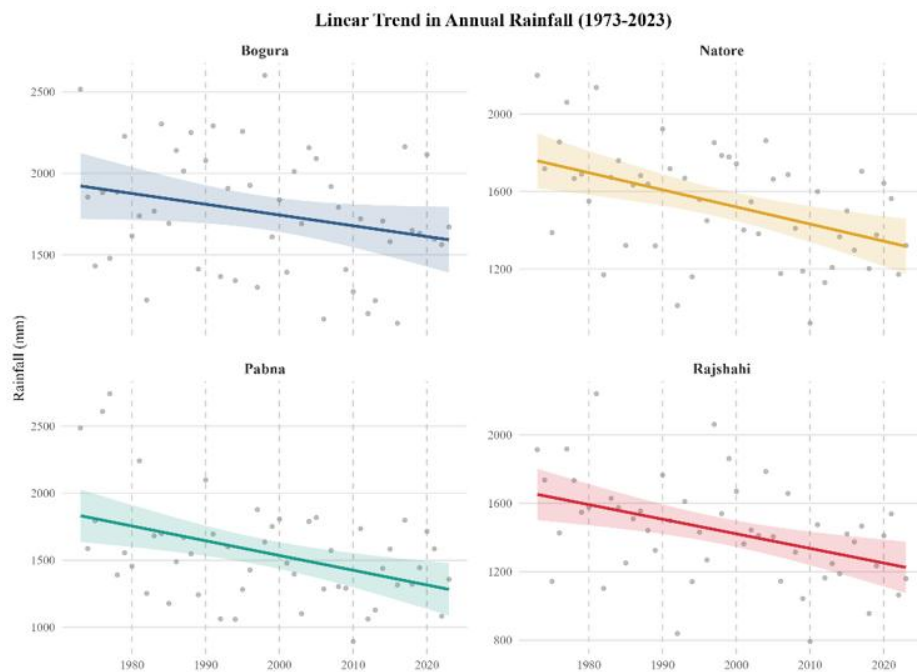


Fig.4.2. Linear trend in annual rainfall for Rajshahi, Pabna, Natore and Bogura, (1973-2023).

The strongest visual decline appears in Rajshahi and Pabna, while Bogura and Natore also show downward tendencies. This broadly agrees with earlier observations of decreasing or spatially variable rainfall in western Bangladesh, although recent studies show that trends vary by period and data source: Shahriyar et al. (2026) reported a significant increase in rainfall for Rajshahi City using NASA POWER data from 1990-2024, whereas Esha et al. (2026) found predominantly decreasing trends at several Barind stations using BWDB data from 2000-2024. The present results therefore should not be interpreted as evidence of a single regional rainfall trend. The four-period maps complement the long-term linear trends by showing that rainfall fluctuated considerably

between periods rather than declining uniformly, with formal statistical testing providing further evidence in Section 4.2.

4.1.2 Spatial and Temporal Variation of Temperature

Temperature exhibited a markedly different spatial pattern from rainfall. Throughout the four periods, Rajshahi consistently occupied the highest temperature class, approximately 25.0–26.5°C, whereas Pabna, Natore, and Bogura generally remained within the lower class of approximately 22.0–23.5°C. This difference remained relatively stable throughout the study periods, indicating a persistent spatial temperature gradient among the districts. The maps therefore show that Rajshahi remained distinctly warmer than the other three districts throughout the study period, with no major shift in the overall spatial classification.

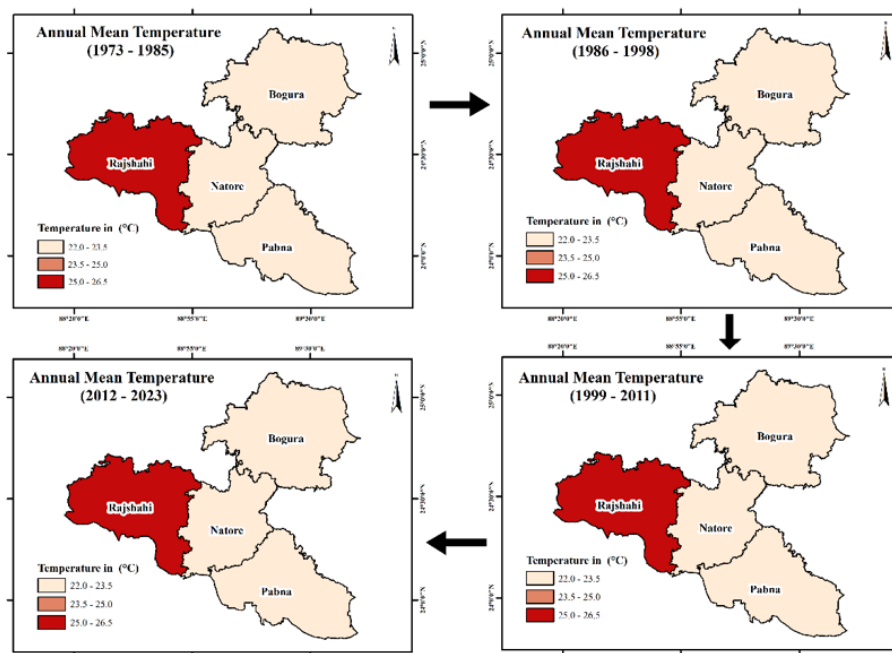


Fig.4.3. Spatial distribution of annual mean temperature for 1973-1985, 1986-1998, 1999-2011 and 2012-2023.

Unlike rainfall, the temperature maps show relatively little change in spatial classification across the four periods. Rajshahi remained distinctly warmer, while the other three districts stayed comparatively cooler, indicating a persistent spatial temperature pattern despite year-to-year variation in annual values. This consistency suggests that the relative temperature differences among the districts remained largely stable throughout the study

period. However, the linear trend plots reveal differences in the temporal direction of temperature change among the districts.

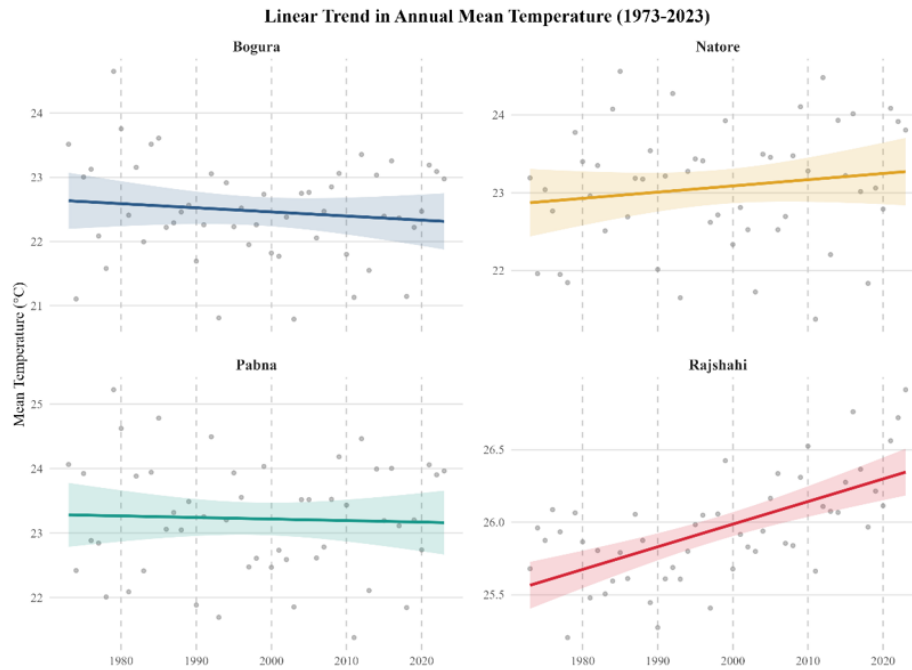


Fig.4.4. Linear trend in annual mean temperature for Rajshahi, Pabna, Natore and Bogura, (1973-2023).

Rajshahi shows a clear positive linear tendency, Natore a weaker upward trend, Pabna an almost flat to slightly declining trend, and Bogura a comparatively weak downward tendency. Thus, the temperature results do not show uniform warming across the four districts; both the magnitude and direction of the observed long-term change vary spatially. Khan et al. (2019) similarly reported spatial differences in temperature trends across Bangladesh despite an overall warming tendency in several temperature variables. Rajshahi's persistently higher temperature classification may reflect its distinctive regional climatic setting, although the present analysis does not establish the underlying mechanism; therefore, this gradient is interpreted as a spatial pattern rather than being attributed to a specific environmental factor.

4.1.3 Spatial and Temporal Variation of Relative Humidity

Relative humidity showed one of the clearest temporal shifts among the four climatic variables. During 1973-1985, Rajshahi, Natore, and Pabna were mainly within the 72-75% class, while Bogura was in the 75-78% class. By 1999-2011, the 78-80% class had become dominant across all four districts, indicating a clear increase in the spatially mapped humidity levels over time.

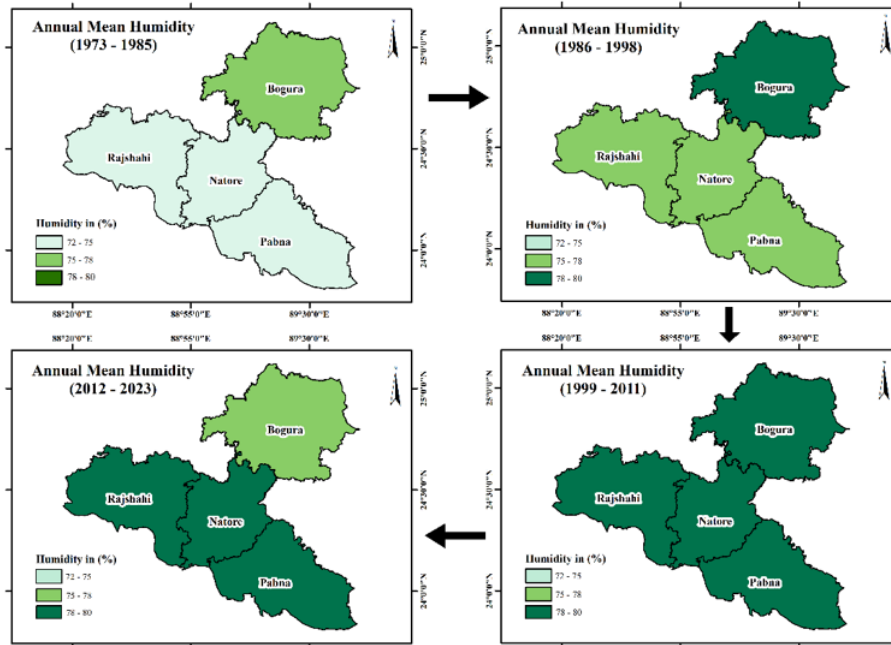


Fig.4.5. Spatial distribution of annual mean relative humidity for 1973-1985, 1986-1998, 1999-2011 and 2012-2023.

In 2012-2023, Rajshahi, Natore, and Pabna remained in the 78-80% class, while Bogura shifted back to 75-78%, representing a spatial reversal despite the overall shift toward higher humidity classes. The linear trend plots nevertheless show an overall positive trend in annual mean humidity across all four districts.

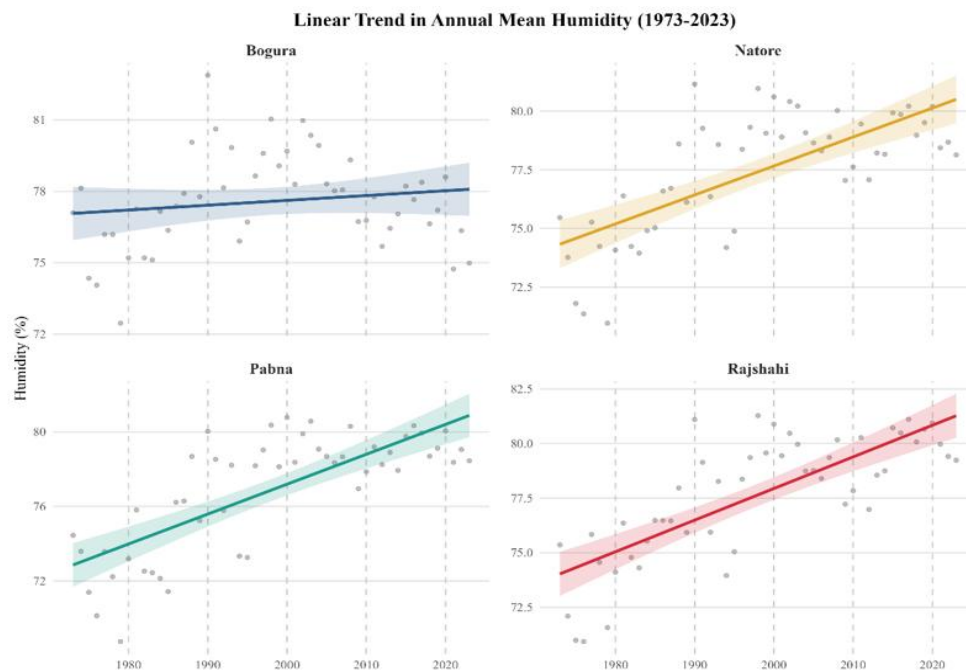


Fig.4.6. Linear trend in annual mean relative humidity in selected districts from (1973-2023).

The upward tendency is strongest in Natore, Pabna, and Rajshahi, while Bogura shows a much weaker increase, indicating that humidity did not rise uniformly across the study area. This increase is noteworthy alongside the declining rainfall pattern, as higher humidity does not necessarily imply higher rainfall; atmospheric moisture is also influenced by temperature, evaporation, vegetation, and other processes. Fattah et al. (2023) similarly reported substantial spatial and temporal variability in relative humidity across Bangladesh and its relationship with other climatic variables. This contrast is relevant to the correlation analysis in Section 4.5, suggesting that the relationship between rainfall and atmospheric moisture is not necessarily straightforward at the annual scale.

4.1.4 Spatial and Temporal Variation of Wind Speed

Wind speed showed considerable temporal variation and a different spatial pattern from both rainfall and humidity. Pabna and Natore held the highest wind-speed class in 1973-1985, with Rajshahi intermediate and Bogura lowest; this spatial contrast narrowed through 1986-1998 as most districts shifted toward the intermediate category.

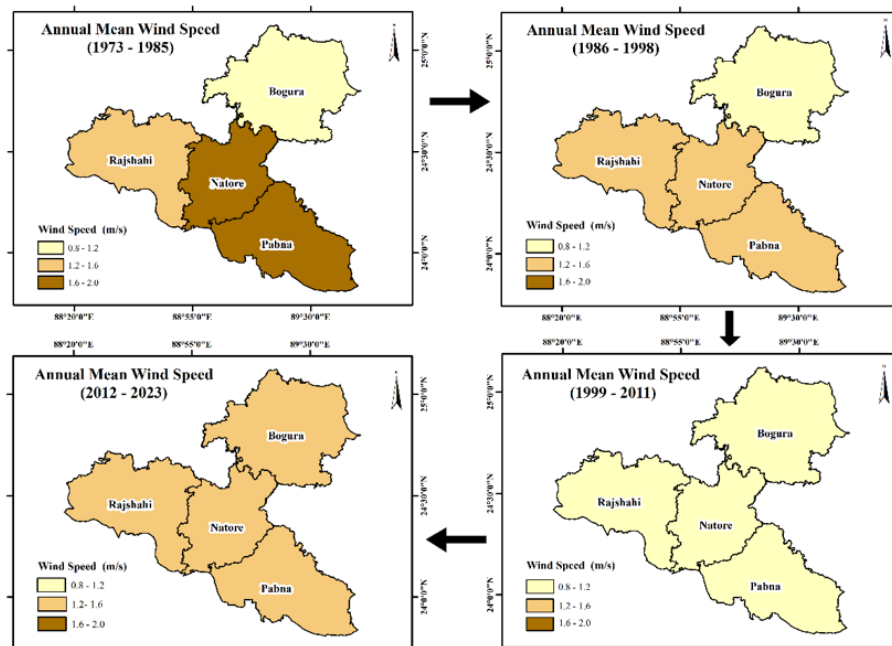


Fig.4.7. Spatial distribution of annual mean wind speed for 1973-1985, 1986-1998, 1999-2011 and 2012-2023.

By 1999-2011 all four districts had fallen to the lowest wind-speed class, before some recovery toward the intermediate class in the final period, 2012-2023. The spatial maps therefore suggest a substantial region-wide reduction in wind speed toward the late

1990s/early 2000s followed by partial recovery, though the linear trends show this recovery was not uniform among the districts.

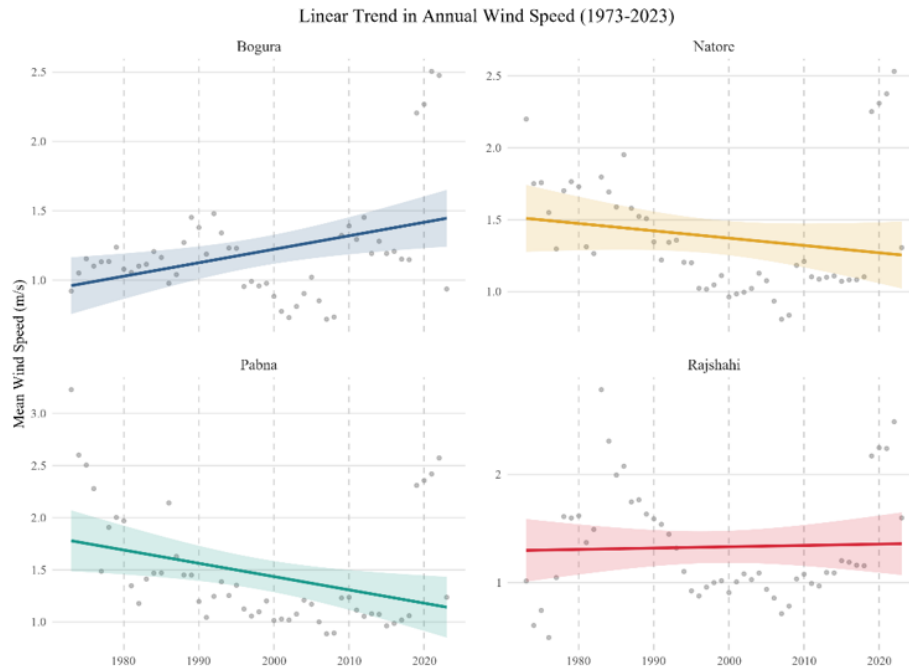


Fig.4.8. Linear trend in annual mean wind speed for Rajshahi, Pabna, Natore and Bogura, (1973-2023).

Bogura shows an overall positive linear tendency, Pabna and Natore negative tendencies, and Rajshahi a comparatively weak, nearly stable direction. These differences indicate that wind-speed behavior varied considerably among the four districts over the study period. The four districts therefore do not share a common long-term wind-speed trajectory, so a single regional statement of increase or decrease would not adequately represent the results. Khan et al. (2019) reported decreasing wind-speed tendencies in several parts of Bangladesh, particularly in the northwest, and the present district-level results indicate that long-term wind-speed behavior may be more heterogeneous than a single regional trend suggests.

4.1.5 Land Use/Land Cover Change

LULC analysis was conducted for 1985, 1998, 2011, and 2023 using four classes: water bodies, vegetation, built-up areas, and bare land. The spatial distributions of these classes are presented separately for Rajshahi, Pabna, Natore, and Bogura, allowing changes in major land-cover patterns to be compared across the four districts.

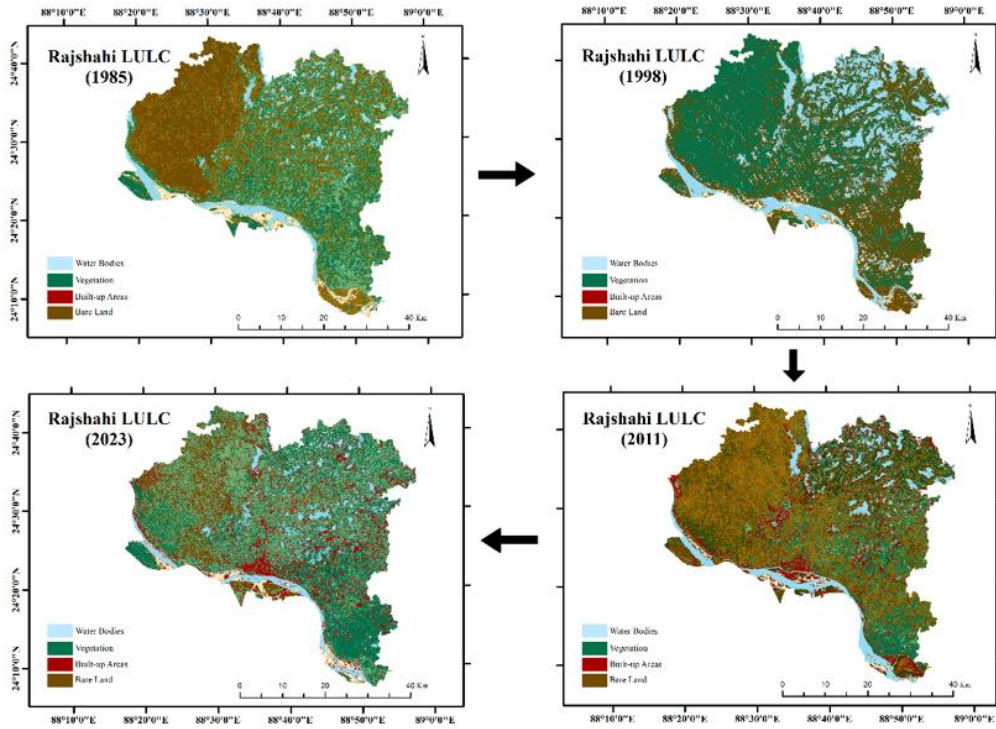


Fig.4.9. LULC change in Rajshahi, 1985, 1998, 2011 and 2023.

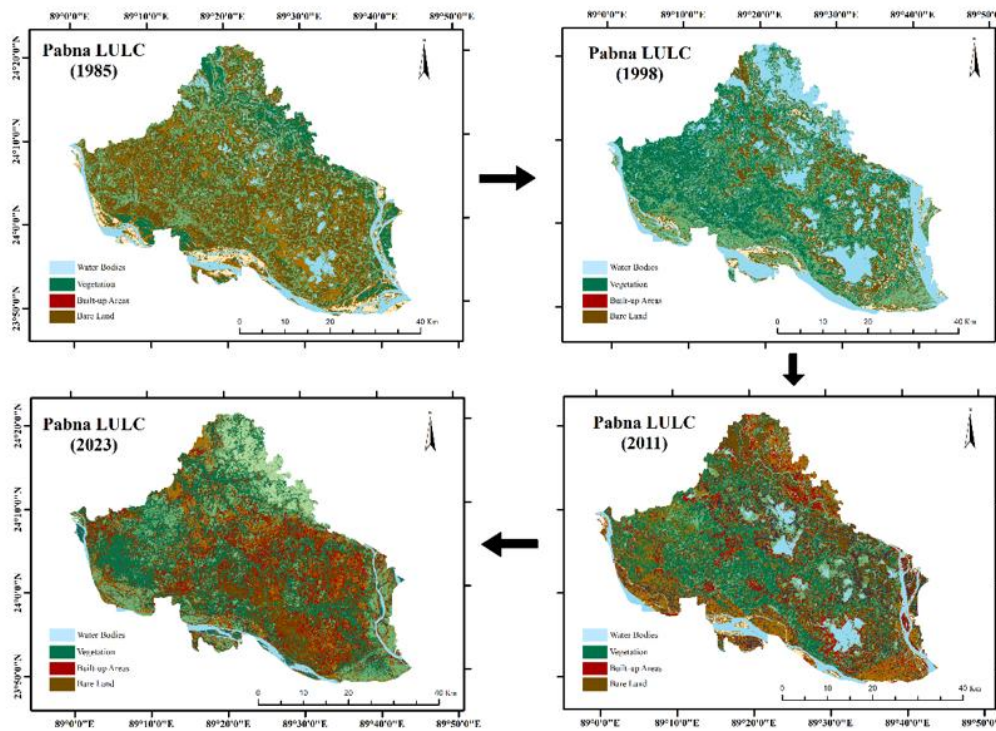


Fig.4.10. LULC change in Pabna, 1985, 1998, 2011 and 2023.

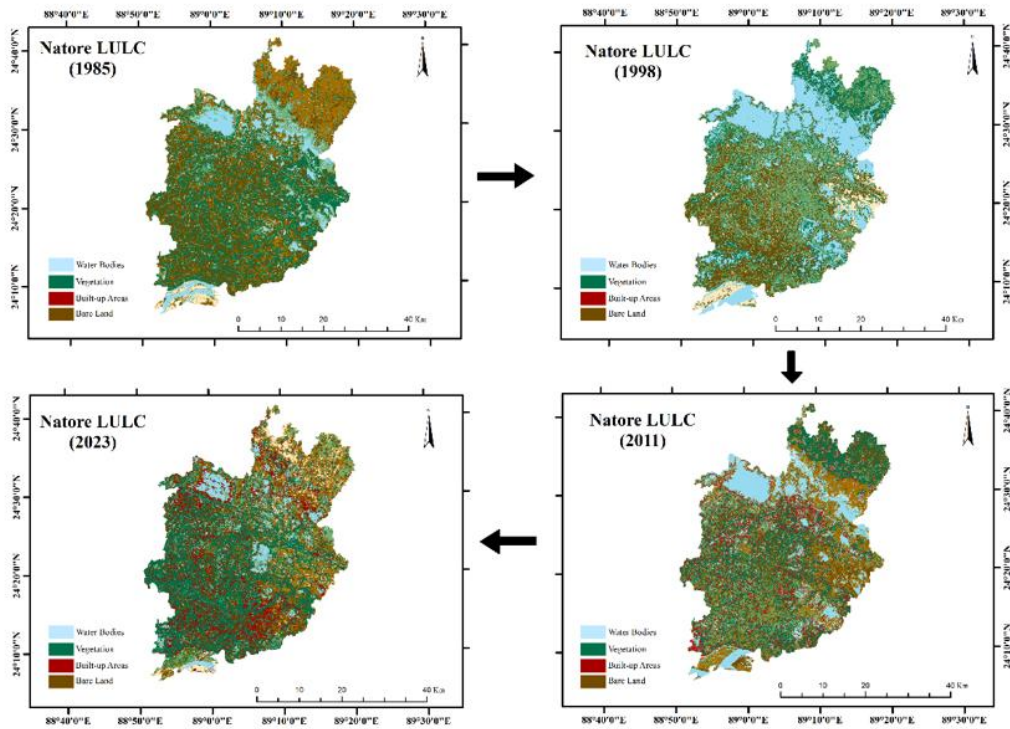


Fig.4.11. LULC change in Natore, 1985, 1998, 2011 and 2023

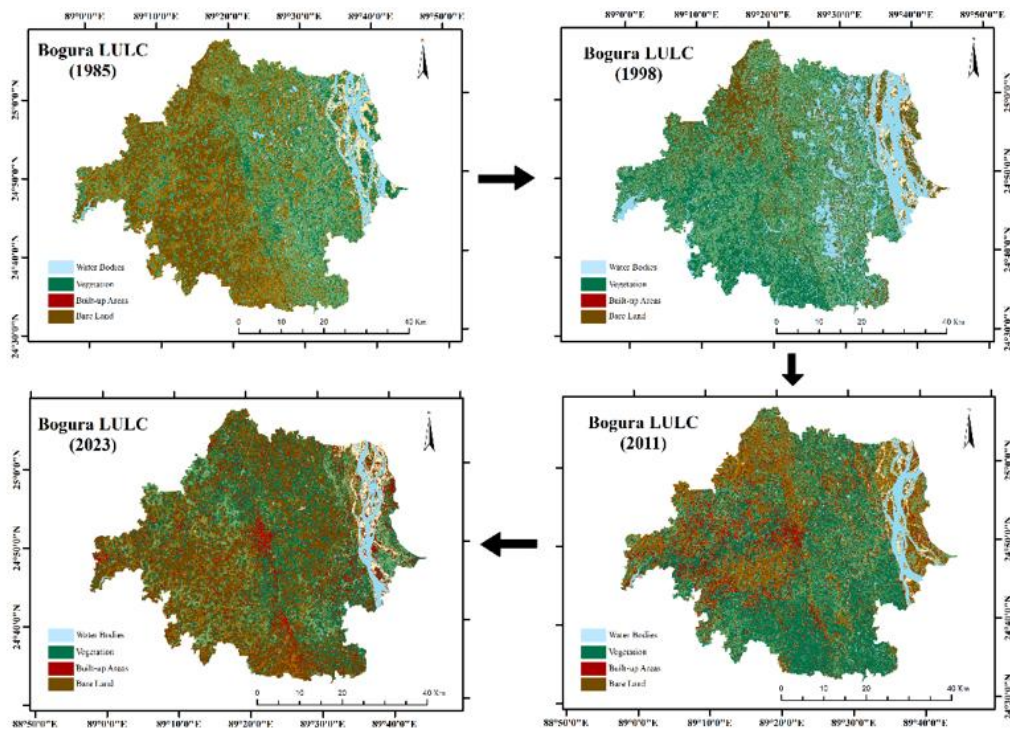


Fig.4.12. LULC change in Bogura, 1985, 1998, 2011 and 2023.

The percentage distribution of the LULC classes indicates substantial changes over the four observation years.

Table 4.1: Percentage distribution of LULC classes in Rajshahi, Pabna, Natore and Bogura, (1985-2023).

District	Year	Water Bodies (%)	Vegetation (%)	Built-up Areas (%)	Bare Land (%)
Rajshahi	1985	7.43	44.15	0.00	48.42
	1998	25.24	50.13	0.00	24.63
	2011	7.54	35.41	12.26	44.80
	2023	9.23	64.34	14.18	12.24
Pabna	1985	11.34	36.96	0.00	51.70
	1998	24.15	57.98	0.00	17.88
	2011	8.39	49.55	12.06	30.00
	2023	2.69	50.12	8.39	38.81
Natore	1985	9.41	41.84	0.00	48.76
	1998	25.03	43.56	0.00	31.40
	2011	2.45	45.60	7.63	44.33
	2023	7.07	48.66	14.73	29.54
Bogura	1985	6.05	43.64	0.00	50.31
	1998	19.49	61.07	0.00	19.43
	2011	4.28	57.51	12.95	25.26
	2023	3.91	38.99	12.06	45.04

In Rajshahi, vegetation covered 44.15% of the area in 1985 and increased to 50.13% in 1998. It then declined to 35.41% in 2011 before rising considerably to 64.34% in 2023. Bare land showed an overall decline, falling from 48.42% in 1985 to 12.24% in 2023. Water bodies also changed considerably, increasing from 7.43% in 1985 to 25.24% in 1998, followed by a decline in 2011 and a slight increase in 2023. Built-up land was not classified in 1985 or 1998 but accounted for 12.26% in 2011 and 14.18% in 2023, indicating a substantial increase in the classified built-up area during the later observation years. Pabna also showed considerable changes in its LULC composition. Vegetation increased from 36.96% in 1985 to 57.98% in 1998, then declined to 49.55% in 2011 and remained relatively similar at 50.12% in 2023. Bare land decreased sharply from 51.70%

in 1985 to 17.88% in 1998 but increased again to 30.00% in 2011 and 38.81% in 2023. Water bodies increased from 11.34% in 1985 to 24.15% in 1998, before falling to 8.39% in 2011 and 2.69% in 2023. Built-up land accounted for 12.06% in 2011 and 8.39% in 2023. In Natore, the changes were comparatively moderate for vegetation, which increased from 41.84% in 1985 to 48.66% in 2023. Bare land declined from 48.76% to 29.54% over the same period, although an increase occurred in 2011. Water bodies increased substantially from 9.41% in 1985 to 25.03% in 1998, then declined to 2.45% in 2011 before recovering to 7.07% in 2023. Built-up land was not classified in 1985 or 1998 but increased to 7.63% in 2011 and 14.73% in 2023, giving Natore the highest classified built-up proportion among the four districts in 2023. Bogura followed a different pattern. Vegetation increased from 43.64% in 1985 to 61.07% in 1998, followed by a decline to 57.51% in 2011 and 38.99% in 2023. Bare land decreased sharply from 50.31% in 1985 to 19.43% in 1998, but increased considerably afterward, reaching 45.04% in 2023. Water bodies also declined, from 19.49% in 1998 to 3.91% in 2023. Built-up land accounted for 12.95% in 2011 and 12.06% in 2023.

Overall, the four districts did not follow a common LULC trajectory. Rajshahi and Natore generally show declines in bare land alongside increases in vegetation, whereas Bogura shows a marked reversal after 2011, with vegetation declining and bare land increasing. Pabna also experienced considerable fluctuations in vegetation, bare land, and water bodies. These differences highlight the strong spatial variation in LULC dynamics across the study area.

4.1.6 Relationship Between Climatic Variation and LULC Change

The climatic and LULC results were compared to examine whether the observed environmental changes provide a possible context for the spatial and temporal climatic patterns identified in the study. The comparison indicates several notable co-occurring patterns, but it does not establish direct causal relationships between LULC change and climatic trends. The clearest common climatic pattern is the general decline in annual rainfall accompanied by an increase in annual mean relative humidity across the four districts. At the same time, all four districts show the appearance of classified built-up areas in the 2011 and 2023 LULC maps, while vegetation and bare land underwent substantial changes. This temporal overlap suggests that changes in land surface characteristics may be relevant when interpreting local climatic conditions, particularly

variables such as humidity and temperature. However, the relationship is not sufficiently consistent to attribute the climatic trends directly to LULC change. Rajshahi and Natore fit this pattern most closely: both show rising vegetation alongside rising humidity and declining rainfall over the same period. Because several LULC classes changed simultaneously in each district, however, it is not possible to isolate the contribution of any single class to the observed climatic pattern. Bogura provides an important contrast. Although rainfall showed a downward linear tendency and humidity increased, vegetation declined considerably between 1998 and 2023, while bare land increased to 45.04% in 2023.

Built-up land remained around 12% in the 2011 and 2023 classifications. Thus, the same climatic tendencies observed elsewhere occurred despite a substantially different LULC trajectory. This weakens the argument for a simple direct relationship between a particular LULC change and the observed climatic trends. These comparisons indicate that LULC change may provide useful environmental context, but it cannot be considered the sole explanation for the observed climatic variations. Climatic variables are influenced by atmospheric circulation, regional moisture transport, seasonal processes and other factors that were not explicitly modelled in the present analysis. Moreover, the climatic variables represent district-level averages, whereas the LULC classifications represent spatial distributions within each district. Therefore, a direct one-to-one comparison between the percentage of a LULC class and a climatic trend should be interpreted cautiously. The LULC classification itself also requires some qualifications. No built-up area was detected in the 1985 and 1998 classifications for any of the four districts. This should not be interpreted as evidence that built-up land was completely absent during those years. The absence may partly reflect the spatial resolution and quality of the historical satellite imagery, the relatively dispersed nature of settlements at the time, and the difficulty of distinguishing small or mixed built-up features from surrounding land covers. Therefore, the 0.00% values for those two years should be treated as a limitation of the classification rather than an exact representation of historical settlement extent.

Similarly, the relatively high built-up proportions detected in Natore and Pabna in 2011 and 2023 should be interpreted with caution. Differences in image quality, spectral characteristics and classification performance may have resulted in some built-up surfaces being classified more extensively than their actual distribution (Olofsson et al., 2014; Foody, 2002). This is particularly relevant when comparing the built-up percentages

across years. Consequently, the LULC results are most appropriate for identifying broad spatial and temporal changes, rather than treating every percentage-point difference as an exact measure of real-world land conversion. Despite these limitations, the LULC analysis provides useful supporting evidence for interpreting the climatic results. The major finding is not that a particular LULC class caused a particular climatic trend, but that substantial changes in land surface composition occurred during the same long-term period in which the climatic variables also changed. The different LULC trajectories among the four districts, together with their different temperature and wind-speed trends, further indicate that local environmental conditions may contribute to spatial heterogeneity in climatic behavior. This interpretation is consistent with previous research in northwestern Bangladesh. Siddik et al. (2022) demonstrated that LULC changes can influence environmental processes such as groundwater recharge, while Xu et al. (2020) showed that LULC dynamics in Bangladesh are spatially heterogeneous and influenced by multiple factors. The present findings extend this perspective by placing LULC change alongside long-term climatic observations, while avoiding a direct causal attribution that cannot be established from the current dataset.

4.2 Long-term Climatic Trends

To assess whether the observed long-term changes in the selected climatic parameters represented statistically supported trends, the annual climatic series for 1973-2023 were examined using the Mann-Kendall (MK) test and Sen's slope estimator. The analysis was conducted separately for Rajshahi, Pabna, Natore, and Bogura. The MK test was used to identify the direction and statistical significance of monotonic trends, while Sen's slope was used to quantify the rate of change per year. A significance level of 0.05 was adopted; therefore, trends with $p < 0.05$ were considered statistically significant. Unlike the linear trend lines presented in Section 4.1, which were used primarily to illustrate the general direction of change, the present analysis provides formal statistical evidence regarding trend significance and magnitude. This distinction is important because a visible upward or downward slope does not necessarily indicate a statistically significant long-term change. The analysis was based on the complete 51-year annual series rather than the four shorter periods used for spatial comparison. The annual rainfall series was derived from yearly accumulated rainfall, whereas annual temperature, relative humidity, and wind speed were calculated as the means of their respective monthly observations.

Table 4.2: Mann-Kendall and Sen's slope results for rainfall, temperature, relative humidity, and wind speed.

Parameter	District	Mean	SD	Sen Slope	MK Tau	MK P-value	MK Sig
Wind Speed (m/s/yr)	Bogura	1.2	0.39	0.0032	0.118	0.223	ns
	Natore	1.38	0.42	-0.0129	-0.312	0.0013	**
	Pabna	1.46	0.55	-0.0141	-0.391	1.00E-04	***
	Rajshahi	1.33	0.52	0.0012	0.018	0.8582	ns
Temperature (°C/yr)	Bogura	22.47	0.79	-0.0031	-0.038	0.6966	ns
	Natore	23.07	0.78	0.0083	0.114	0.2389	ns
	Pabna	23.22	0.88	5.00E-04	0.0039	0.9741	ns
	Rajshahi	25.96	0.37	0.0153	0.431	0	***
Humidity (%/yr)	Bogura	77.58	2.02	0.0175	0.081	0.4028	ns
	Natore	77.42	2.59	0.1145	0.436	0	***
	Pabna	76.87	3.16	0.1397	0.488	0	***
	Rajshahi	77.65	2.81	0.1341	0.548	0	***
Rainfall (mm/yr)	Bogura	1757.57	374.03	-7.037	-0.185	0.0553	ns
	Natore	1538.32	284.84	-8.9969	-0.308	0.0015	**
	Pabna	1557.59	384.52	-7.8387	-0.209	0.0307	*
	Rajshahi	1438.92	297.03	-8.7083	-0.314	0.0012	**

4.2.1 Long-term trend in rainfall

Rainfall showed a predominantly decreasing tendency across the four study districts during 1973-2023, although the strength and statistical significance of the decline varied spatially. The Sen's slope estimates were negative for all four districts, ranging from -7.04 mm/year in Bogura to -9.00 mm/year in Natore. Rajshahi showed a slope of -8.71 mm/year, while Pabna showed -7.84 mm/year. Thus, the estimated rates correspond approximately to declines of 70.4, 90.0, 78.4, and 87.1 mm per decade in Bogura, Natore, Pabna, and Rajshahi, respectively. The MK results indicate that the decreasing tendency was statistically significant in Natore, Pabna, and Rajshahi, but not in Bogura. Natore recorded a Kendall's tau of -0.3082 with $p = 0.0015$, while Rajshahi showed a slightly stronger negative association in rank terms, with $\tau = -0.3145$ and $p = 0.0012$. Pabna also showed a significant decreasing trend ($\tau = -0.2094$, $p = 0.0307$). In contrast, Bogura had $\tau = -0.1859$, but its p -value of 0.0553 was marginally above the adopted 0.05 significance threshold. Therefore, its negative tendency cannot be classified as statistically significant under the criterion used in this study.

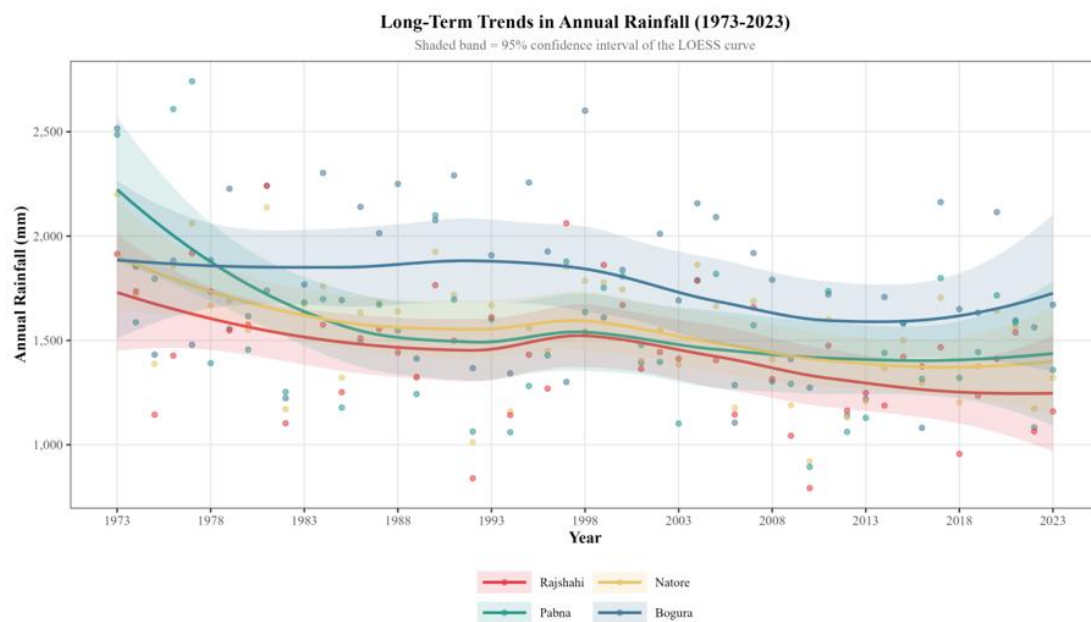


Fig.4.13. Long-term trend in annual rainfall across the four study districts, 1973-2023.

The results provide stronger evidence of rainfall decline than the visual trend assessment alone. The spatial difference is also important. The statistically significant decline in Rajshahi is consistent with earlier findings from northwestern Bangladesh. Ferdous and Baten (2012), using a long-term record covering the former Rajshahi Division, reported a decreasing annual rainfall tendency for Rajshahi while increasing tendencies were

observed in some other northwestern locations. Their findings therefore support the present observation that rainfall trends cannot necessarily be generalized across the whole northwestern region. The more recent studies focused specifically on the Rajshahi-Barind region also provide useful context, although their results are not directly equivalent to the present analysis because they used different periods, datasets, and spatial frameworks. Shahriyar et al. (2026), for example, examined rainfall variability in Rajshahi City for 1990-2024 using NASA POWER data, while Esha et al. (2026) assessed annual rainfall across the Barind Tract using a network of stations for 2000-2024. These shorter and differently structured datasets demonstrate the spatial and temporal variability of rainfall in the region, whereas the present analysis extends the annual record back to 1973 and compares four districts within a common framework. The non-significant result for Bogura should also be considered carefully. Its negative Sen's slope is relatively similar in magnitude to those of the other districts, but the MK test does not provide sufficient evidence to reject the null hypothesis of no monotonic trend at the 5% level. This difference illustrates why the magnitude of a fitted trend should not be interpreted independently of its statistical significance.

4.2.2 Long-term trend in temperature

Temperature exhibited a different pattern from rainfall. The estimated Sen's slopes were positive in Natore, Pabna, and Rajshahi, while Bogura showed a very small negative slope. However, only Rajshahi displayed a statistically significant increasing trend over the study period. Rajshahi recorded a mean annual temperature of 25.96°C and a Sen's slope of +0.0153°C/year, equivalent to approximately +0.153°C per decade. The MK test produced a Kendall's tau of 0.4315 with $p < 0.001$, indicating a statistically significant increasing trend. In contrast, Natore had a positive Sen's slope of +0.0083°C/year, but the corresponding MK result was not significant ($\tau = 0.1146$, $p = 0.2389$). Pabna showed an almost negligible positive slope of +0.0005°C/year, with $p = 0.9741$. Bogura showed a slight negative slope of -0.0031°C/year, but this was also statistically non-significant ($p = 0.6966$).

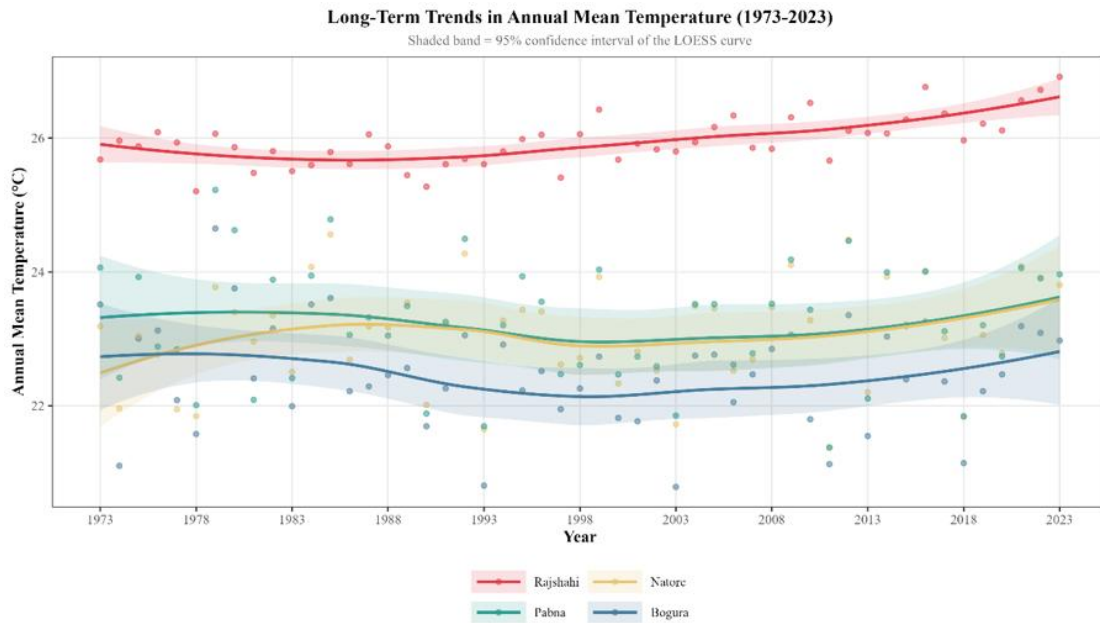


Fig.4.14. Long-term trend in annual mean temperature across the four study districts, (1973-2023).

The temperature results therefore indicate a clear spatial contrast. Rajshahi experienced a statistically supported long-term warming tendency, whereas the other three districts did not show significant monotonic changes over the same period. This distinction is important because the positive slopes observed in Natore and Pabna might otherwise be interpreted as evidence of regional warming simply from their direction. The MK results show that such an interpretation would be too broad. The result is broadly compatible with earlier evidence that climatic trends in northwestern Bangladesh vary between locations. Ferdous and Baten (2012) identified differences in temperature and other climatic variables among Rajshahi, Rangpur, and Dinajpur, demonstrating that regional averages may conceal important station-level differences. The present four-district comparison extends this spatial perspective by examining the same 51-year period and applying the same statistical procedure to each district. The stronger warming signal in Rajshahi may also help explain why its annual mean temperature is distinctly higher than the other districts in the dataset. However, the trend analysis alone does not establish the physical causes of this difference. Possible influences such as regional atmospheric processes, local environmental conditions, land-cover characteristics, and differences in station surroundings would require separate attribution analysis and therefore are beyond what can be concluded from the present results.

4.2.3 Long-term trend in relative humidity

Relative humidity showed the most consistent increase across the four districts, although the magnitude varied considerably. Sen's slopes were +0.0175, +0.1145, +0.1397, and +0.1341 percentage points/year for Bogura, Natore, Pabna, and Rajshahi, respectively, equivalent to approximately 0.18, 1.15, 1.40, and 1.34 percentage points per decade. Statistical results distinguish Bogura from the other districts: its weak positive trend was non-significant ($\tau = 0.0816$, $p = 0.4028$), whereas Natore ($\tau = 0.4369$), Pabna ($\tau = 0.4886$), and Rajshahi ($\tau = 0.5482$) showed statistically significant trends, with $p < 0.001$ for all three.

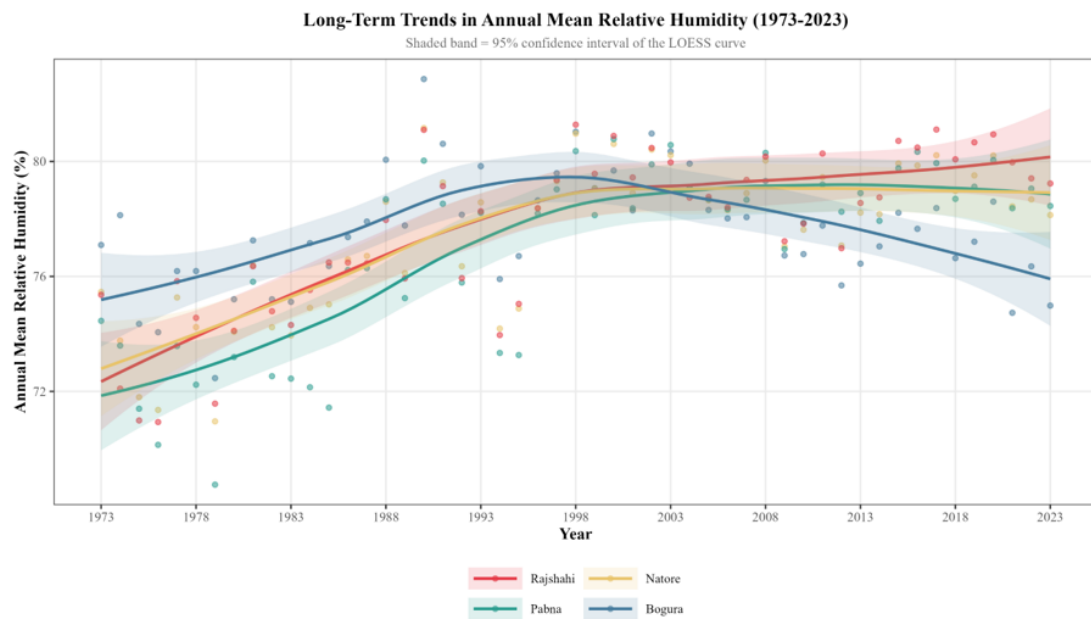


Fig.4.15. Long-term trend in annual mean relative humidity across the four study districts, (1973-2023).

The relative humidity results are particularly interesting when considered alongside the annual series. The LOESS curves show that humidity did not increase at a constant rate: Pabna, Natore, and Rajshahi increased mainly during the earlier and middle periods, followed by more limited changes in later years, while Bogura increased earlier and then gradually declined after around the turn of the century. Thus, the significant positive MK results for three districts represent long-term overall tendencies rather than continuous year-to-year increases. This distinction is important because the Mann-Kendall test assesses the overall tendency of the full series, whereas the smoothed curves show shorter-term variations and departures from a simple monotonic pattern. The two therefore provide complementary rather than conflicting evidence. The broader humidity

increases in Rajshahi, Pabna, and Natore also adds context to the temperature results, particularly in Rajshahi, where both temperature and humidity increased significantly. This does not indicate a direct causal relationship between the two variables; their statistical association is examined separately through the correlation analysis in Section 4.5.

4.2.4 Long-term trend in wind speed

Wind speed exhibited the weakest and most spatially inconsistent trend among the four climatic parameters. The Sen's slope estimates were positive for Bogura and Rajshahi but negative for Natore and Pabna. Bogura recorded a slope of approximately $+0.0032$ m/s/year, while Rajshahi showed $+0.0012$ m/s/year. In contrast, Natore and Pabna recorded negative slopes of -0.0129 m/s/year and -0.0141 m/s/year, respectively. The MK results indicate that the decreasing tendencies in Natore and Pabna were statistically significant. Natore recorded $\tau = -0.3123$ with $p = 0.0013$, while Pabna recorded $\tau = -0.391$ with $p < 0.001$. Bogura's positive tendency was not significant ($\tau = 0.1186$, $p = 0.223$), and Rajshahi likewise showed no statistically significant trend ($\tau = 0.0180$, $p = 0.8582$).

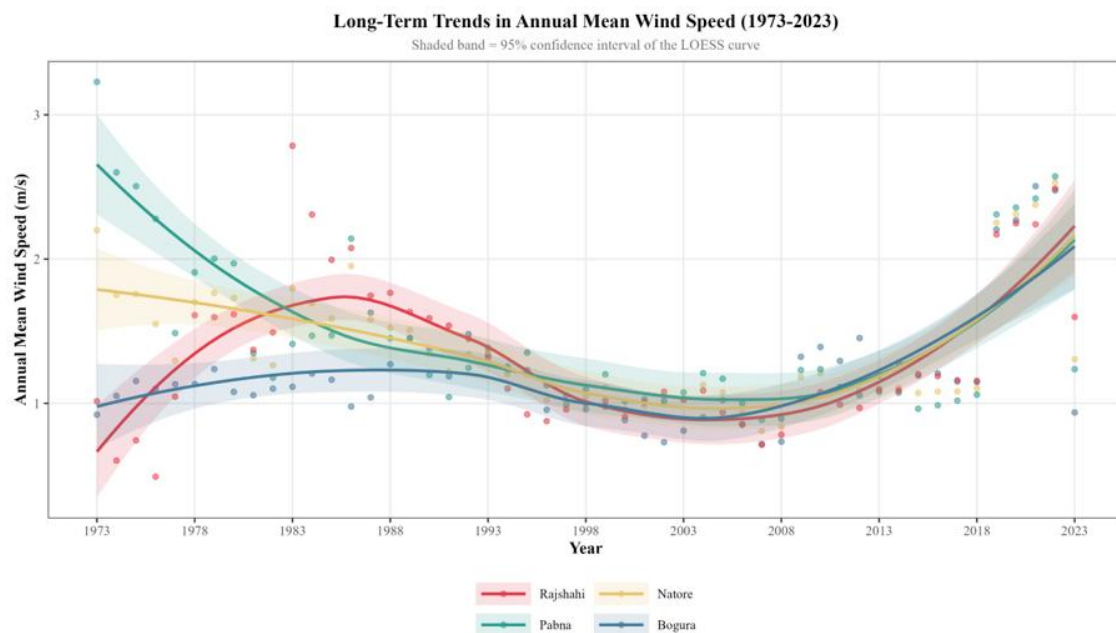


Fig.4.16. Long-term trend in annual mean wind speed across the four study districts, 1973-2023.

The results therefore suggest that changes in wind speed were not regionally uniform. The statistically significant declines in Natore and Pabna contrast with the weak and non-significant tendencies observed in Bogura and Rajshahi. This spatial contrast is important

because wind speed is influenced by local and regional atmospheric conditions and may also be affected by changes in the surroundings of meteorological stations. Therefore, the present results are best interpreted as district-level temporal tendencies rather than evidence of a single regional wind-speed regime. The contrast between wind speed and the other climatic variables also demonstrates the value of including multiple parameters in the analysis. Whereas rainfall showed significant declines in three districts and relative humidity showed significant increases in three districts, wind speed produced a different spatial pattern. A single climatic variable would therefore provide only a partial representation of the long-term climatic behavior of the study region.

4.2.5 Comparative assessment of long-term climatic trends

The combined results demonstrate that the four climatic parameters did not change in a uniform manner across Rajshahi, Pabna, Natore, and Bogura. The strongest spatial consistency was observed for rainfall and relative humidity, although their directions were opposite. Rainfall showed negative Sen's slopes in all four districts and statistically significant decreases in three, whereas relative humidity showed positive slopes in all four districts and statistically significant increases in three. Temperature and wind speed displayed greater spatial differentiation, with statistically significant warming detected only in Rajshahi and significant wind-speed declines restricted to Natore and Pabna.

Table 4.3: Summary of long-term climatic trends during 1973-2023.

Climatic parameter	Bogura	Natore	Pabna	Rajshahi
Rainfall	Decreasing, ns	Decreasing, significant	Decreasing, significant	Decreasing, significant
Temperature	Decreasing, ns	Increasing, ns	Increasing, ns	Increasing, significant
Relative humidity	Increasing, ns	Increasing, significant	Increasing, significant	Increasing, significant
Wind speed	Increasing, ns	Decreasing, significant	Decreasing, significant	Increasing, ns

Table 4.3 summarizes these results across all four variables and districts. The clearest pattern is Bogura's consistent exceptionalism: rainfall, temperature, relative humidity and wind speed all show directional tendencies in the fitted series, yet none reaches the adopted significance threshold, the only district where this is true for every parameter. This reinforces the spatial heterogeneity identified in earlier studies of northwestern Bangladesh; Ferdous and Baten (2012), for example, reported contrasting climatic trends among locations within the broader region rather than a uniform regional response. The most robust signals overall are the rainfall decline in Natore, Pabna and Rajshahi; the humidity increase in the same three districts; the temperature increase in Rajshahi alone; and the wind-speed decline in Natore and Pabna.

4.3 Seasonal and Monthly Climatic Variability

The seasonal and monthly analysis was conducted to characterize the typical annual cycle of rainfall, temperature, relative humidity, and wind speed across Rajshahi, Pabna, Natore, and Bogura. The analysis used the complete monthly record from 1973 to 2023. Following the seasonal classification described in Chapter 3, December–February was treated as winter, March–May as pre-monsoon, June–September as monsoon, and October–November as post-monsoon. Seasonal rainfall was calculated as the sum of monthly rainfall, whereas temperature, relative humidity, and wind speed were represented by the mean of the corresponding monthly values. Monthly climatology was obtained by averaging observations for the same calendar month across the 51-year record. The results show a distinct annual climatic cycle across all four districts, although the magnitude of individual variables differs spatially. Rainfall is strongly concentrated during the monsoon season, temperature follows a pronounced warm-season cycle, relative humidity generally increases during the rainy period, and wind speed reaches its highest values mainly during the pre-monsoon and early monsoon months. The overall seasonal structure is similar among the districts, but several differences are evident in the magnitude and timing of the monthly peaks.

Table 4.4: Seasonal mean characteristics of rainfall, temperature, relative humidity, and wind speed across the four study districts, (1973-2023).

District	Season	Temperature Mean	Rainfall Mean	Humidity Mean	Wind Speed Mean
Rajshahi	Winter	19.1	32	75.2	1.02
Rajshahi	Pre- monsoon	28.2	232.1	67.9	1.63
Rajshahi	Monsoon	29.6	1047.7	85.5	1.56
Rajshahi	Post- monsoon	25.5	127.1	80.2	0.9
Pabna	Winter	18.9	34.8	74.2	0.98
Pabna	Pre- monsoon	28.1	294	67.8	1.87
Pabna	Monsoon	29.4	1094.3	84.4	1.81
Pabna	Post- monsoon	25.5	134.5	79.6	0.88
Natore	Winter	19.2	32.3	74.7	0.97
Natore	Pre- monsoon	28	268.8	68.4	1.74
Natore	Monsoon	29.5	1103.1	85	1.68
Natore	Post- monsoon	25.6	134.2	79.8	0.87
Bogura	Winter	19.7	29.4	74.1	0.88
Bogura	Pre- monsoon	27.5	324.7	70.6	1.52
Bogura	Monsoon	29.3	1252.1	84.7	1.42
Bogura	Post- monsoon	26	151.4	79	0.77

4.3.1 Seasonal variability of climatic parameters

Table 4.4 summarizes the seasonal mean characteristics of rainfall, temperature, relative humidity and wind speed across the four districts. Rainfall showed the strongest seasonal contrast among the four climatic parameters: in all four districts, the monsoon season was overwhelmingly the dominant rainfall period, while winter received very little rainfall, and Bogura consistently recorded the highest seasonal rainfall throughout the wetter part of the year, with its monsoon rainfall approximately 204 mm greater than Rajshahi's. This spatial difference extends across the pre-monsoon and post-monsoon periods as well as the monsoon peak, indicating it is not restricted to a single season. The overall seasonal pattern is a pronounced transition from relatively dry winter conditions to increasing rainfall during pre-monsoon, a strong monsoon maximum, and a rapid reduction during post-monsoon.

Temperature showed a much narrower seasonal range than rainfall but a clear annual cycle, with the lowest seasonal temperatures occurring during winter and the warmest conditions concentrated in the pre-monsoon and monsoon periods, when all four districts converge to about half a degree of each other (29.3-29.6°C). This differs from rainfall in one important respect: while rainfall changes very sharply between dry and wet seasons, temperature changes more gradually across the year. Relative humidity showed a seasonal pattern broadly opposite to temperature during the transition from the dry to wet part of the year, declining to its lowest values during the pre-monsoon season before increasing sharply during the monsoon, when Rajshahi recorded the highest seasonal humidity of any district at 85.5%. This close timing between the humidity maximum and the rainfall maximum is notable when the two seasonal results are considered together, though the relationship should not be interpreted as evidence of direct causation from these seasonal averages alone; both variables are responding to the broader seasonal climatic regime, and their statistical association is examined separately in Section 4.5.

Wind speed displayed a different seasonal pattern from the other variables, with the highest values generally occurring during the pre-monsoon and monsoon periods rather than at the rainfall peak, and Pabna consistently recording the highest seasonal wind speed of the four districts. Wind speed then declined considerably after the monsoon, reaching its lowest value during post-monsoon and winter. This seasonal pattern was broadly consistent across the four districts, although the magnitude of wind speed differed among them. Thus, unlike rainfall and humidity, wind speed does not reach its annual maximum during the period of highest rainfall in every district; its strongest conditions are concentrated around the pre-monsoon and early wet-season period.

4.3.2 Monthly Climatology of Rainfall

The monthly rainfall climatology provides a clearer view of the annual rainfall cycle than the four-season averages alone. Across all four districts, rainfall remained very low from November through February, began to increase during March and April, rose sharply in May, and reached its highest levels during the monsoon. The rainfall pattern therefore shows a clear seasonal concentration, with most of the annual rainfall occurring during the monsoon months.

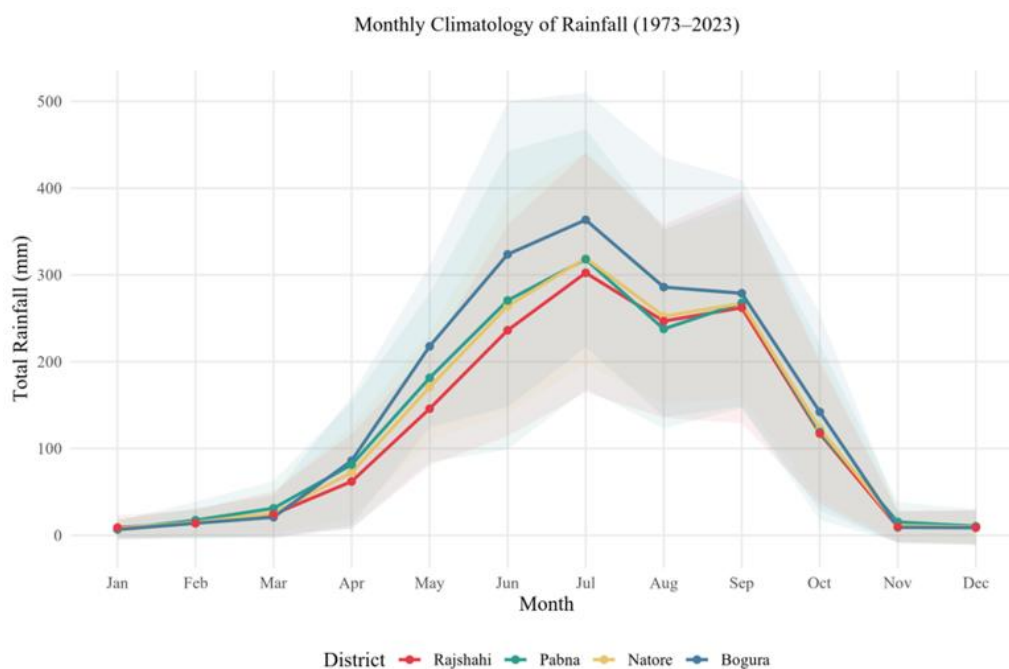


Fig.4.17. Monthly climatology of rainfall across Rajshahi, Pabna, Natore, and Bogura, (1973-2023).

The monthly values show that July was the wettest month in all four districts. Rainfall reached 363.5 mm in Bogura, 319.5 mm in Natore, 318.1 mm in Pabna, and 302.4 mm in Rajshahi during July. August and September also remained relatively wet, but rainfall began to decline after the maximum of July. The transition into the wet season was already evident in May. Rainfall increased from only 20.9-31.2 mm in March to 62.0-86.0 mm in April, and then to 145.9-217.8 mm in May across the four districts. This rapid increase indicates that the main rainfall season is preceded by a substantial pre-monsoon buildup rather than beginning abruptly in June. The monthly pattern also shows that Bogura generally experienced greater rainfall during the wettest months. Its July rainfall of 363.5 mm was approximately 61 mm higher than Rajshahi's July value. The difference was also evident in June, when Bogura recorded 323.7 mm, compared with 236.3 mm in Rajshahi. After September, rainfall decreased sharply. October rainfall remained moderate, ranging from 117.3 mm in Rajshahi to 142.2 mm in Bogura, but November values fell below 16 mm in all districts, and December rainfall was below 11 mm. Thus, the monthly climatology identifies a distinct rainfall cycle consisting of a dry winter, gradual pre-monsoon increase, strong monsoon maximum centered on July, and rapid post-monsoon decline. This monthly pattern is consistent with the seasonal results but provides a more precise indication of the timing and magnitude of the peak of the rainfall.

4.3.3 Monthly Climatology of Temperature

The monthly temperature cycle was more gradual than the rainfall cycle. Temperatures were lowest during January and increased steadily through the pre-monsoon months before reaching their highest values around June–August, followed by a gradual decline toward winter.

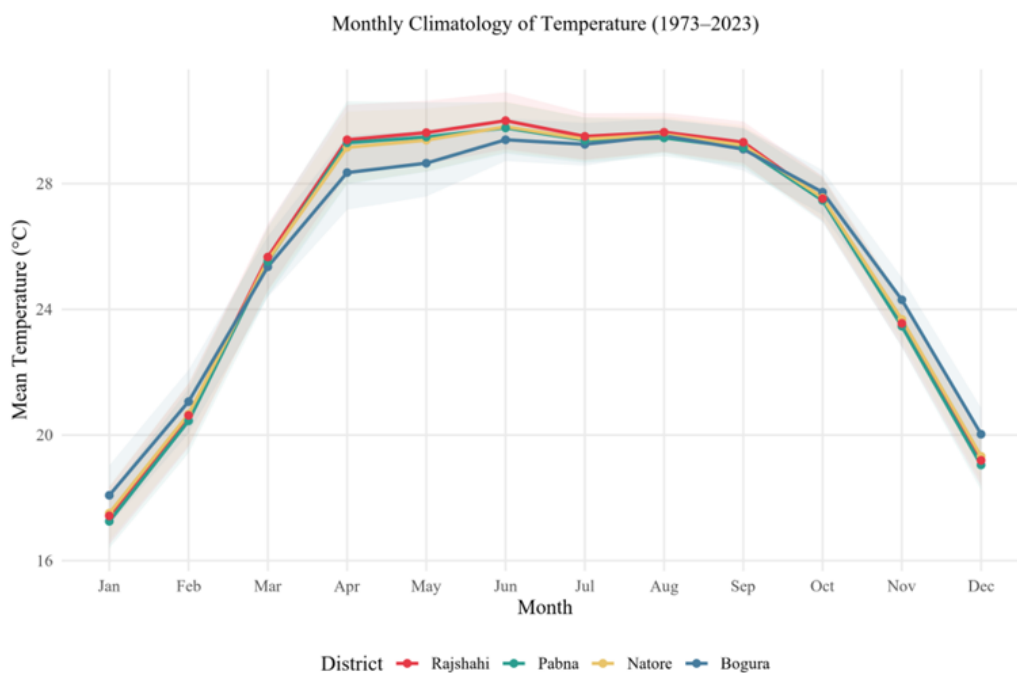


Fig.4.18. Monthly climatology of mean temperature across Rajshahi, Pabna, Natore, and Bogura, (1973-2023).

January was the coolest month in all four districts. Mean January temperatures ranged from 17.25°C in Pabna to 18.08°C in Bogura. Temperature increased through February and March, reaching approximately 25-26°C by March. The strongest warming within the annual cycle occurred between March and May. By May, mean temperatures had reached 29.62°C in Rajshahi, 29.48°C in Pabna, 29.38°C in Natore, and 28.65°C in Bogura. The highest monthly temperature differed slightly in timing among the districts. Rajshahi reached its maximum climatological temperature in June at 30.00°C, while Natore and Pabna also recorded their highest values around the June-May period. Bogura remained comparatively cooler, with a June mean of 29.40°C. After June, temperatures remained relatively stable through August and September before declining more noticeably in October and November. By December, temperatures had fallen to approximately 19.0-20.0°C across the four districts. The temperature cycle therefore shows a relatively broad warm period extending from approximately April through September rather than a single

sharply defined monthly peak. This contrasts with rainfall, whose annual maximum is much more concentrated around July.

4.3.4 Monthly climatology of relative humidity

Relative humidity displayed a pronounced seasonal cycle, with the lowest values occurring during the pre-monsoon period and the highest values during the monsoon.

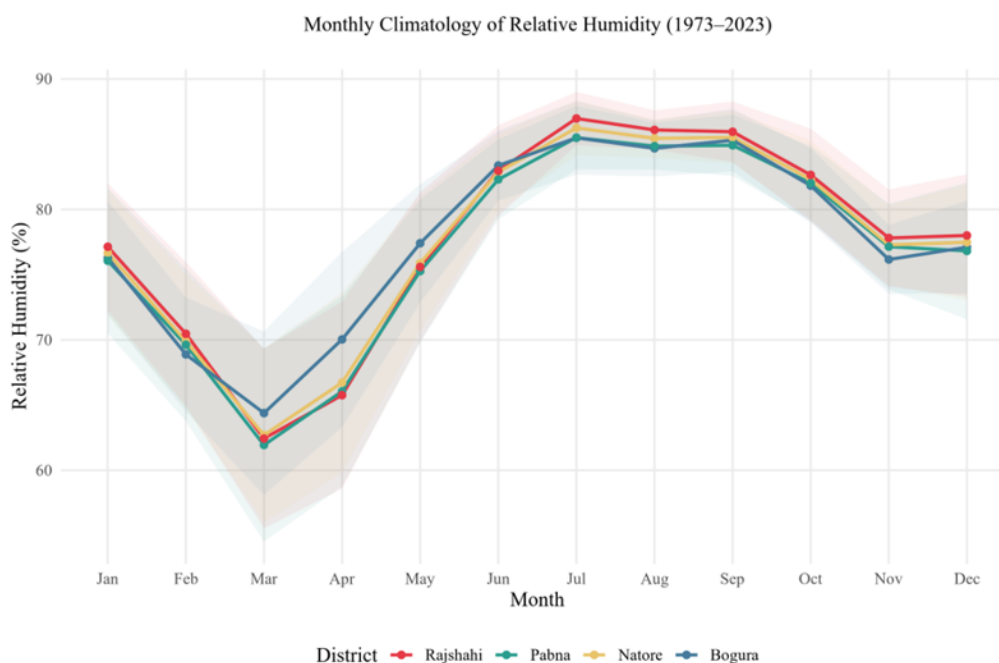


Fig.4.19. Monthly climatology of mean relative humidity across Rajshahi, Pabna, Natore, and Bogura, (1973-2023).

Humidity decreased from the winter months into March. The lowest monthly values occurred in March, when mean relative humidity was 64.39% in Bogura, 62.70% in Natore, 61.94% in Pabna, and 62.44% in Rajshahi. The decline in humidity coincided with the rise in temperature during the pre-monsoon period. After March, however, humidity increased rapidly. By May, values had risen to approximately 75-77%, and by June they exceeded 82% in all four districts. The highest humidity occurred in July. Rajshahi recorded the highest monthly value at 86.97%, followed by Natore at 86.25%, Pabna at 85.50%, and Bogura at 85.48%. Humidity remained high through August and September before beginning to decline in October. The annual humidity cycle therefore has a relatively clear minimum during the late pre-monsoon period and maximum during the monsoon. The close timing of the humidity maximum with the rainfall maximum

further emphasizes the strong seasonal control of atmospheric moisture in the study region.

4.3.5 Monthly climatology of wind speed

Wind speed showed the greatest difference in its monthly cycle compared with rainfall, temperature, and relative humidity. The highest values occurred mainly during the pre-monsoon and early monsoon months, followed by a steady decline during the later monsoon and post-monsoon periods.

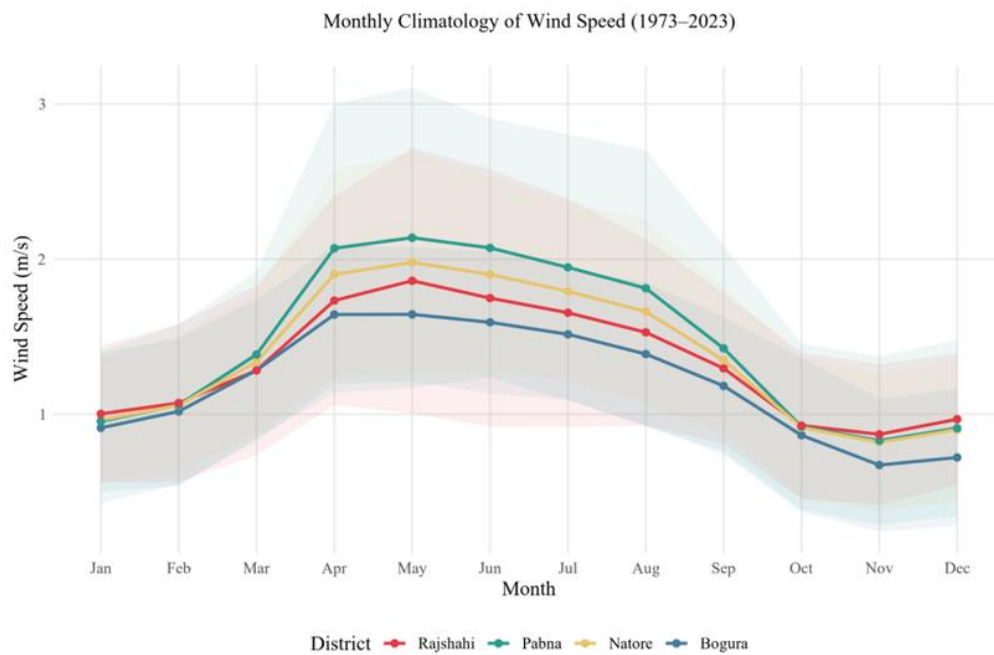


Fig.4.20. Monthly climatology of mean wind speed across Rajshahi, Pabna, Natore, and Bogura, (1973-2023).

Wind speed increased from winter into the pre-monsoon season. In April, values reached 2.07 m/s in Pabna, 1.90 m/s in Natore, 1.73 m/s in Rajshahi, and 1.64 m/s in Bogura. Pabna maintained the highest values in May and June, reaching 2.14 m/s in May and 2.07 m/s in June. In Rajshahi, the monthly maximum occurred in May (1.86 m/s), while Natore reached its maximum in April at approximately 1.98 m/s. Bogura showed a lower overall wind-speed regime, with a maximum of approximately 1.64 m/s in April–May. Wind speed subsequently decreased through the latter part of the monsoon. By October, the values had fallen below 1 m/s in all four districts, and the lowest monthly values occurred in November, ranging from 0.67 m/s in Bogura to 0.87 m/s in Rajshahi. The annual wind-speed cycle therefore differs from the rainfall and humidity cycles. Its maximum is reached before or around the beginning of the main monsoon period,

followed by a gradual decline. This pattern is also reflected in the seasonal results, where pre-monsoon wind speeds were generally higher than post-monsoon and winter values.

4.3.6 Comparative interpretation of seasonal and monthly variability

The combined seasonal and monthly results demonstrate that the four climatic parameters follow distinct but interconnected annual cycles across the study area. Rainfall is strongly concentrated during the monsoon, temperature increases from winter to warmer months, relative humidity reaches its highest value during the wet season, and wind speed peaks primarily during the pre-monsoon and early monsoon periods. The rainfall cycle is the most sharply seasonal. July represents the climatological rainfall maximum in all four districts, with Bogura recording the highest July rainfall at 363.5 mm and Rajshahi the lowest at 302.4 mm. This spatial difference is consistent with the seasonal analysis, in which Bogura recorded the highest total monsoon rainfall. Temperature, in contrast, has a broader warm period. Temperatures rise rapidly from March through May and remain high through September before declining toward winter. The highest monthly temperature recorded in the climatological series was approximately 30.0°C in Rajshahi during June. The relatively small differences among districts indicate that the annual temperature cycle is more spatially consistent than the rainfall cycle. Relative humidity follows a pattern closely associated with the seasonal moisture regime. Its minimum occurs around March, when temperatures are already increasing but rainfall remains comparatively low. Humidity then rises sharply with the onset of wetter conditions and reaches its maximum in July. This seasonal behavior is particularly relevant when considered with the long-term trend results in Section 4.2, where significant increases in annual mean relative humidity were identified for Rajshahi, Pabna, and Natore. Wind speed shows a different timing, reaching its highest monthly value before or around the beginning of the monsoon. Pabna recorded the highest monthly wind speed among the four districts, reaching 2.14 m/s in May, while the lowest values occurred during November. This indicates that the strongest winds do not coincide exactly with the maximum annual rainfall, despite both variables showing substantial seasonal variability.

Taken together, the results indicate that seasonal variability is a major component of the climatic characteristics of the four districts. Bogura is distinguished by comparatively high rainfall during the wet season, Rajshahi by relatively high temperature and humidity during several months, and Pabna by comparatively high pre-monsoon wind speeds. This

seasonal structure is important for interpreting the extreme precipitation results in the next section, since a significant annual trend does not imply uniform change across every month.

4.4 Extreme Precipitation and GEV Analysis

Extreme precipitation was examined using the Annual Maximum Series (AMS) of monthly rainfall for the four study districts over the 1973-2023 period. For each year, the highest monthly rainfall value was selected to represent the annual maximum event. This approach focuses on the upper end of the rainfall distribution and allows changes in the magnitude of extreme monthly rainfall to be examined separately from changes in annual rainfall totals. It also provides a consistent basis for comparing the behavior of the largest monthly rainfall events among the four districts. The resulting AMS series therefore captures the most extreme monthly rainfall conditions within each year and provides the basis for the subsequent extreme-value analysis. Since the analysis was based on monthly rather than daily observations, the resulting extremes represent annual maximum monthly rainfall and should be interpreted accordingly. The AMS series were first assessed for their long-term trend and were then fitted with a Generalized Extreme Value (GEV) distribution to estimate rainfall associated with different return periods.

4.4.1 Annual Maximum Rainfall and Its Long-Term Trend

The annual maximum rainfall series showed substantial year-to-year variation across all four districts throughout the 51-year study period. The magnitude of the annual maximum events varied considerably from one year to another, with several years showing rainfall values well above the general level of the series. This variability was particularly evident in Rajshahi and Pabna, where a few relatively high annual maximum events occurred during the earlier and middle portions of the record. Similar fluctuations were also observed in Natore and Bogura, although the magnitude and timing of the higher events differed among the districts. Despite this interannual variability, the fitted trend lines indicate an overall downward tendency in annual maximum rainfall in all four districts. Thus, the long-term decline occurred alongside considerable year to year variability rather than a steady decrease throughout the record.

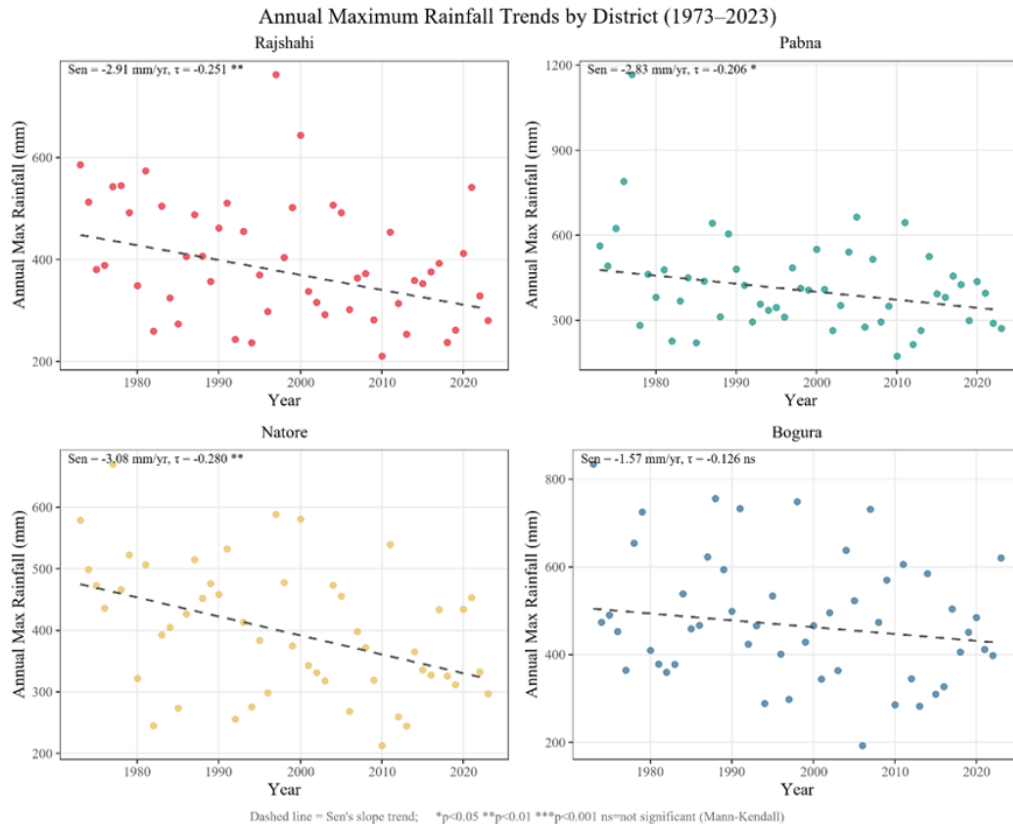


Fig.4.21. Long-term trends in annual maximum rainfall across Rajshahi, Pabna, Natore, and Bogura, (1973–2023).

The Mann-Kendall and Sen's slope results support the downward tendency in three of the four districts. Rajshahi recorded a Sen's slope of -2.91 mm/year (Kendall's tau = -0.251), Natore -3.08 mm/year (tau = -0.280), and Pabna -2.83 mm/year (tau = -0.206), with all three trends statistically significant. Bogura showed a smaller decline of -1.57 mm/year, but its Kendall's tau of -0.126 was not significant. Thus, although all four districts had negative slopes, evidence of a systematic decline in annual maximum rainfall was weaker in Bogura. The AMS trends broadly follow the declining pattern found for annual rainfall in Section 4.2, although their magnitudes differ because annual rainfall represents the total yearly accumulation, whereas AMS represents only the largest monthly rainfall event in each year. The negative AMS trends therefore indicate that the largest monthly rainfall events generally declined in Rajshahi, Natore, and Pabna, but this should not be taken to mean that all forms of extreme rainfall declined uniformly. Natore had the steepest estimated decline, followed by Rajshahi and Pabna, while Bogura showed the weakest trend. The non-significant result for Bogura means that its negative slope alone does not establish a systematic long-term decline, consistent with the weaker statistical evidence found for its annual rainfall trend. These results point to spatial differences in

extreme-rainfall behavior rather than a single regional pattern. The downward AMS trends should also be considered alongside the strong seasonal concentration of rainfall identified in Section 4.3. July was the wettest month in all four districts, and most annual rainfall occurred during the monsoon period, making changes in annual maximum monthly rainfall particularly relevant to high-rainfall events during the wet season. However, the present analysis does not identify the physical or environmental processes responsible for these changes. It is also important to distinguish the trend results from the return-level analysis: a significant decreasing trend in observed AMS does not by itself determine the probability or magnitude of rare future events. The GEV analysis provides a separate probabilistic assessment of the upper tail of the AMS distribution and estimates extreme rainfall magnitudes for specified return periods.

4.4.2 GEV Model Fit and Goodness of Fit

The AMS series for each district was fitted using the Generalized Extreme Value distribution. The fitted parameters and goodness-of-fit statistics are presented in Table 4.5. The estimated GEV parameters differed among the districts, reflecting differences in the distribution and magnitude of annual maximum rainfall.

Table 4.5: GEV parameter estimates and goodness-of-fit statistics for annual maximum rainfall.

District	μ	σ	ξ	NLLH	AIC	AD statistic	AD p-value
Bogura	423.16	125.87	-0.127	323.63	653.26	0.225	0.982
Natore	358.98	95.84	-0.172	308.48	622.97	0.352	0.894
Pabna	351.04	116.41	0.069	324.96	655.93	0.142	0.999
Rajshahi	345.36	95.92	-0.031	312.58	631.16	0.321	0.921

The estimated location parameter was highest for Bogura ($\mu = 423.16$ mm), followed by Natore (358.98 mm), Pabna (351.04 mm), and Rajshahi (345.36 mm). Bogura also had the largest scale parameter ($\sigma = 125.87$ mm), indicating greater dispersion of annual maximum values, while Pabna, Natore, and Rajshahi had values of 116.41, 95.84, and 95.92 mm, respectively. The shape parameter was negative for Bogura (-0.127), Natore (-0.172), and Rajshahi (-0.031) but positive for Pabna (0.069), indicating differences in the upper-tail behavior of the fitted GEV distributions and, consequently, in the estimated return levels. Here, μ denotes the location parameter, σ the scale parameter, ξ the shape

parameter, NLLH the negative log-likelihood, AIC the Akaike Information Criterion, and AD the Anderson-Darling statistic.

Q-Q diagnostic assessment: The fitted GEV distributions were further evaluated using quantile–quantile (Q–Q) plots, which compare the empirical AMS quantiles with the theoretical quantiles expected from the fitted GEV distributions. A close alignment of the points with the reference line indicates that the fitted distribution represents the observed data reasonably well, whereas systematic departures from the line indicate differences between the observed and theoretical distributions.

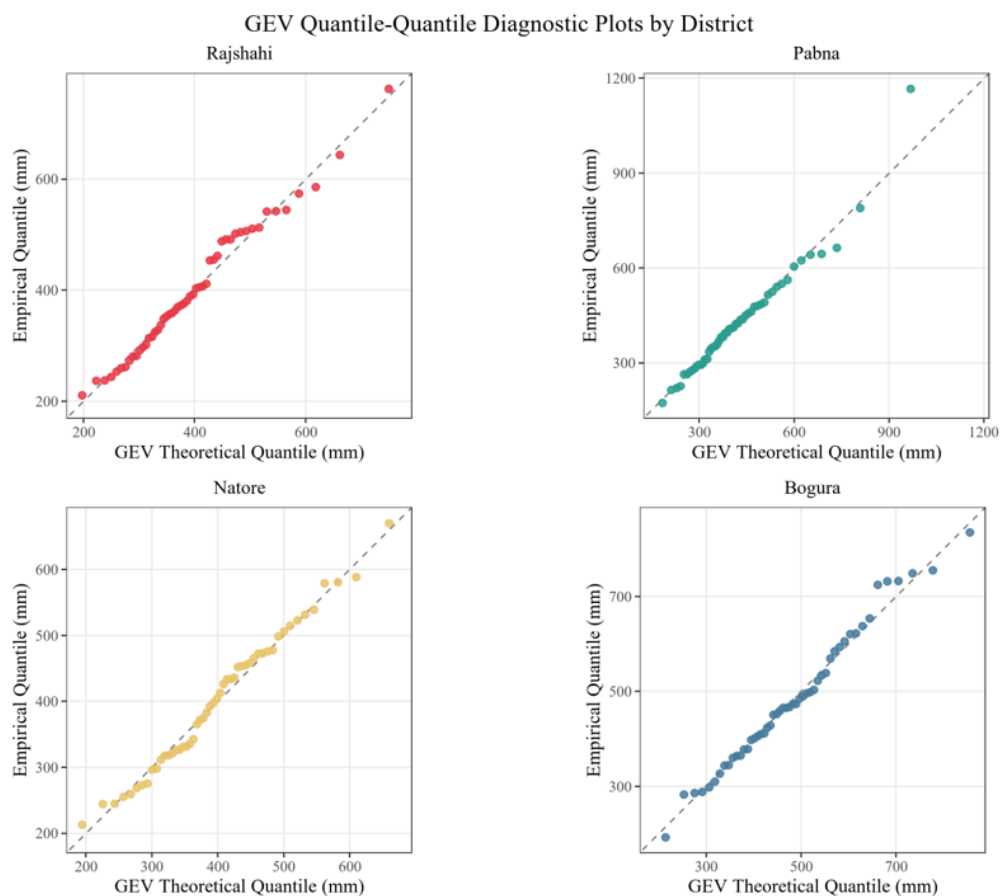


Fig.4.22. GEV Q-Q diagnostic plots for Rajshahi, Pabna, Natore, and Bogura.

The Q-Q plots show generally good agreement between the observed AMS values and the fitted GEV distributions across all four districts, particularly through the central part of the distributions. Rajshahi follows the reference line relatively closely, while Natore and Bogura show some deviations at the lower and upper ends. Pabna shows the most notable departure at the highest observed value, where the largest empirical value lies considerably above the corresponding theoretical quantile. This reflects an unusually high

AMS value and suggests greater uncertainty in the far upper tail, which is also consistent with Pabna's relatively large, long return period estimates. Overall, the plots support the GEV model for describing the main structure of the AMS series, although departures near the extremes are important because the upper tail is used to estimate long return periods. The Q–Q results should therefore be considered alongside the confidence intervals when interpreting these estimates.

The Anderson-Darling (AD) results also indicate an acceptable GEV fit for all four districts. The AD statistics were 0.225 for Bogura, 0.352 for Natore, 0.142 for Pabna, and 0.321 for Rajshahi, with corresponding p-values of 0.982, 0.894, 0.999, and 0.921, respectively. Since all p-values were well above 0.05, there was no statistical evidence to reject the fitted GEV distributions. This indicates that the differences between the AMS values observed and the fitted distributions were not statistically significant according to the AD test. Pabna had the smallest AD statistic and highest p-value, indicating the closest agreement according to this test, while Bogura also showed a very small AD statistic and high p-value; Natore and Rajshahi likewise showed no evidence of poor fit. The AIC values ranged from 622.97 for Natore to 655.93 for Pabna. Because the same GEV model and number of observations were used for all districts, these values describe the relative likelihood-based fit within their respective datasets and do not indicate differences in climatic conditions. Taken together with the Q–Q plots, the AD results provide consistent evidence that the fitted GEV distributions adequately represent the observed AMS series. Overall, both the graphical and statistical diagnostics support using the GEV model to estimate rainfall return levels for the four districts.

4.4.3 Return-Level Estimates of Extreme Rainfall

The fitted GEV distributions were used to estimate rainfall return levels for 2-, 5-, 10-, 25-, 50-, and 100-year return periods. These estimates quantify the magnitude of extreme monthly rainfall associated with progressively lower exceedance probabilities under the historical distribution. The results, together with their corresponding confidence intervals, are presented in Table 4.6 and illustrate both the increase in estimated rainfall with longer return periods and the growing uncertainty attached to rarer events.

Table 4.6: GEV-based return-level estimates of annual maximum rainfall.

District	Return period (years)	Estimated rainfall (mm)	Lower CI (mm)	Upper CI (mm)
Bogura	2	468.2	426.6	514.5
	5	595.0	536.4	644.4
	10	669.5	600.7	725.5
	25	753.9	656.0	840.5
	50	810.3	689.1	932.1
	100	861.4	715.5	1027.3
Natore	2	393.0	362.7	424.4
	5	485.7	446.4	519.7
	10	537.9	491.6	576.4
	25	594.9	533.3	650.0
	50	631.6	553.5	707.9
	100	663.8	567.8	768.9
Pabna	2	394.3	354.6	440.7
	5	535.1	471.7	602.3
	10	634.6	543.3	735.6
	25	767.9	616.6	959.5
	50	872.7	662.3	1170.7
	100	981.9	708.8	1427.1
Rajshahi	2	380.3	349.5	413.1
	5	486.0	441.0	527.0
	10	554.0	492.7	609.1
	25	637.6	543.6	724.7
	50	698.2	574.9	823.1
	100	757.0	602.4	935.1

The estimated return levels increased progressively with return period in all four districts. For the 2-year return period, the estimated rainfall ranged from 380.3 mm in Rajshahi to 468.2 mm in Bogura. At the 10-year return period, the estimates increased to 554.0 mm in Rajshahi, 537.9 mm in Natore, 634.6 mm in Pabna, and 669.5 mm in Bogura. The increasing return levels indicate that larger rainfall events are associated with progressively lower exceedance probabilities under the fitted GEV distributions. The

spatial differences become more pronounced at longer return periods. During the 25-year return period, the estimated rainfall was 594.9 mm in Natore, 637.6 mm in Rajshahi, 753.9 mm in Bogura, and 767.9 mm in Pabna. At the 50-year return period, Pabna had the highest estimated value (872.7 mm), followed by Bogura (810.3 mm), Rajshahi (698.2 mm), and Natore (631.6 mm). The 100-year return-level estimates show the clearest contrast among the districts. Pabna produced the highest estimate at 981.9 mm, followed by Bogura at 861.4 mm, Rajshahi at 757.0 mm, and Natore at 663.8 mm. Thus, although Pabna did not have the highest 2-year estimate, its return-level estimates increased more rapidly with increasing return period. This indicates a heavier upper tail in the fitted distribution compared with the other districts, which is consistent with its positive estimated shape parameter.

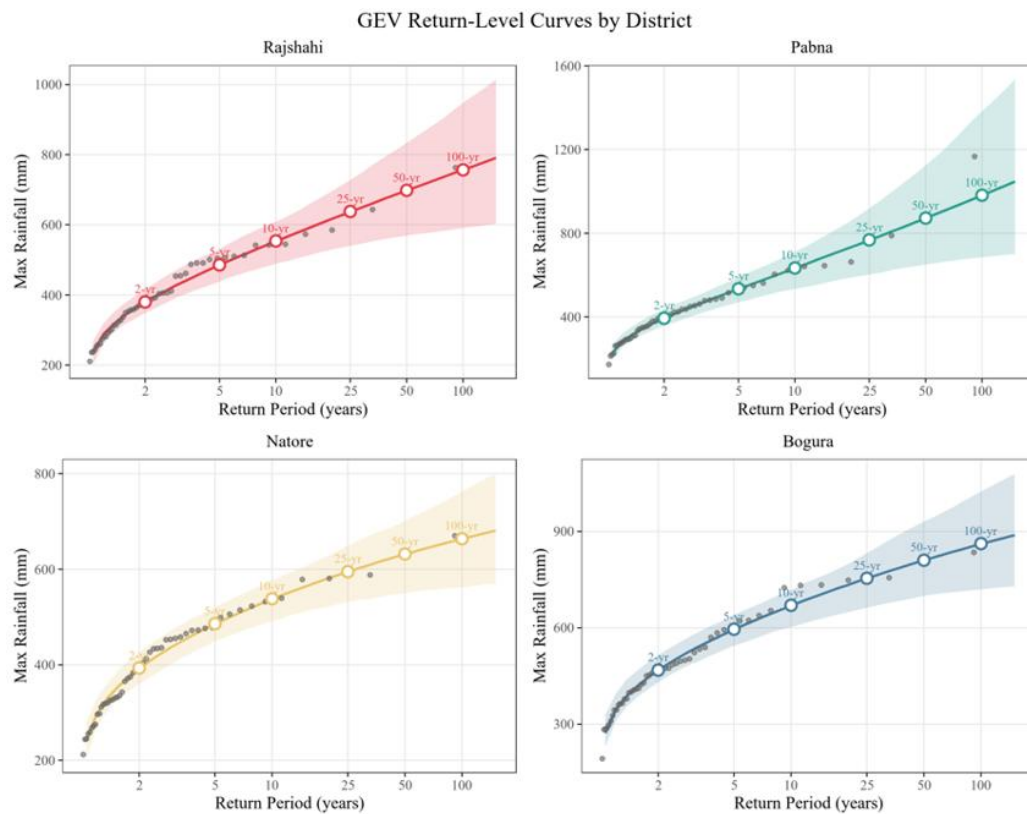


Fig.4.23. GEV return-level curves for the four districts.

The confidence intervals become wider as the return period increases, particularly for the longer return periods. This pattern is expected because estimates of very rare events are based on progressively less information in the observed 51-year record. The uncertainty is especially pronounced for Pabna, where the 100-year estimate of 981.9 mm has a lower confidence limit of 708.8 mm and an upper confidence limit of 1427.1 mm. Bogura also

shows considerable uncertainty at the 100-year level, with an estimated return level of 861.4 mm and a confidence interval from 715.5 to 1027.3 mm. The widening confidence intervals are important when interpreting the long return periods. The 100-year values should therefore not be treated as precise predictions of a rainfall amount that will occur in a particular year. Rather, they represent model-based estimates of the magnitude associated with a 100-year return period, with substantial uncertainty arising from the limited length of the observational record. The uncertainty becomes particularly relevant when extrapolating beyond the range of frequently observed annual maximum events.

Overall, the return-level analysis indicates considerable spatial variation in extreme rainfall characteristics across the study area. Bogura generally showed relatively high return levels in the shorter and intermediate return periods, while Pabna produced the highest estimates at the longer return periods. Rajshahi and Natore showed comparatively lower return levels throughout the analysis. These differences suggest that the four districts do not have identical extreme-rainfall distributions despite being located within the same broader climatic region. The GEV results also complement the AMS trend analysis. The observed AMS series showed decreasing trends in Rajshahi, Natore, and Pabna, while the fitted GEV distributions indicate that substantial extreme rainfall events remain possible within the historical climatic regime. The two results therefore address different aspects of extreme precipitation: the trend analysis describes the direction of change in the observed annual maxima, whereas the GEV analysis characterizes the probability and magnitude of extreme events across different return periods. Together, they provide a more complete assessment of extreme precipitation behavior in the four study districts.

4.5 Correlation Among Climatic Parameters

The relationships between temperature, rainfall, relative humidity, and wind speed were examined using Pearson, Spearman, and Kendall correlation coefficients. Pearson correlation was used to assess linear association, while Spearman and Kendall correlations provided rank-based measures of monotonic association. Using the three approaches together helped determine whether the observed relationships were consistent across different correlation measures. The results show that the strength and direction of the relationships varied considerably among the four districts. Rather than a single common correlation structure, each district showed a somewhat different combination of

significant and non-significant associations. The regional mean results also showed that some relationships remained evident after combining the district level series, although this regional pattern did not fully represent the differences observed among individual districts.

4.5.1 Overall Correlation Pattern

The regional correlation results indicate that the most consistent relationships were associated with relative humidity. Temperature showed a significant negative relationship with relative humidity, with Pearson's $r = -0.307$ ($p = 0.0285$), Spearman's $\rho = -0.313$ ($p = 0.0254$), and Kendall's $\tau = -0.213$ ($p = 0.0277$). Relative humidity also showed a negative association with wind speed, particularly in the rank-based analyses. The corresponding coefficients were $r = -0.297$ ($p = 0.0346$), $\rho = -0.553$ ($p = 2.52 \times 10^{-5}$), and $\tau = -0.402$ ($p = 3.20 \times 10^{-5}$).

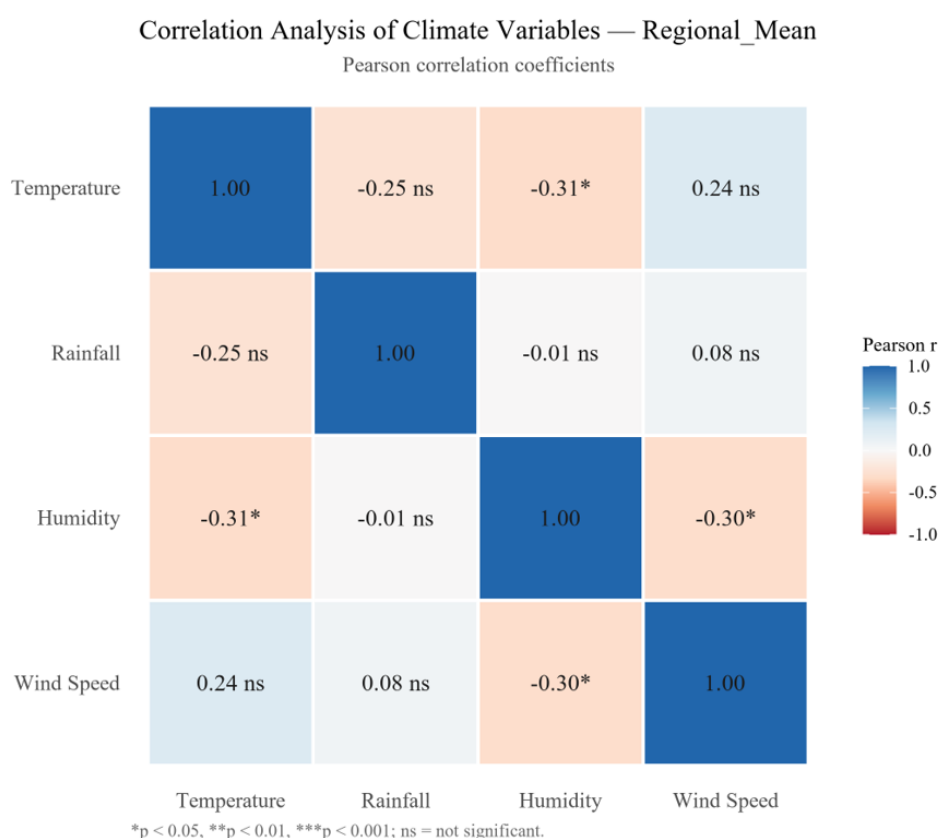


Fig.4.24. Regional mean correlation heatmap.

In contrast, the regional mean series did not show statistically significant relationships between temperature and rainfall, rainfall and relative humidity, or rainfall and wind speed. The temperature-rainfall coefficients were negative but non-significant across all

three methods ($r = -0.247$, $\rho = -0.264$, $\tau = -0.158$). Rainfall–humidity correlations were close to zero for Pearson and remained non-significant in the rank-based tests. Rainfall–wind speed correlations were also weak and non-significant. Temperature–wind speed showed a weak positive tendency, but this was not statistically significant. The regional results therefore suggest that the clearest associations in the combined series were those involving relative humidity, particularly its negative relationship with temperature and wind speed. However, these regional averages should not be taken as representative of all four districts because the district-level results show substantial spatial differences.

4.5.2 District-Level Relationships among Climatic Variables

The district-level correlation results provide a more detailed picture of the climatic relationships. Several associations that were weak or absent in the regional analysis became statistically significant at individual districts, while some relationships changed direction between districts. The relationship between temperature and rainfall was negative in most districts, although its statistical significance varied spatially. In Natore, temperature showed a statistically significant negative association with rainfall across all three correlation methods. Pearson's r was -0.323 ($p = 0.021$), while Spearman's ρ and Kendall's τ were -0.340 ($p = 0.0146$) and -0.229 ($p = 0.0177$), respectively. A stronger negative relationship was observed in Rajshahi. The Pearson and Spearman coefficients were both approximately -0.433 , with p -values of 0.00149 and 0.00152 , respectively, while Kendall's τ was -0.312 ($p = 0.00123$). The agreement among the three methods indicates that the negative association was not limited to a particular statistical measure. In Pabna, temperature and rainfall also showed a negative association, but the relationship was not statistically significant. The Pearson coefficient was -0.223 ($p = 0.116$), with similar non-significant results from Spearman and Kendall correlations. In Bogura, the relationship was essentially absent, with Pearson's $r = 0.030$ ($p = 0.834$), Spearman's $\rho = -0.038$ ($p = 0.790$), and Kendall's $\tau = -0.042$ ($p = 0.667$).

Thus, the temperature–rainfall relationship was most evident in Rajshahi and Natore, while it was weak or non-significant in Pabna and Bogura. This spatial difference is consistent with the broader finding that climatic variability was not identical across the four districts.

The temperature–relative humidity relationship showed one of the clearest spatial contrasts in the analysis. A significant negative association was found in Bogura, where

Pearson's r was -0.497 , Spearman's ρ was -0.431 , and Kendall's τ was -0.295 , with all three relationships statistically significant. A similar negative relationship occurred in Pabna, where Pearson's $r = -0.360$, Spearman's $\rho = -0.365$, and Kendall's $\tau = -0.258$. All three were statistically significant. The pattern was different in Natore. The coefficients remained negative, but they were relatively weak and non-significant: Pearson's $r = -0.159$, Spearman's $\rho = -0.210$, and Kendall's $\tau = -0.146$. In Rajshahi, the association changed direction, with positive but non-significant coefficients of 0.253 , 0.270 , and 0.174 for Pearson, Spearman, and Kendall correlations, respectively.

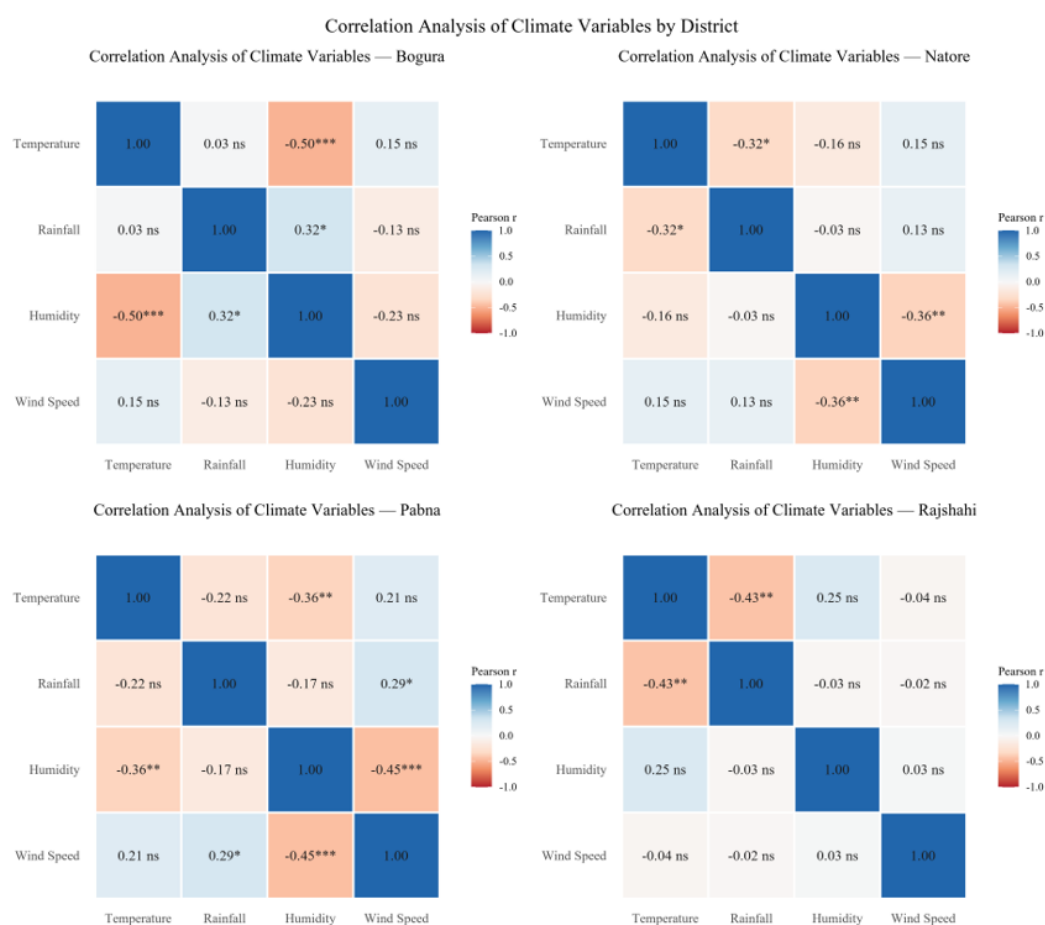


Fig.4.25. Pearson correlation heatmaps for the four districts.

The district-level results therefore do not support a single uniform temperature-humidity relationship across the study area. The negative relationship was strongest in Bogura and was also evident in Pabna, whereas Natore showed only a weak negative tendency and Rajshahi showed a weak positive tendency. This contrast is important because the regional mean series produced a significant negative relationship. The regional result therefore represents an overall tendency that partly masks the differences among individual districts. Relative humidity and wind speed also showed substantial spatial

variation. In Natore, the relationship was significantly negative, with Pearson's $r = -0.363$, Spearman's $\rho = -0.535$, and Kendall's $\tau = -0.400$. The rank-based coefficients were particularly strong compared with the Pearson coefficient, indicating that the negative association was more evident when the variables were considered in terms of their ranks.

The relationship was even stronger in Rajshahi. Pearson's r was -0.451 , while Spearman's ρ and Kendall's τ were -0.580 and -0.436 , respectively. All three were highly significant. In contrast, Bogura showed a negative tendency that did not reach statistical significance in the three methods. Pearson's r was -0.233 ($p = 0.0991$), while Spearman's $\rho = -0.299$ ($p = 0.0328$) and Kendall's $\tau = -0.196$ ($p = 0.0423$). Although the rank-based coefficients were statistically significant at the 5% level, the relationship was weaker than those observed in Natore and Rajshahi. Pabna showed almost no relationship between relative humidity and wind speed, with coefficients close to zero for all three methods. The contrast among districts indicates that the association between atmospheric moisture and wind speed was particularly pronounced in Natore and Rajshahi, while it was much weaker in Pabna. The regional mean result nevertheless retained a significant negative association, especially in the rank-based measures.

Most of the remaining variable pairs showed weak or statistically non-significant relationships. Temperature and wind speed, for example, did not show significant associations in any of the four districts. The same was generally true for rainfall and humidity, although Bogura showed a significant positive relationship between these variables. In Bogura, Pearson's $r = 0.319$ ($p = 0.0226$), Spearman's $\rho = 0.360$ ($p = 0.00947$), and Kendall's $\tau = 0.248$ ($p = 0.0103$). Rainfall and wind speed were also mostly weakly associated. A statistically significant positive Pearson correlation was observed in Rajshahi, with $r = 0.288$ ($p = 0.0405$), but the Spearman and Kendall coefficients were not significant. This difference between the methods suggests that the evidence for a consistent monotonic relationship is limited. In Pabna, rainfall and wind speed showed almost no association under any of the three approaches, while rainfall and humidity were similarly close to zero. Overall, these results indicate that rainfall did not maintain a consistent relationship with either humidity or wind speed across the study region.

4.5.3 Temperature-Relative Humidity Relationship

Because temperature and relative humidity showed one of the clearest and most spatially contrasting relationships, their association was examined further using scatterplots.

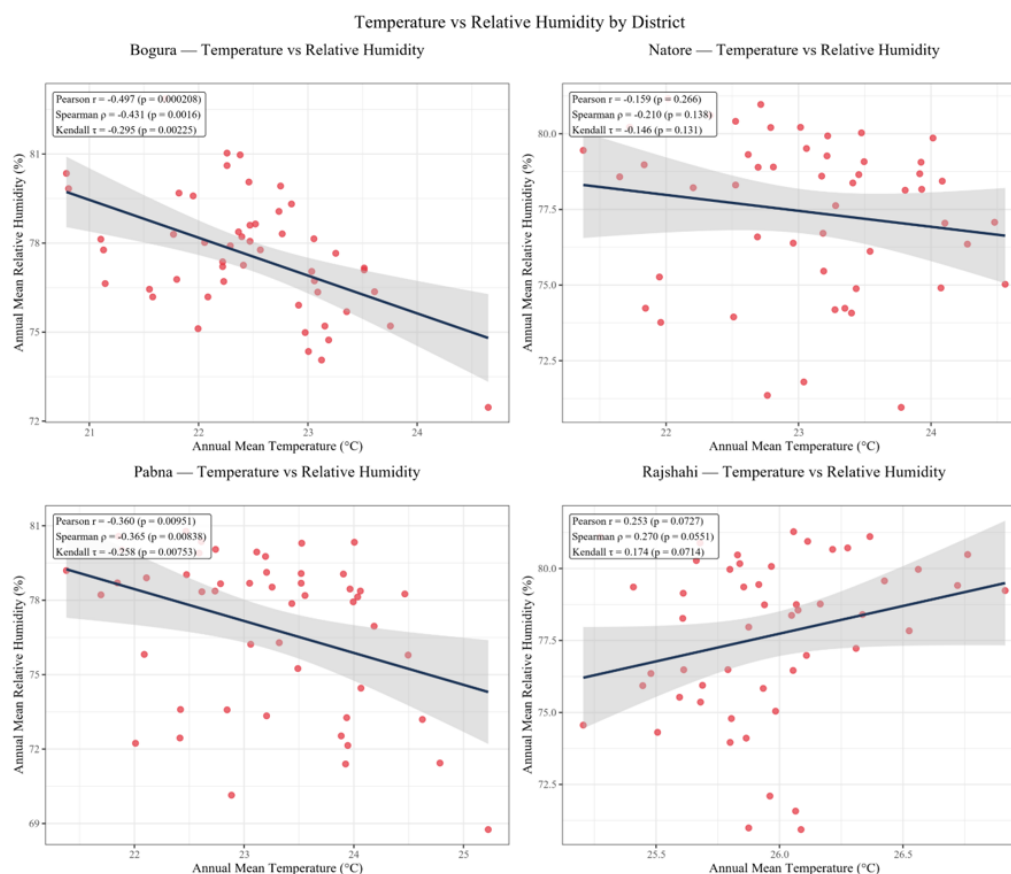


Fig.4.26. Relationship between annual mean temperature and relative humidity in the four study districts.

The scatterplots show that the direction of the relationship differed among the districts. Bogura displayed the clearest downward pattern, consistent with its relatively strong negative correlation. Pabna also showed a clear negative tendency, although the association was weaker than in Bogura. In Natore, the points were more dispersed, and the fitted relationship was weakly negative. Rajshahi differed from the other districts, showing a weak upward tendency, although the corresponding correlation was not statistically significant. The statistical results support these visual differences. In Bogura, the three correlation coefficients ranged from -0.295 to -0.497 , all statistically significant. Pabna similarly showed significant negative coefficients ranging from -0.258 to -0.365 . Natore's coefficients ranged from -0.146 to -0.210 and were not significant, while Rajshahi showed positive coefficients ranging from 0.174 to 0.270 , also without

statistical significance. The different behavior among the districts is important when interpreting the regional result. The regional mean series showed a significant negative association between temperature and relative humidity, but this does not mean that the same relationship occurred equally in every district. Instead, the regional result reflects the combined behavior of the four locations, while the district-level results reveal considerable spatial heterogeneity. This finding also complemented the temporal results presented in the earlier sections of the chapter. The long-term analyses showed changes in both temperature and relative humidity, but the correlation results demonstrate that these variables did not respond in the same way across all districts. A correlation coefficient describes how the two variables varied together over the study period; it does not by itself establish that changes in one variable directly caused changes in the other. Therefore, the observed temperature–humidity associations are best interpreted as statistical relationships within the climatic records of the respective districts.

4.5.4 Comparison of Correlation Methods

Although Pearson, Spearman, and Kendall correlations were calculated for all variable pairs, the three methods generally produced the same direction of association. This agreement strengthens the interpretation of relationships where significance was maintained across all three methods. For example, the negative temperature–rainfall relationship in Natore and Rajshahi remained significant regardless of the correlation method used. The same consistency was observed for temperature–humidity in Bogura and Pabna and for humidity–wind speed in Natore and Rajshahi. There were also cases where the strength or significance differed among methods. The clearest example is the humidity–wind speed relationship in Natore, where Pearson's r was -0.363 , compared with a stronger Spearman coefficient of -0.535 and Kendall coefficient of -0.400 . A similar pattern occurred in the regional mean series, where the Pearson coefficient for humidity–wind speed was -0.297 , compared with -0.553 for Spearman and -0.402 for Kendall. Another example is the rainfall–wind speed relationship in Rajshahi. Pearson correlation indicated a significant positive association, whereas neither Spearman nor Kendall correlation was significant. This suggests that the apparent linear association was not sufficiently consistent in rank order to be supported by the two non-parametric measures. Taken together, the three correlation approaches provide a more cautious interpretation than relying on Pearson correlation alone. Relationships that remained

significant across Pearson, Spearman, and Kendall tests can be regarded as the more robust associations in the dataset, while relationships detected by only one method should be interpreted with greater caution.

4.5.5 Discussion of the Overall Relationship among Climatic Variables

The correlation analysis demonstrates that the four climatic parameters did not vary together in a uniform manner across northwestern Bangladesh, with the clearest contrast being the temperature-humidity relationship discussed in Section 4.5.3. This spatial inconsistency is the central reason the district-level analysis matters here: the regional mean series produced significant negative associations for temperature-humidity and humidity-wind speed, but averaging across districts concealed the contrasting, and in Rajshahi's case opposite-signed, behavior found at individual locations. Overall, the correlation analysis supports the presence of several important interrelationships among the climatic parameters, but it does not indicate a single dominant relationship applicable to the entire study area. The findings are better characterized by spatially variable associations, with temperature–humidity, temperature–rainfall, and humidity–wind speed emerging as the principal relationships requiring attention. Most other variable pairs were weak or statistically non-significant.

4.6 Integrated Discussion of Major Findings

The analyses presented in the preceding sections provide different perspectives on climatic variability across Rajshahi, Pabna, Natore, and Bogura. The spatial maps describe how climatic conditions differ across the study area, the trend analysis shows how these conditions have changed over time, the seasonal and monthly analysis identifies the periods in which variability is most pronounced, and the extreme rainfall analysis examines changes in the upper end of the rainfall distribution. The LULC assessment provides an additional environmental context, while the correlation analysis shows how the climatic variables vary in relation to one another. Considered together, these findings indicate that climatic change in the study area is not represented by a single uniform trend. Rather, it is characterized by spatial differences, strong seasonality, changing long-term behaviour, and varying relationships among climatic variables.

4.6.1 From Spatial Patterns to Long-Term Change

The spatial and trend analyses converge on the same conclusion from different angles: the climatic evolution of the study area has been spatially uneven rather than following one regional signal. This is consistent with Kamruzzaman et al. (2018) and Khan et al. (2019), who found comparable spatial inconsistency elsewhere in western Bangladesh. The comparison with the two most recent, direct local studies illustrates why these matters in practice: Shahriyar et al. (2026) found an increasing rainfall trend for Rajshahi City (NASA POWER data, 1990-2024), while Esha et al. (2026) found predominantly decreasing rainfall across Barind Tract stations (BWDB data, 2000-2024). Neither result is wrong; the difference simply reflects data source, period and spatial scale, which is precisely why this study's longer, common 1973-2023 framework across four districts is useful, since it removes those confounds and still finds the same underlying spatial inconsistency.

4.6.2 Seasonality, Rainfall Regime and Extreme Events

The long-term rainfall findings become more meaningful when considered alongside the monthly and seasonal climatology. Rainfall in all four districts is strongly concentrated during the monsoon, while temperature, relative humidity and wind speed follow their own seasonal cycles. The seasonal analysis therefore shows that annual values represent the combined effect of substantially different conditions throughout the year. This is consistent with previous studies showing that climatic trends in Bangladesh vary according to both location and season. Khan et al. (2019) reported differences in climatic changes among seasons and regions, while Shahriyar et al. (2026) found that recent rainfall changes in Rajshahi were concentrated mainly between March and October, with the monsoon making the largest contribution to annual rainfall. The present study similarly demonstrates that annual trends need to be interpreted within the normal seasonal cycle rather than considered independently of it. The seasonal structure also provides the necessary context for interpreting the extreme rainfall results. The annual rainfall trend describes the behaviour of total rainfall, whereas the Annual Maximum Series and GEV analysis focus on the largest monthly rainfall total occurring within each year. These two analyses therefore address different aspects of rainfall variability. A declining annual rainfall tendency does not necessarily mean that large rainfall events have disappeared or that the probability of substantial rainfall in a particular year is

negligible. This distinction is also important in relation to existing literature. Recent studies by Shahriyar et al. (2026) and Esha et al. (2026) provide valuable evidence of rainfall trends in Rajshahi and the Barind Tract, respectively, but neither study fitted a GEV model to an Annual Maximum Series or produced return-level estimates. Thus, their findings describe changes in ordinary rainfall behavior rather than the probability structure of extreme rainfall. The present study extends the rainfall assessment by examining the upper tail of the distribution through AMS–GEV analysis. At the same time, the extreme rainfall results should be interpreted within the limitations imposed by the monthly dataset. The AMS in this study represents the maximum monthly rainfall total recorded in each year, rather than the maximum daily or sub-daily rainfall event. Consequently, the estimated return levels describe extreme monthly rainfall totals and should not be interpreted as estimates of extreme daily precipitation. This distinction is particularly important because much of the wider extreme-precipitation literature is based on daily or sub-daily observations.

4.6.3 Climate and Landscape Change

The climatic findings can also be considered alongside the observed changes in land use and land cover. The LULC analysis showed substantial changes in the composition and distribution of land-cover classes across the study area, particularly involving built-up, agricultural and vegetated areas. These changes indicate that the land surface surrounding the climatic environment has also undergone considerable transformation during the study period. Previous research provides useful regional context for this observation. Xu et al. (2020) documented substantial LULC transitions across Bangladesh and showed that different land-cover changes have different underlying drivers. More locally, Siddik et al. (2022) reported an 80.3% increase in impervious built-up areas and a 16.4% decrease in vegetated land cover in northwestern Bangladesh between 2006 and 2016. Their results also demonstrated that LULC changes can influence hydrological processes. The LULC changes observed in the present study therefore provide an important environmental context for interpreting the climatic results. However, spatial agreement between land-cover transformation and climatic change cannot by itself establish causality. The present analysis does not isolate the effects of LULC change from atmospheric circulation, regional climate variability and other possible drivers. LULC change is consequently

interpreted here as supporting environmental context rather than a direct cause of the observed climatic trends, consistent with the caution emphasized in previous research.

4.6.4 Interrelationships among Climatic Variables

The correlation analysis adds another dimension to the integrated interpretation by showing that temperature, rainfall, relative humidity and wind speed did not vary independently. The relationships differed among the four districts, which is consistent with the spatial heterogeneity identified in the earlier analyses. This finding agrees with Khan et al. (2019), who showed that relationships among climatic variables can vary across regions and seasons. Fattah et al. (2023) provided more direct support for the multi-method approach used in the present study by examining relationships among temperature, rainfall, relative humidity and wind speed using Pearson, Spearman and Kendall correlations. Their work also demonstrated the value of considering multiple correlation measures when assessing climatic interrelationships. The present results extend this approach to the four neighboring districts. The strongest and most consistent associations were found for selected temperature–humidity and humidity–wind speed relationships, while temperature–rainfall relationships were particularly evident in Rajshahi and Natore. However, the direction and strength of several relationships differed between districts. This suggests that climatic interactions are not spatially uniform even within a relatively confined region. These relationships should be interpreted as statistical associations rather than direct physical causes. Correlation analysis cannot determine whether a change in one climatic variable directly produced a change in another, because several variables may respond simultaneously to atmospheric and environmental processes that were not included in the analysis. This interpretation is consistent with the limitations discussed by Fattah et al. (2023).

4.6.5 Integrated Interpretation and Comparison with Previous Studies

When the findings are considered together, a consistent picture emerges. The four districts share the same broad seasonal climatic cycle, but they differ in their spatial characteristics, the magnitude of long-term changes, the behavior of extreme rainfall, and the relationships among climatic variables. Rainfall shows a predominantly decreasing long-term tendency, whereas temperature does not exhibit an equally uniform pattern across the districts. Relative humidity and wind speed also display spatially different

behavior, reinforcing the importance of examining multiple climatic variables within the same framework. The results further show that climatic change cannot be described adequately through annual trends alone. The seasonal analysis demonstrates when climatic variability is concentrated, while the AMS-GEV analysis shows that the behavior of extreme rainfall represents a separate dimension of the rainfall regime. The LULC analysis adds an environmental perspective to the observed changes, and the correlation analysis demonstrates that the climatic variables themselves are interconnected. This integrated approach addresses an important limitation identified in literature. Existing studies have generally focused on individual variables, selected locations, shorter periods, or analytical techniques. The literature review showed that no study identified for the present research had combined a common long-term assessment of rainfall, temperature, relative humidity and wind speed across Rajshahi, Pabna, Natore and Bogura with trend analysis, seasonal climatology, extreme monthly rainfall through GEV modelling, LULC assessment and multi-method correlation analysis. The present study therefore does not suggest that one factor explains all observed climatic changes. Instead, the results point to a spatially heterogeneous, strongly seasonal and interconnected climatic system, occurring alongside substantial changes in the land surface. This broader interpretation is more consistent with the complexity of climatic change than relying on any single trend, map, extreme-value estimate, or correlation coefficient.

Chapter 5

Conclusion and Recommendations

5.1 Major Findings

The five objectives set out in Chapter 1 called for an integrated assessment of spatial variation, long-term trends, seasonal behavior, extreme rainfall, and relationships among climatic variables across Rajshahi, Pabna, Natore, and Bogura. Chapter 4 examined these aspects separately, but the results point to a common pattern: climatic behavior across this part of northwestern Bangladesh is not uniform. Each objective reaches this conclusion from a different perspective. The spatial and LULC analysis (Objective 1) showed that rainfall, temperature, relative humidity, and wind speed each had distinct spatial patterns across the four districts, and these patterns also changed across the four periods considered. Rajshahi remained in the highest temperature class throughout the 1973-2023 period, generally 25.0-26.5°C compared with 22.0-23.5°C in the other districts. Rainfall showed almost the opposite pattern. Bogura and Pabna occupied the higher rainfall classes during the earlier period, whereas by 2012-2023 Rajshahi and Natore had shifted to the lowest class. Relative humidity showed an opposing pattern to rainfall over the same period, moving from the 72-75% class to 78-80% in most districts. This pattern is also reflected in the correlation results discussed later. The LULC assessment for 1985, 1998, 2011, and 2023 showed that these climatic changes occurred alongside substantial changes in land cover. In Rajshahi, vegetated land increased from 44.15% to 64.34%, while built-up land accounted for 14.18% in 2023. Bogura followed a different pattern, with vegetation declining after 1998 and bare land reaching 45.04%. Since several LULC classes changed simultaneously within each district, and because no built-up land was identified in the 1985 and 1998 classifications, which is more likely related to limitations in the imagery and classification resolution than a complete absence of built-up land, the LULC findings are treated here as environmental context rather than direct evidence of the causes of climatic change.

The trend analysis (Objective 2) provided stronger statistical evidence that the spatial differences were not simply a result of the mapping approach. Among the four districts, only Rajshahi showed a statistically significant warming trend, with a Sen's slope of +0.0153°C/year (MK $p < 0.001$). Natore, Pabna, and Bogura also showed positive slopes,

but their trends were not statistically significant. Rainfall, on the other hand, declined in all four districts. The decline was statistically significant in Natore, Pabna, and Rajshahi, with Sen's slopes of -8.99 , -7.84 , and -8.71 mm/year, respectively. In Bogura, the decline was slightly weaker at -7.04 mm/year and narrowly missed the 0.05 significance level ($p = 0.0553$). Relative humidity increased significantly in the same three districts where rainfall declined significantly, while wind speed decreased significantly only in Natore and Pabna. Bogura again stood apart: all four climatic variables showed directional tendencies in their fitted series, but none of the trends reached statistical significance. This contrast is important because it demonstrates why examining all four districts is necessary. A study based on only two or three locations could easily have overlooked this difference in climatic behavior.

Objective 3 placed these annual results within their seasonal cycle. Rainfall was strongly concentrated in the monsoon across all four districts, ranging from 1047.7 mm in Rajshahi to 1252.1 mm in Bogura, while winter rainfall was only about 29–35 mm across the study area. This means that the annual rainfall trend largely reflects changes during the four monsoon months rather than an even change throughout the year. Temperature, relative humidity, and wind speed followed their own seasonal patterns, with their highest and lowest values occurring at different times of the year. The seasonal results therefore add an important point to the trend analysis in Objective 2: the observed annual changes should not be assumed to be distributed evenly across all months.

The extreme-precipitation analysis (Objective 4) showed that annual maximum monthly rainfall declined during 1973–2023 in all four districts. The decline was statistically significant in Rajshahi, Natore, and Pabna, but not in Bogura, following the same general pattern observed for total annual rainfall. The GEV models provided acceptable fits for all four districts according to the Anderson–Darling test. However, the estimated return levels became increasingly different among the districts at longer return periods. The 100-year return level ranged from 663.8 mm in Natore to 981.9 mm in Pabna, while the confidence interval for Pabna alone extended from 708.8 to 1427.1 mm. Interestingly, Pabna did not have the highest 2-year return level. That was observed in Bogura, at approximately 468 mm, compared with 380.3 mm in Rajshahi. The relatively faster increase in Pabna's estimated return levels at longer return periods suggests a heavier upper tail in its fitted distribution. The wide confidence intervals are also important to consider. A 51-year record provides limited information about an event expected to occur

once in 100 years, so the long-return-period estimates presented in Section 4.4 should be interpreted with appropriate caution.

Objective 5 further showed how differently the climatic variables can behave from one district to another. The regional mean series showed a significant negative relationship between temperature and relative humidity ($r = -0.307$, $p = 0.0285$), but the district-level results were quite different. The relationship was significantly negative in Bogura ($r = -0.497$) and Pabna ($r = -0.360$), non-significant in Natore, and positive, although not statistically significant, in Rajshahi ($r = 0.253$). A similar spatial contrast appeared in the relationship between relative humidity and wind speed. The relationship was strongly negative in Rajshahi ($r = -0.451$) and Natore ($r = -0.363$) but was essentially absent in Pabna. These differences show why the regional means should not be treated as a direct representation of the individual districts. When districts differ not only in the strength of a relationship but also in its direction, averaging them together can hide the spatial variation that is central to the study.

5.2 Contributions

Chapter 2 identified a clear gap in existing literature. None of the studies reviewed for this thesis brought rainfall, temperature, relative humidity, and wind speed together across Rajshahi, Pabna, Natore, and Bogura within a common 1973-2023 framework that combined trend analysis, seasonal climatology, extreme-value analysis, LULC assessment, and multi-method correlation. The present study was designed around this gap, and the results in Chapter 4 show what this combined approach adds beyond the findings of individual studies. Ferdous and Baten (2012) and Khan et al. (2019) had already shown that climatic trends in northwestern Bangladesh vary from one location to another. The present study extends this understanding using a longer and more recent record, covering 1973-2023 rather than 1961-2010 and 1988-2017, respectively. It also examines four climatic variables within the same analytical framework, allowing their spatial and temporal behavior to be considered together rather than compared across separate studies.

The contribution becomes more specific when the results are compared with the two recent studies most closely related to the study area. Shahriyar et al. (2026) and Esha et al. (2026) provided useful evidence of rainfall trends for Rajshahi and the wider Barind Tract, respectively, but neither study applied GEV modelling to an annual maximum

series. Their findings therefore describe rainfall trends without addressing the magnitude and recurrence of extreme rainfall events. The AMS-GEV analysis in the present study extends this work by estimating return levels and their associated uncertainty, providing an additional perspective on extreme rainfall behavior in the same broader region. Fattah et al. (2023) demonstrated the usefulness of combining Pearson, Spearman, and Kendall correlations at the national scale. The present study applies the same multi-method approach at the district level and shows that the relationships among climatic variables can differ considerably even between neighboring districts. The LULC component provides a more limited but useful contribution. It does not demonstrate that land-cover change caused the observed climatic trends, nor was it designed to establish such a causal relationship. Instead, it documents and quantifies changes in the physical landscape of the four districts over the study period. This provides an environmental context for interpreting the climatic trends and correlations, recognizing that the landscape itself also changed substantially during the same decades in which the climatic variables were changing.

5.3 Recommendations

Because the main finding of this study is that climatic behavior differs among four districts within the same broader region, the most direct recommendation is that water-resource and agricultural planning in northwestern Bangladesh should consider district-level differences rather than treating the Barind region as a uniform climatic unit. Rajshahi, for example, showed both significant warming and a significant decline in rainfall, while Bogura showed directional changes in all four climatic variables without any reaching statistical significance. Applying the same planning assumptions to both districts could therefore overlook important local differences. Groundwater already plays an important role in irrigation across the region, and the significant decline in wind speed together with increasing humidity observed in Natore and Pabna also suggests that assumptions about evapotranspiration and irrigation scheduling based on older regional averages may need to be reconsidered at the district level.

The extreme-rainfall findings have a more specific implication for drainage and flood-preparedness planning. Although annual rainfall declined in three of the four districts, the return-level analysis shows that large monthly rainfall totals remain possible. This is particularly relevant for Pabna, where the estimated return levels increased considerably

during longer return periods. Infrastructure and drainage planning should therefore not be based only on the relatively lower rainfall amounts associated with short return periods. Several aspects of the present analytical framework could also be improved in future studies. First, testing for serial correlation before applying the Mann-Kendall test would strengthen the trend analysis and address a limitation shared with some of the studies reviewed in Chapter 2, including Esha et al. (2026). Second, obtaining a direct meteorological record for Natore would remove the need to estimate its climatic values through IDW interpolation from surrounding stations. Finally, the relationship between LULC change and climatic trends could be examined more formally in future work. The present study treats them as parallel lines of evidence, but a study incorporating additional environmental variables and an appropriate attribution or modelling framework could investigate whether the observed land-cover changes contributed to the climatic patterns identified here.

5.4 Limitations of the Study

Several limitations should be considered when interpreting the results of this study. The most important one concerns the temporal resolution of the rainfall data. Daily meteorological records were not available for the full 1973-2023 period, so the Annual Maximum Series was constructed from monthly rainfall totals rather than daily maximum. As a result, the GEV return levels presented in this study represent extreme monthly rainfall totals and should not be directly compared with studies based on daily extreme rainfall, which make up much of the existing extreme-precipitation literature. A second limitation is related to the spatial coverage of the meteorological data. Direct station observations were available for Rajshahi, Pabna, and Bogura, but no direct station record was available for Natore. The climatic series for Natore was therefore estimated to use Inverse Distance Weighted interpolation from the surrounding stations. The results for Natore should consequently be interpreted as estimates influenced by the interpolation method rather than as direct observations equivalent to those available for the other three districts. The study also does not attempt to establish causal relationships. The correlation analysis identifies statistical associations between climatic variables but does not explain the mechanisms behind those relationships. Similarly, the LULC analysis was used as environmental context for the climatic changes rather than as a direct explanatory variable. Therefore, the results cannot determine whether the observed climatic and land-

cover changes are causally related, or whether both are influenced by other processes beyond the scope of this study, such as regional atmospheric circulation or moisture transport. Another limitation concerns the Mann–Kendall trend analysis. The test was applied without a preceding formal assessment of serial correlation in the annual series. Yue, Pilon, and Cavadias (2002) demonstrated that positive serial correlation can inflate the likelihood of detecting a significant trend when using the standard Mann-Kendall test, and Yue and Wang (2002, 2004) proposed pre-whitening and effective-sample-size approaches to address this issue. Kamruzzaman et al. (2018) showed that ignoring lag-1 autocorrelation can influence the significance of detected trends. The present study did not apply an autocorrelation correction and therefore shares this limitation with several of the recent local studies discussed in Chapter 2.

Finally, some uncertainty is associated with the LULC classification. Built-up land was not detected in the 1985 or 1998 imagery for any of the districts. This most likely reflects limitations related to the resolution of the historical Landsat imagery and the dispersed nature of settlement during the earlier periods rather than a complete absence of built-up land. In addition, agricultural land could not always be reliably separated from vegetation and bare land during classification. The LULC percentages reported in Chapter 4 should therefore be interpreted primarily as an indication of broad land-cover changes over time, rather than as exact measurements of individual land conversions.

5.5 Future Research Directions

The limitations identified above also point to several clear directions for future research. If daily or sub-daily rainfall data becomes available for the four districts, the extreme-precipitation analysis could be repeated using a conventional daily Annual Maximum Series. This would make the resulting return-level estimates more directly comparable with the wider extreme-precipitation literature and would remove the monthly-data limitation of the present study. Similarly, obtaining a direct meteorological record for Natore would strengthen the district-level comparison by replacing the interpolated series with observed data. The LULC and climatic-trend results could also be taken a step further through a formal attribution analysis. Rather than treating land-cover change and climatic trends as separate findings, future studies could incorporate additional variables, such as regional atmospheric circulation indices or land-surface temperature, to examine whether changes in land cover have contributed measurably to local climatic conditions.

Extending the climatic record and repeating the Mann-Kendall and Sen's slope analyses at regular intervals would also help determine whether the currently non-significant trends in Bogura become more pronounced as additional years of data are added, or whether the district continues to differ from the other three. Finally, the same district-level approach could be extended to neighboring districts beyond Rajshahi, Pabna, Natore, and Bogura. A broader assessment would help determine whether the spatial differences identified in this study are characteristics of these four districts or reflect a wider pattern of climatic variability across northwestern Bangladesh.

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